ICMA214

Ordinary Differential Equations

Methods, Analysis and Applications

JUSTIN COPELAND

Taught by Chatchawan Panraksa

T1_2022-2023

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1 | Basic Mathematical Models

Class Logistics

This class covers linear first order equations and their applications, nonlinear first order equations, second order linear equations and their applications, higher order linear equations, and more.

Instructor

Associate Professor Dr. Chatchawan Panraksa (`chatchawan.pan@mahidol.edu`)

Grading Scheme

Item	Grade Percentage
2 Written Quizzes	20%
Group Project	10%
Midterm	35%
Final	35%

Exam Information

One piece of A4 paper will be allowed for notes for quizzes and exams

- **Midterm**: Saturday, 29th October 2022, 8:00-9:50AM
- **Final**: Thursday, 8th December 2022, 8:00-9:50AM

Textbook

W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 11th Edition, 2017.

Differentiable Equations

A **differential equation** refers to any equation that deals with derivatives. Why should we care about these sorts of equations? Many things that occur in the natural world involve changing rates. A differential equation that describes a physical processes is known as a **mathematical model**. Some examples of physical phenomena involving rates of change include

- Motion of fluids (e.g. The Navier-Strokes Equation)
- Motion of mechanical systems (e.g. a swinging pendulum)
- Flow of current in electrical circuits
- Dissipation of heat in solid objects
- Seismic waves
- Population dynamics

Free Fall

Suppose we wanted to formulate a differentiable equation describing motion of an object falling[1]. We know from Newton's second law that

$$F = ma \tag{1}$$

Alternatively, acceleration can be written as the change in velocity with respect to time. We can rewrite (1) to reflect this by substituting $a = \frac{dv}{dt}$:

$$F = m\left(\frac{dv}{dt}\right) \tag{2}$$

We also know our upward and downward forces. The force of gravity $F = mg$ acts as our downward force. The force of air resistance $F = \gamma v$ acts as our upward force. Thus, it follows that for any falling object

$$m\left(\frac{dv}{dt}\right) = mg - \gamma v$$

Taking $g = 9.8\text{m/sec}^2$, $m = 10\text{kg}$ and $\gamma = 2\text{kg/sec}$, we obtain

$$\frac{dv}{dt} = 9.8 - 0.2v \tag{3}$$

With our differential equation above, for any velocity v, we can find the corresponding value of $\frac{dv}{dt}$. Hooray, we've just modeled something using a differential equation!

Mice and Owls

Consider a mouse population that reproduces at a rate proportional to the current population – let us say 0.5 mice per month[2]. If owls are present, they eat the mice. Suppose that the owls eat 15 mice per day on average. This is a prime opportunity for us to use differential equations to describe mouse population. If we denote time by t and the mouse population at time t by $p(t)$, we can express the population by the equation

$$\frac{dp}{dt} = rp \tag{4}$$

where r is the **rate constant** or **growth rate** of the population.

With the unwritten assumption that the number of owls remains constant, if we count thirty days in a month, that would be a total of 450 mice eaten per month. An equation to describe the mouse population could look something like the following

$$\frac{dp}{dt} = 0.5p - 450 \tag{5}$$

[1] In the atmosphere near sea level
[2] Assuming no owls present

This is of a more general form of the population equation

$$\frac{dp}{dt} = rp - k \tag{6}$$

in which the growth rate r and predation rate k are positive constants.

Direction Fields

By plugging in various values for a differentiable equation, we can generate a graphical representation of the equation itself. This is known as a **direction field**. Consider equation (3) which we derived for free fall earlier. In a sense, we can trace the equation across various velocities to equilibrium by plotting t against v'.

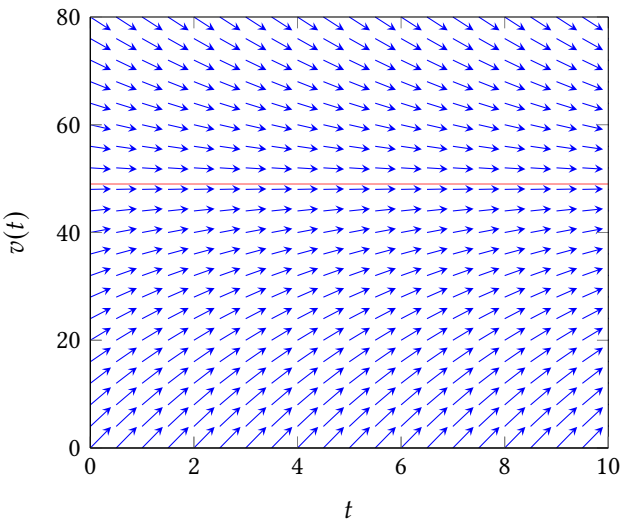


Figure 1.1: A direction field for the differential equation $dv/dt = 9.8 - 0.2v$

Arrows give tangent lines to solution curves and indicate where the solution is increasing and decreasing, as well as by how much. Horizontal solution curves are known as **equilibrium solutions**. We can do this for equation (5) we obtained for the mice and owls as well:

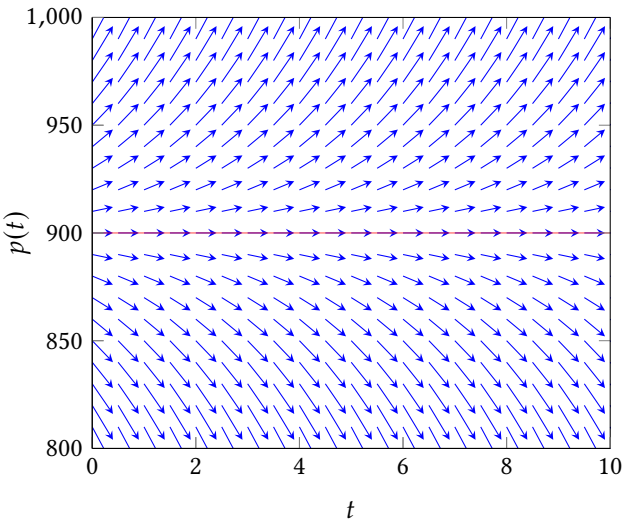


Figure 1.2: A direction field for the differential equation $dp/dt = 0.5p - 450$

Direction fields are very valuable tools in studying the solutions of differential equations of the form

$$\frac{dy}{dt} = f(t, y)$$

where f is a given function of the two variables t and y, and is sometimes called the **rate function**.

Equilibrium Solutions

In general, for a differentiable equation of the form

$$y' = ay - b$$

We can find equilibrium solutions by setting $y' = 0$ and solving for y

$$y(t) = \frac{b}{a}$$

Take for example, the free fall differentiable equation above

$$\begin{aligned} v' &= 9.8 - 0.2v \\ 0 &= 9.8 - 0.2v \\ v &= \frac{9.8}{0.2} \\ v &= 49 \text{ m/s} \end{aligned}$$

Steps in Constructing Mathematical Models Using Differential Equations

In order to apply differential equations to the variety of fields in which they are useful, we must first dream up an appropriate equation that can actually describe the problem being investigated. Each

problem is different, and requires different knowledge from different fields[3]. Here are some pro tips to aid in the construction of a successful model

1. Identify the independent and dependent variables and assign letters to represent them. It is often the case that the independent variable is time.
2. Choose the units of measurement for each variable. The choice of units may seem arbitrary but some choices are more logical and easier to work with than others.
3. Articulate the basic principle that underlies or governs the problem you are investigating. This may be a widely recognized physical law – such as Newton's second law of motion – or it may be a more specific assumption based on your own observations. May require a bit of familiarity with whatever field the problem originates from.
4. Express the principle or law from the previous step in terms of the variables chosen in the first step. This may lead to the introduction of physical constants or parameters, as well as the determination of appropriate values for them. It can also involve the use of auxiliary or intermediate variables that are related to the primary variables.
5. Make sure your equation is dimensionally consistent by checking that units are the same.
6. Obtain one or more differential equations as the result.

Solutions of Some Differential Equations

So far, we have derived the differential equations

$$m\frac{dv}{dt} = mg - \gamma v \tag{1}$$

and

$$\frac{dp}{dt} = rp - k \tag{2}$$

Equation (1) models a falling object and equation (2) models a population of field mice preyed on by owls. Both of these equations are of the general form

$$\frac{dy}{dt} = ay - b$$

We were able to draw some qualitative conclusions about the behaviors of the solutions of both equations by examining their associated direction fields. However, to answer any questions of a quantitative nature, we must find the solutions to the equations themselves. Consider the equation we came up with for the mice and the owls

$$\frac{dp}{dt} = 0.5p - 450$$

[3]In the case of the examples above, one required knowledge from physics and the other from ecology

One method of solving it could be as follows

$$\begin{aligned}
\frac{dp}{dt} &= 0.5p - 450 \\
\frac{dp}{dt} &= 0.5(p - 900) \\
\frac{dp/dt}{(p - 900)} &= \frac{1}{2} \\
\int \frac{dp}{p - 900} &= \int \frac{1}{2}\, dt \\
\ln|p - 900| &= 0.5t + C \\
|p - 900| &= e^{0.5t + C} \\
p - 900 &= \pm e^{C} e^{t/2}
\end{aligned}$$

Or in other words

$$p = 900 + ce^{t/2}$$

where $c = \pm e^{C}$ is an arbitrary (nonzero) constant. This equation would thus have infinitely many solutions. It is often the case that we would like to focus on a single member of the infinitely many solutions by specifying the value of the arbitrary constant. Most often, this is done by specifying a point that the must lie on the graph of the solution. Suppose we required to be 850 at time $t = 0$. Symbolically

$$p(0) = 850$$

Plugging this into our solution above, we obtain

$$850 = 900 + ce^{0/2}$$

Thus, $c = -50$.

Initial Value Problem

The addition of a condition used to determine c is an example of an **initial condition**. A differential equation combined with an initial condition forms an **initial value problem**. Let us now consider the more general problem consisting of the differential equation

$$\frac{dy}{dt} = ay - b$$

with the initial condition

$$y(0) = y_0$$

where y_0 is an arbitrary initial value. If $a \neq 0$ and $y \neq \frac{b}{a}$, we can rewrite the equation in a similar fashion to the mice and owls solution above

$$\begin{aligned}
\frac{dy}{dt} &= ay - b \\
\frac{dy}{dt} &= a(y - \frac{b}{a}) \\
\frac{dy/dt}{y - \frac{b}{a}} &= a \\
\frac{dy}{y - \frac{b}{a}} &= a\, dt
\end{aligned}$$

After integrating both sides, we find that

$$\ln\left|y(t) - \frac{b}{a}\right| = at + C$$

Taking the exponential of both sides, we obtain

$$y(t) = \frac{b}{a} + ce^{at} \tag{a}$$

where $c = \pm e^C$ is again, an arbitrary constant. We can see that $c = 0$ corresponds to the equilibrium solution $y(t) = \frac{b}{a}$. However, we must also take into account our initial condition:

$$\begin{aligned} y(0) &= y_0 \\ y_0 &= \frac{b}{a} + ce^0 \\ c &= y_0 - \frac{b}{a} \end{aligned}$$

So the solution of our initial value problem is

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a}\right)e^{at} \tag{b}$$

For $a \neq 0$, the equation (a) contains all possible solutions of $\frac{dy}{dt} = ay - b$ and is called the **general solution**. To relate the solution (b) to our mouse and owls equation, we need only replace a with the growth rate r and replace b by the predation rate k, assuming that $r, k > 0$.

Classification of Differential Equations

The following are several useful ways of classifying differential equations. They will assist in the selection of appropriate solution methods and description of their properties.

Ordinary and Partial Differential Equations

One important distinction is based on whether or not the function being examined depends on a single independent variable or on several. For the former case, where only ordinary derivatives appear in the equation, it is said to be an **ordinary differential equation**. In the second case the derivatives are partial derivatives, and the equation is said to be a **partial differential equation**.

All equations discussed so far have been ordinary differential equations. Some examples of partial differential equations are:

$$\begin{aligned} \alpha^2 \frac{\partial^2 u(x,t)}{\partial x^2} &= \frac{\partial^2 u(x,t)}{\partial t} \quad \text{heat equation} \\ a^2 \frac{\partial^2 u(x,t)}{\partial x^2} &= \frac{\partial^2 u(x,t)}{\partial t^2} \quad \text{wave equation} \end{aligned}$$

Systems of Differential Equations

Another classification of differential equations depends on the number of unknown functions that are involved. If there are two or more unknown functions, then a system of equations is required. For

example, predator-prey equations have the form

$$\frac{dx}{dt} = ax - \alpha xy$$
$$\frac{dy}{dt} = -cy + \gamma xy$$

where $x(t)$ and $y(t)$ are the respective populations of predator and prey species and constants a, c, α, γ depend on the species being studied. In some applications, it is possible to encounter very large systems of equations containing hundreds, or even thousands, of differential equations.

Order of Differential Equations

The **order** of a differential equation is the order of the highest derivative that appears in the equation. The equations covered so far have all been first order equations. An second order differential equation would be something like the following:

$$y'' + 3y' - 2t = 0$$

We will be studying differential equations for which the highest derivative can be isolated

$$y^{(n)}(t) = f\left(t, y, y', y'', y''', \ldots, y^{n-1}\right)$$

Linearity of Differential Equations

One of the most important classifications of differential equations is whether or not they are linear or non-linear. The ordinary differential equation

$$F\left(t, y, y', y'', y''', \ldots, y^{n}\right) = 0$$

is said to be **linear** if F is a linear function of the variables $y, y', \ldots, y^{(n)}$. A general linear ordinary differential equation of order n has the form

$$a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \cdots + a_n(t)y = g(t)$$

Essentially, in a differential equation, if the variables of the function and their derivatives are multiplied only by constants, the equation is linear[4]. A more in-depth explanation is that any differential equation that can be reduced to the form

$$Ly = f$$

where L is some linear operator, is a linear differential equation. If we had the following equation

$$4\frac{d^2y}{dx^2} - \sin(x)\frac{dy}{dx} + 5y = x^3e^x$$

It is linear because we could write it in the following manner

$$\underbrace{\left(4\frac{d^2}{dx^2} - \sin(x)\frac{d}{dx} + 5\right)}_{L} y = \underbrace{x^3e^x}_{f(x)}$$

[4]That is to say, the variables and their derivatives always appear as a simple first power

2 | First-Order Differential Equations

We will now be examining more first-order linear differential equations of the form

$$\frac{dy}{dt} f(t, y)$$

where f is a given function of two variables. Any differentiable function $y = \phi(t)$ that satisfies this equation for all t in some interval is called a solution. Our goal is to determine whether such functions exist, and to develop methods for finding them.

Method of Integrating Factors

In chapter 1, we discussed a restricted type of first order linear differential equation with constant coefficients, such as

$$\frac{dy}{dt} = -ay + b$$

where a and b are provided constants. Now we would like to consider the most general first-order linear differential equation, which is obtained by replacing the coefficients a and b above by arbitrary functions of t. In standard form

$$\frac{dy}{dt} + p(t)y = g(t)$$

In some instances, it is more convenient to write the equation in the form

$$P(t)\frac{dy}{dt} + Q(t)y = G(t)$$

where P, Q and G are given. In some cases, it's possible to solve a first order differential equation by mere integration, such as the example below.

Example

Solve the differential equation

$$(4 + t^2)\frac{dy}{dt} + 2ty = 4t$$

The left hand side of the equation is a result of the product rule from derivation. It can be rewritten as

$$(4 + t^2)\frac{dy}{dt} + 2ty = \frac{d}{dt}\left((4 + t^2)y\right)$$

So the equation becomes

$$\frac{d}{dt}\left((4 + t^2)y\right) = 4t$$

Although y is unknown, we can integrate both sides with respect to t, and obtain

$$(4 + t^2)y = 2t^2 + C$$

Solving for y, we get

$$y = \frac{2t^2 + C}{4t^2}$$

However, most first order linear equations cannot be solved as easily as this, as their left hand sides are not the derivative of the product of y and some function. Luckily, Leibniz discovered that if a differential equation is multiplied by some function $\mu(t)$, then we can obtain an equation that would be solvable in the manner above. The function $\mu(t)$ is called an **integrating factor**.

Integrating Factor for Variable Coefficients

To find a general method for determining the integrating factor, we can begin with an example. Suppose we had the equation

$$\frac{dy}{dt} + \frac{1}{2}y = \frac{1}{2}e^{t/3}$$

We would begin by multiplying an equation by a function $\mu(t)$, which is yet to be determined, giving us

$$\mu(t)\frac{dy}{dt} + \frac{1}{2}\mu(t)y = \frac{1}{2}\mu(t)e^{t/3}$$

We would like to choose $\mu(t)$ such that the left hand side of the equation is the derivative of the product of $\mu(t)y$. It should be the case that

$$\frac{d}{dt}(\mu(t)y) = \mu(t)\frac{dy}{dt} + \frac{d\mu(t)}{dt}y$$

To make this a reality, we would have to choose a $\mu(t)$ such that

$$\frac{d}{dt}\mu(t) = \frac{1}{2}\mu(t)$$

What well-known function in Calculus has its derivative equal to one half times the original function? In this case, it would have to be the exponential function e^x. In this case

$$\mu(t) = e^{t/2}$$

A more robust explanation would be

$$\begin{aligned} \frac{d}{dt}\mu(t) &= \frac{1}{2}\mu(t) \\ \frac{1}{\mu(t)}\frac{d}{dt}\mu(t) &= \frac{1}{2} \end{aligned}$$

Which is equivalent to

$$\begin{aligned} \frac{d}{dt}\ln|\mu(t)| &= \frac{1}{2} \\ \ln|\mu(t)| &= \frac{1}{2}t + C \\ \mu(t) &= ce^{t/2} \end{aligned}$$

However, in this instance we will choose c to be 1 as we do not require the most general form. With $\mu(t) = e^{\frac{t}{2}}$, we can solve the original equation as follows

$$\begin{aligned}
\frac{dy}{dt} + \frac{1}{2}y &= \frac{1}{2}e^{t/3} \\
e^{t/2}\frac{dy}{dt} + \frac{1}{2}e^{t/2}y &= \frac{1}{2}e^{5t/6} \\
\frac{d}{dt}[e^{t/2}y] &= \frac{1}{2}e^{5t/6} \\
e^{t/2}y &= \frac{1}{2}\int e^{5t/6}\,dt
\end{aligned}$$

We substitute $u = \frac{5t}{6}$ so $\frac{du}{dt} = \frac{5}{6}$ and $dt = \frac{6}{5}\,du$

$$\begin{aligned}
e^{t/2}y &= \frac{1}{2}\cdot\frac{6}{5}\int e^{u}\,du \\
e^{t/2}y &= \frac{3}{5}e^{5t/6} + C
\end{aligned}$$

Giving us the general solution

$$y = \frac{3}{5}e^{t/3} + Ce^{-t/2}$$

Integrating Factor for Variable RHS

We can also extend our method of integrating factors to equations of the form

$$\frac{dy}{dt} + ay = g(t)$$

where a and $g(t)$ are a constant and given function, respectively. Similarly to the variable coefficient case, we find that our integrating factor $\mu(t)$ must satisfy the following case

$$\frac{d\mu}{dt} = a\mu$$

However, in this case, our integrating factor would be $\mu(t) = e^{at}$. Multiplying our function by this, we obtain

$$e^{at}\frac{dy}{dt} + ae^{at}y = e^{at}g(t)$$

Or equivalently

$$\frac{d}{dt}(e^{at}y) = e^{at}g(t)$$

By integrating both sides, we find that

$$e^{at}y = \int e^{at}g(t)\,dt + C$$

If $g(t)$ is a simple function, it is possible for us to evaluate the integral. However, if it is more complicated, it may be necessary to leave it in integral form. In this case

$$y = e^{-at}\int_{t_0}^{t} e^{as}g(s)\,ds + ce^{-at}$$

In this instance, s denotes the integration variable to distinguish it from our independent variable t, and a value t_0 is chosen as the lower limit for integration.

Integrating Factors for General First Order Linear Equations

Let us now return to our general first-order linear differential equation

$$y' + p(t)y = g(t)$$

where p and g are given functions. Following the same method, if we multiply the equation by an undetermined function $\mu(t)$, we obtain

$$\mu(t)\frac{dy}{dt} + p(t)\mu(t)y = \mu(t)g(t)$$

We can see that the left hand side is the derivative of the product $\mu(t)y$ provided that $\mu(t)$ satisfies the following

$$\frac{d\mu(t)}{dt} = p(t)\mu(t)$$

Assuming temporarily that $\mu(t)$ is positive

$$\begin{aligned} \frac{d\mu(t)}{dt} &= p(t)\mu(t) \\ \frac{1}{\mu(t)}\frac{d\mu(t)}{dt} &= p(t) \\ \ln|\mu(t)| &= \int p(t)\,dt + k \end{aligned}$$

By choosing the arbitrary constant k to be zero, we obtain the simplest possible function for μ, which is

$$\mu(t) = e^{\int p(t)\,dt}$$

As we assumed previously, it is in fact the case that $\mu(t)$ is positive for all t. Returning to our earlier equation, we have

$$\begin{aligned} \mu(t)\frac{dy}{dt} + p(t)\mu(t)y &= \mu(t)g(t) \\ \frac{d}{dt}(\mu(t)y) &= \mu(t)g(t) \\ \mu(t)y &= \int \mu(t)g(t)\,dt + C \end{aligned}$$

Sometimes the integral can be evaluated in terms of elementary functions, however, it is not generally the case that this is possible. So the general solution becomes

$$y = \frac{\int \mu(t)g(t)\,dt + c}{\mu(t)}, \quad \text{where } \mu(t) = e^{\int p(t)\,dt}$$

Differences Between Linear and Nonlinear Equations

First order linear and nonlinear differential equations differ in a number of ways, including

- The theory describing existence and uniqueness of solutions, as well as corresponding domains.
- The ability to be expressed in terms of a general solution.

- Whether or not there is an explicitly defined solution, of which linear equations hold but nonlinear equations typically do not.

For both kinds of equations, numerical and graphical construction of solutions is very valuable. However, questions about the existence and uniqueness of solutions can be answered, depending on the linearity of the equation. For linear equations, the answers to these questions are given by the following theorem

Existence and Uniqueness Theorem for First-Order Linear Equations

If the functions p and g are continuous on an open interval $I : \alpha < t < \beta$ containing the point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the differential equation

$$y' + p(t)y = g(t)$$

for each t in I, and that also satisfies the initial condition

$$y(t_0) = y_0$$

where y_0 is an arbitrary prescribed initial value.

The theorem states that the given initial value problem has a solution and also that the problem has only one solution. It also states that the solution can be discontinuous or fail to exist only at points where at least one of p and g is discontinuous. We can replace this theorem with a more general one for nonlinear differential equations

Existence and Uniqueness Theorem for First-Order Nonlinear Equations

Let the functions f and $\partial f/\partial y$ be continuous in some rectangle $\alpha < t < \beta, \gamma < y < \epsilon$ containing the point (t_0, y_0). Then, in some interval $t_0 - h < t < t_0 + h$ containing in $a < t < \beta$, there is a unique solution $y = \phi(t)$ of the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0$$

Autonomous Equations and Population Dynamics

In this section, we examine first order equations of the form

$$y' = f(y)$$

which are known as **autonomous** equations, where the independent variable t does not explicitly appear. This is a "time-independent" equation, in which y' depends on the current quantity of the function $f(y)$. We will learn how geometric methods can be used to obtain qualitative information directly from a differential equation without solving it. One example of an autonomous system is population growth, where the current growth rate is dependent on the current population.

Exponential Growth

Let $y = \phi(t)$ be the population of a given species at time t. To model this population, we can say that the rate of change of y is proportional to the current value of y

$$\frac{dy}{dt} = ry$$

where the rate of growth or decline is r. Assuming that the population is growing ($r > 0$), we can solve the equation subject to the initial condition

$$y(0) = y_0$$

and obtain

$$y = y_0 e^{rt}$$

This model predicts that the population will grow exponentially for all time. This can be observed for some populations in ideal conditions but only for limited periods of time, as ideal conditions cannot continue indefinitely.

Logistic Growth

To take account of the fact that the growth rate actually depends on the population, we can replace the constant r in the previous equation by a function $h(y)$

$$\frac{dy}{dt} = h(y)y$$

We would like to choose $h(y)$ so that $h(y) \cong r > 0$ when y is small, $h(y)$ decreases as y grows larger and $h(y) < 0$ when y is sufficiently large. The simplest function for this is

$$h(y) = r - ay$$

where $a > 0$. Our differential equation then becomes

$$\frac{dy}{dt} = (r - ay)y, \quad r, a > 0$$

This is known as the **Verhulst equation** or the **logistic equation**. It is often convenient to write the logistic equation in the equivalent form

$$\frac{dy}{dt} = r\left(1 - \frac{y}{K}\right)y$$

where $K = \frac{r}{a}$. The constant r is called the **intrinsic growth rate** and K represents the **carrying capacity** of the population. We can visualize the equation using a direction field with $r = 1$ and $K = 10$.

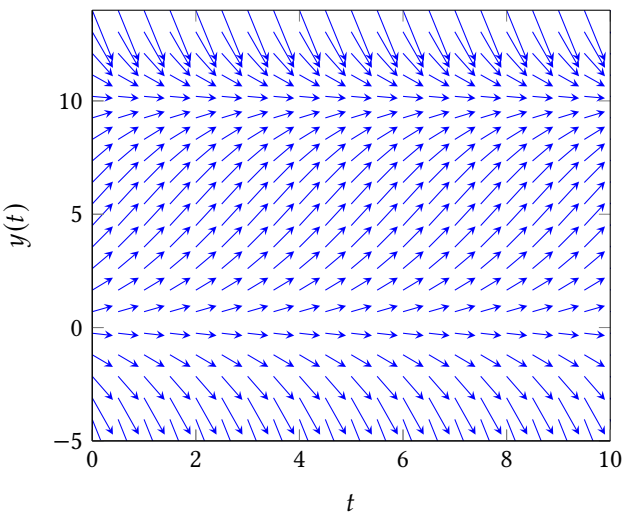


Figure 2.1: A direction field for the logistic equation with $r = 1$ and $K = 10$

Note how two equilibrium solutions are clearly present.

$$1.\ y = \phi_1(t) = 0$$
$$2.\ y = \phi_2(t) = K$$

we can see the behavior of the solutions as well. At $y = 0$, things appear to be **unstable**. Any value close to this will try to escape. At $y = 10$, it is **asymptotically stable**.

The Logistic Equation

In order to better understand the nature of solutions to autonomous equations, we can start by graphing $f(y)$ against y. In the case of logistic growth, that means graphing the following function

$$f(y) = r\left(1 - \frac{y}{K}\right)y$$

and analyzing its graph using calculus.

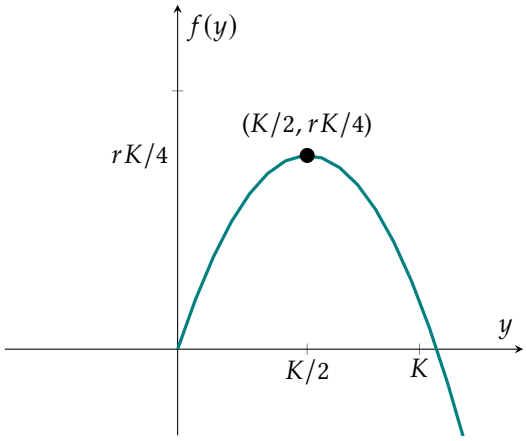


Figure 2.2: A graph of the logistic function

We can see that y will increase up until we reach the point $(K/2, rK/4)$, and then proceed until K, where the population growth becomes negative. When $\frac{dy}{dt} \cong 0$ when $y \cong 0$ or $y \cong K$, so y is relatively flat there and gets steep as y moves away from 0 or K.

To solve the logistic equation, provided that $y \neq 0$ and $y \neq K$, we can rewrite the logistic ODE as

$$\frac{dy}{(1 - y/K)y} = rdt$$

Expanding the left side using partial fractions, we obtain

$$\frac{1}{(1 - y/K)y} = \frac{A}{1 - y/K} + \frac{B}{y}$$

So

$$1 = Ay + B(1 - y/K)$$
$$B = 1, A = 1/K$$

Thus, the logistic equation can be rewritten as

$$\left(\frac{1}{y} + \frac{1/K}{1 - y/K}\right) dy = rdt$$

Integrating the above result, we obtain

$$\ln|y| - \ln|1 - \frac{y}{K}| = rt + C$$

If $0 < y_0 < K$, then $0 < y < K$ (we don't start with a negative population) and hence

$$\ln y - \ln\left(1 - \frac{y}{K}\right) = rt + C$$

Rewriting this using properties of logarithms

$$\begin{aligned} \ln\left[\frac{y}{1 - y/K}\right] &= rt + C \\ \frac{y}{1 - y/K} &= e^{rt+C} \\ \frac{y}{1 - y/K} &= Ce^{rt} \end{aligned}$$

Or

$$y = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}}$$

As our time increases, the e^{-rt} term will become zero as $t \longrightarrow \infty$. The equation itself will converge to K, just as we stated earlier.

Critical Threshold Equation

Consider the following modification of the logistic ODE

$$\frac{dy}{dt} = -r(1 - \frac{y}{T})y, r > 0$$

The graph is given below

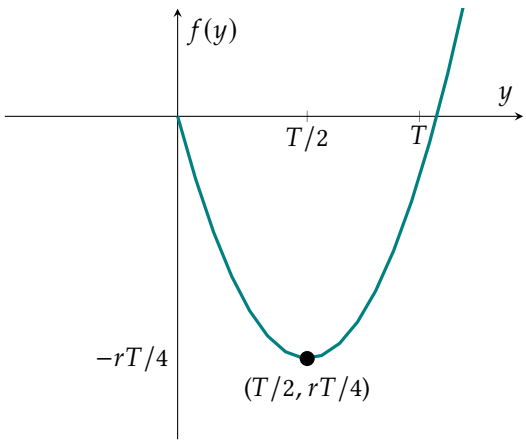


Figure 2.3: A graph of a modified version of the logistic equation

Now we have changed K to something called T, our **threshold**. T is a threshold value for y_0, in that the population dies off or grows unbounded, depending on which side of T the initial value starts. It can be shown that the solution to the threshold equation

$$\frac{dy}{dt} = -r(1 - \frac{y}{T})y, \quad r > 0$$

is

$$y = \frac{y_0 T}{y_0 + (T - y_0)e^{rt}}$$

Separable Equations

Previously, we used direct integration to solve first-order linear differential equations of the form

$$\frac{dy}{dt} = ay + b$$

where a and b are constants. We can show that this process is applicable to a much larger class or nonlinear differential equations. We will be using x rather than t to denote the independent variables in this section. Consider the general first order differential equation

$$\frac{dy}{dx} = f(x, y)$$

If

$$\frac{dy}{dx} = g(x) \cdot h(y)$$

we consider the equation to be **separable** – i.e. we can factor $f(x, y)$ into parts. This can be rewritten in the form

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

It is always possible to do this by setting $M(x, y) = -f(x, y)$ and $N(x, y) = 1$, but there may be other ways as well. Such an equation is said to be **separable** because if it is written in the differential form

$$M(x)\, dx + N(y)\, dy = 0$$

then terms involving each variable may be placed on opposite sides of the equation, and the equation itself can be solved by integrating the functions M and N.

Exact Equations and Integrating Factors

Consider a first order ordinary differential equation of the form

$$M(x, y)\, dx + N(x, y)\, dy = 0$$

Or

$$\frac{dy}{dx} = \frac{-M(x, y)}{N(x, y)}$$

We would like to find a function ψ, whose derivative with respect to x (ψ_x) is M, and whose derivative with respect to y (ψ_y) is N. In other words, a function ψ such that

$$\psi_x(x, y) = M(x, y)$$
$$\psi_y(x, y) = N(x, y)$$

If it is the case that

$$\frac{\partial}{\partial y}M = \frac{\partial}{\partial x}N$$

we say that our our equation is **exact**.

Example

Take for example, the following equation

$$2xy\,dx + (1 + x^2)\,dy = 0$$

In this case, we can see that

$$\begin{aligned} M &= 2xy \\ N &= (1 + x^2) \end{aligned}$$

To test for exactness, we need to compare M_y and N_x – that is, the derivative of M with respect to y and the derivative of N with respect to x. For M

$$\begin{aligned} \frac{\partial}{\partial y}M &= \frac{\partial}{\partial y}(2xy) \\ &= 2x\frac{\partial y}{\partial y} \\ &= 2x \end{aligned}$$

For N

$$\begin{aligned} \frac{\partial}{\partial x}N &= \frac{\partial}{\partial x}(1 + x^2) \\ &= 2x \end{aligned}$$

Therefore, the equation is exact, as

$$\frac{\partial}{\partial x}M = 2x = \frac{\partial}{\partial y}N$$

In other words, there exists a function $\psi(x, y)$ such that

$$\psi_x = M, \psi_y = N$$

To determine our ψ, we need only to integrate M or N with respect to x or y, respectively.

$$\begin{aligned} \psi(x, y) &= \int \psi_x\,dx \\ &= \int M\,dx \\ &= \int \psi_y\,dy \\ &= \int N\,dy \end{aligned}$$

We know from M that

$$\begin{aligned}\psi &= \int M\,dx \\ &= \int 2xy\,dx \\ &= y\int 2x\,dx \\ &= y(x^2 + C_1)\end{aligned}$$

We also know from N

$$\begin{aligned}\psi &= \int N\,dy \\ &= \int (1 + x^2)\,dy \\ &= (1 + x^2)\int dy \\ &= (1 + x^2)y + C_2\end{aligned}$$

Since both results are ψ, then they must be equal

$$y(x^2 + C_1) = (1 + x^2)y + C_2$$

We can see by eye that $C_1 = 1$, but alternatively, we can expand everything out

$$yx^2 + C_1 y = y + yx^2 + C_2$$

So C_1 must be 1, giving us

$$\psi(x, y) = yx^2 + y + C_2$$

One could also think of C_2 as being zero, but seeing as how we are finding the solution to a differential equation, having

$$\psi(x, y) = C$$

is fine. Since all we know is that $\psi_x = M$ and $\psi_y = N$, it is perfectly possible for ψ itself to be a constant. So, the solution would be in the form

$$\psi(x, y) = yx^2 + y + C \qquad \text{(implicit form)}$$

And we can solve for y

$$y = \frac{C}{x^2 + 1}$$

Exact Equation Steps

To solve an exact equation, we must begin with an equation in the form

$$M\,dx + N\,dy = 0$$

We first check if the equation is exact by comparing

$$\frac{\partial}{\partial y}M = \frac{\partial}{\partial x}N$$

If the above is true, this guarantees we can find a ψ such that

$$\psi = \int M\,dy \text{ and } \psi = \int N\,dx$$

By integrating both and equating the answers, we can solve for an equation

$$\psi(x, y) = C$$

Which is our solution.

Integrating Factor for Non-Exact Equations

If we have an equation in the form

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

But the equation is not exact, or

$$M_y \neq N_x$$

It is possible to turn the equation into an exact one by multiplying it with an integrating factor μ, similar to linear first-order differential equations. To find our function μ, we assume that the integrating factor we are looking for is of the form $\mu(x)$ or $\mu(y)$.

If we view $\mu = \mu(x)$ – as a function of x – when we multiply it to the equation, we obtain

$$\underbrace{\mu(x)M}_{M_1}\,dx + \underbrace{\mu(x)N}_{N_1}\,dy = 0$$

Assuming, of course, that the equation is not already exact, otherwise we would have our work cut out for us. We will assume that our new equation

$$M_1\,dx + N_1\,dy = 0$$

is exact. It follows that

$$\psi_x = \mu M$$
$$\psi_y = \mu N$$

Since the equation is exact, that means

$$\psi_{xy} = \mu\frac{\partial M}{\partial y} = \psi_{yx} = \underbrace{\mu_x N + \mu N_x}_{\text{Product Rule}}$$

Since they are equal

$$\mu\frac{\partial M}{\partial y} = \mu_x N + \mu N_x$$

We can now solve for μ

$$
\begin{aligned}
\mu\frac{\partial M}{\partial y} &= \mu_x N + \mu N_x \\
\mu_x N &= \mu M_y - \mu N_x \\
\mu_x &= \frac{\mu(M_y - N_x)}{N} \\
\frac{\frac{d\mu}{dx}}{\mu} &= \int \frac{M_y - N_x}{N} \\
\frac{d\mu}{\mu} &= \left(\frac{M_y - N_x}{N}\right) dx \\
\ln|\mu| &= \int \left(\frac{M_y - N_x}{N}\right) dx \\
\ln|\mu| &= \int g(x)\, dx \\
\mu &= e^{\int g(x)\, dx}
\end{aligned}
$$

3 Second-Order Linear Differential Equations

So far we have explored first-order linear differential equations. However, second-order linear differential equations are essential to analyze any areas of mathematical physics – e.g. fluid mechanics, heat conduction, wave motion and electromagnetic phenomena.

Second-Order Linear Differential Equations

Most second-order ordinary differential equations have the form

$$\frac{d^2y}{dt^2} = f\left(t, y, \frac{dy}{dt}\right)$$

where the function f is given. We use t (or x) to denote the independent variable and y to denote the dependent variable. A second-order ODE is said to be **linear** if the function f has the form

$$f\left(t, y, \frac{dy}{dt}\right) = g(t) - p(t)\frac{dy}{dt} - q(t)y$$

where g, p and q are specified functions of the independent variable (but do not depend on y). This form is usually rewritten as the following

$$y'' + p(t)y' + q(t)y = g(t)$$

where the primes denote differentiation with respect to the independent variable. If a second-order ODE is not in this form, it is said to be a **nonlinear** equation, which are quite difficult to approach in an analytical fashion.

Initial Value Problem for Second Order Linear Differential Equations

An initial value problem for a second order linear differential equation consists of a pair of initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_0'$$

where y_0 and y_0' are given initial conditions for y and y' at the initial point t_0. Note that now we also need to identify the slope y_0' of the graph at that point, as well as a particular point (t_0, y_0) through which the graph must pass.

Homogeneity of Second-Order Linear Differential Equations

A second-order linear ODE

$$y'' + p(t)y' + q(t)y = g(t)$$

is said to be **homogeneous** if the term $g(t)$ is zero for all t. Otherwise, the equation is called **nonhomogeneous**. The nonhomogeneous term $g(t)$ is sometimes referred to as the **forcing function**, as in many applications it describes an externally applied force.

Principle of Superposition and The Wronskian Determinant

To discuss general properties of linear differential equations, it's useful to introduce a **differential operator** notation. If p and q are continuous functions on an open interval I, then for any function ϕ that is twice differential on I, we define the differential operator L as follows

$$L[\phi] = \phi''(t) + p(t)\phi'(t) + q(t)\phi(t)$$

As we are dealing with second-order linear homogeneous differential equations, $L[\phi](t) = 0$. It is customary to use the symbol y to denote $\phi(t)$, and to associate the equation with a set of initial conditions

$$L[y] = y'' + p(t)y' + q(t)y = 0, \quad y(t_0) = y_0,\ y'(t_0) = y_0'$$

Assume that y_1 and y_2 are two solutions of the equation above. In other words

$$L[y_1] = y_1'' + py_1' + qy_1 = 0 \tag{$\star$}$$

and the same for y_2. Then, it is possible to generate more solutions by forming linear combinations of y_1 and y_2. This is stated in the theorem below.

Principle of Superposition

If y_1 and y_2 are solutions to the equation

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

then the linear combination $c_1y_1 + c_2y_2$ is also a solution $\forall\ c1, c2$

We can prove this by substituting $y = c_1y_1(t) + c_2y_2(t)$ for y in equation $\star$:

$$\begin{aligned} L[c_1y_1 + c_2y_2] &= [c_1y_1 + c_2y_2]'' + p[c_1y_1 + c_2y_2]' + q[c_1y_1 + c_2y_2] \\ &= c_1y'' + c_2y_2'' + c_1py_1' + c_2py_2' + c_1qy_q + c_2qy_2 \\ &= c_1[y_1'' + py_1' + qy_1] + c_2[y_2'' + py_2' + qy_2] \\ &= c_1L[y_1] + c_2L[y_2] \end{aligned}$$

As $L[y_1] = 0$ and $L[y_2] = 0$, it follows that $L[c_1y_1 + c_2y_2] = 0$ as well. Therefore, regardless of the values of c_1 and c_2, the function

$$y = c_1y_1(t) + c_2y_2(t)$$

satisfies the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

Therefore, for any two solutions y_1 and y_2, we can build an infinite family of solutions, each of the form $y = c_1y_1 + c_2y_2$. However, to determine whether or not all solutions can be written in this manner, we use the **Wronskian Determinant**.

Suppose we wanted to find coefficients such that $y = c_1 y_1 + c_2 y_2$ satisfies the initial conditions

$$y(t_0) = y_0, y'(t_0) = y'_0$$

To do so, we would have to solve the following equations

$$\begin{aligned} c_1 y_1(t_0) + c_2 y_2(t_0) &= y_0 \\ c_1 y'_1(t_0) + c_2 y'_2(t_0) &= y'_0 \end{aligned}$$

Solving the equations, we obtain the following

$$\begin{aligned} c_1 &= \frac{y_0 y'_2(t_0) - y'_0 y_2(t_0)}{y_1(t_0) y'_2(t_0) - y'_1(t_0) y_2(t_0)} \\ c_2 &= \frac{-y_0 y'_1(t_0) + y'_0 y_1(t_0)}{y_1(t_0) y'_2(t_0) - y'_1(t_0) y_2(t_0)} \end{aligned}$$

In terms of determinants

$$c_1 = \frac{\begin{vmatrix} y_0 & y_2(t_0) \\ y'_0 & y'_2(t_0) \end{vmatrix}}{\begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y'_1(t_0) & y'_2(t_0) \end{vmatrix}}, c_2 = \frac{\begin{vmatrix} y_1(t_0) & y_0 \\ y'_1(t_0) & y'_0 \end{vmatrix}}{\begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y'_1(t_0) & y'_2(t_0) \end{vmatrix}}$$

In order for both of these formulas to be valid, the determinant W in the denominator cannot be zero

$$c_1 = \frac{\begin{vmatrix} y_0 & y_2(t_0) \\ y'_0 & y'_2(t_0) \end{vmatrix}}{W}, c_2 = \frac{\begin{vmatrix} y_1(t_0) & y_0 \\ y'_1(t_0) & y'_0 \end{vmatrix}}{W}$$

where

$$W = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y'_1(t_0) & y'_2(t_0) \end{vmatrix} = y_1(t_0) y'_2(t_0) - y'_1(t_0) y_2(t_0)$$

W is called the Wronskian determinant[1] of the solutions y_1 and y_2.

Homogeneous Differential Equations with Constant Coefficients

In this chapter, we will focus on second-order homogeneous linear ordinary differential equations with constant coefficients. In this case, an equation of the form

$$P(t)y'' + Q(t)y' + R(t)y = 0$$

where the functions P, Q and R are constants. In this case, the equation becomes

$$ay'' + by' + cy = 0$$

where a, b and c are given constants. To handle this equation, we begin by seeking exponential solutions of the form $y = e^{rt}$, where r is a parameter to be determined. It then follows that

$$\begin{aligned} y' &= re^{rt} \\ y'' &= r^2 e^{rt} \end{aligned}$$

[1] Or sometimes just the Wronskian

By substituting this into y, y' and y'', we obtain the following equation

$$(ar^2 + br + c)e^{rt} = 0$$

and because $e^{rt} \neq 0$, this condition is only satisfied when the other factor is zero

$$ar^2 + br + c = 0$$

The equation above is known as the **characteristic equation** for the differential equation

$$ay'' + by' + cy = 0$$

If r is a root of the characteristic equation, then $y = e^{rt}$ is a solution of the differential equation above. Since the characteristic equation is a quadratic equation with real coefficients, it has two roots r_1 and r_2, which can be

1. **Real and different** ($r_1 \neq r_2$) – in which case the solutions are

   $$y_1 = c_1 e^{r_1 x}, \quad y_2 = c_2 e^{r_2 x}$$

   And we get the general solution

   $$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

2. **Complex conjugates** ($r_1 \neq r_2$) – in which case the roots are

   $$r_1 = \lambda + i\mu, \quad r_2 = \lambda - i\mu, \qquad \lambda, \mu \in \mathbb{R}$$

   The corresponding solutions for y are now

   $$y_1 = c_1 e^{r_1 x}, \quad y_2 = c_2 e^{r_2 x}$$

   We then obtain the general solution (derived below) as

   $$y(x) = c_1 e^{\lambda x} \cos \mu x + c_2 e^{\lambda x} \sin \mu x$$

3. **Real and repeated** ($r_1 = r_2$) – in which case we will perturb the second solution by x to make it a solution independent of the other

   $$y_1 = e^{rx}, \quad y_2 = xe^{rx}$$

Complex Roots and Euler's Formula

We have seen that when roots for the characteristic equation are complex, solutions for y involve raising the constant e to a complex power. However, what exactly does it mean to do this? The answer is provided by **Euler's Formula**. Without delving too much into details, it basically involves substituting it for t in the Taylor series for e^t about $t = 0$. This works out to be

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n)!} + i \sum_{n=1}^{\infty} \frac{(-1)^{n-1} t^{2n-1}}{(2n-1)!} = \cos t + i \sin t$$

which is a form of Euler's Formula. We can negate t and recall that $\cos(-t) = \cos t$ and $\sin(-t) = -\sin t$ and obtain

$$e^{-it} = \cos t - i \sin t$$

Furthermore, if t is replaced by μt, we obtain a more generalized version of the formula

$$e^{i\mu t} = \cos(\mu t) - i\sin(\mu t)$$

We would now like to extend the definition of the exponential function to include complex exponents of the form $(\lambda + i\mu)t$. Since we would like to have the usual properties of the exponential function to hold, we would like to to be the case that

$$e^{(\lambda+\mu i)t} = e^{\lambda t}e^{\mu i t}$$

Substituting our definition of $e^{i\mu t}$ we obtained earlier in, we get

$$e^{(\lambda+\mu i)t} = e^{\lambda t}(\cos(\mu t) - i\sin(\mu t)) = e^{\lambda t}\cos(\mu t) - ie^{\lambda t}\sin(\mu t)$$

We take this function as the definition of $e^{(\lambda+\mu i)t}$. This gives us the solutions

$$y_1(t) = e^{(\lambda+i\mu)t} = e^{\lambda t}\cos\mu t + ie^{\lambda t}\sin\mu t$$
$$y_2(t) = e^{(\lambda i-\mu)t} = e^{\lambda t}\cos\mu t - ie^{\lambda t}\sin\mu t$$

However, ideally, we'd like to get rid of the imaginary constant. To achieve this, recall that linear combinations of solutions are themselves solutions

$$y_1(t) + y_2(t) = 2e^{\lambda t}\cos\mu t$$
$$y_1(t) - y_2(t) = 2ie^{\lambda t}\sin\mu t$$

Ignoring any potential constants, we obtain the solution

$$y_3(t) = e^{\lambda t}\cos\mu t, \quad y_4(t) = e^{\lambda t}\sin\mu t$$

To check if these solutions are linearly independent, we can take the Wronskian Determinant

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

In this instance, we obtain

$$W = \begin{vmatrix} e^{\lambda t\cos\mu t} & e^{\lambda t}\sin\mu t \\ e^{\lambda t}(\lambda\cos\mu t - \mu\sin\mu t) & e^{\lambda t}(\lambda\sin\mu t + \mu\cos\mu t) \end{vmatrix} = \mu e^{2\lambda t} \neq 0,\ \forall\, t$$

Therefore, the solutions really are linearly independent, and y_3 and y_4 form a fundamental solution set for our ordinary differential equation. The general solution can be expressed as

$$y(t) = c_1e^{\lambda t}\cos\mu t + c_2e^{\lambda t}\sin\mu t$$

Repeated Roots

Recall that when we have repeated roots for a second-order linear homogeneous ordinary differential equation, using the quadratic formula gives one solution

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When $b^2 - 4ac = 0$, $r_1 = r_2 = \frac{-b}{2a}$. So we obtain the solution

$$y_1(t) = ce^{-bt/2a}$$

But we would like a second solution as well. We know that if $y_1(t)$ is a solution, $y_2(t) = cy_1(t)$ is also a solution. Since y_1 and y_2 are linearly dependent, we generalize this approach and multiply by a function v, and determine conditions for which y_2 is a solution

$$y_2(t) = v(t)e^{-bt/2a}$$

We can then take the first and second derivative (as our equation is second-order)

$$\begin{aligned}
y_2(t) &= v(t)e^{-bt/2a} \\
y_2'(t) &= v'(t)e^{-bt/2a} - \frac{b}{2a}v(t)e^{-bt/2a} \\
y_2''(t) &= v''(t)e^{-bt/2a} - \frac{b}{2a}v'(t)e^{-bt/2a} - \frac{b}{2a}v'(t)e^{-bt/2a} + \frac{b^2}{4a^2}v(t)e^{-bt/2a}
\end{aligned}$$

Substituting these derivatives into an ordinary differential equation, we seek a formula for v:

$$\begin{aligned}
&e^{-bt/2a}\left\{a\left[v''(t) - \frac{b}{a}v'(t) + \frac{b^2}{4a^2}v(t)\right] + b\left[v'(t) - \frac{b}{2a}v(t)\right] + cv(t)\right\} = 0 \\
&av''(t) - bv'(t) + \frac{b^2}{4a}v(t) + bv'(t) - \frac{b^2}{2a}v(t) + cv(t) = 0 \\
&av''(t) + \left(\frac{b^2}{4a} - \frac{b^2}{2a} + c\right)v(t) = 0 \\
&av''(t) + \left(\frac{b^2}{4a} - \frac{2b^2}{4a} + \frac{4ac}{4a}\right)v(t) = 0 \Leftrightarrow av''(t) + \left(\frac{-b^2}{4a} + \frac{4ac}{4a}\right)v(t) = 0 \\
&av''(t) - \left(\frac{b^2 - 4ac}{4a}\right)v(t) = 0 \\
&v''(t) = 0 \Rightarrow v(t) = k_3t + k_4
\end{aligned}$$

For the last two steps, we know that $b^2 - 4ac$ is zero because we are dealing with repeated roots. This leaves us with $v''(t) = 0$. This means that $v(t)$ must be no more than a first order equation, as taking the derivative twice must give us zero. To find our general solution, we have

$$\begin{aligned}
y(t) &= k_1e^{-bt/2a} + k_2v(t)e^{-bt/2a} \\
&= k_1e^{-bt/2a} + (k_3t + k_4)e^{-bt/2a} \\
&= c_1e^{-bt/2a} + c_2te^{-bt/2a}
\end{aligned}$$

Thus, the general solution for repeated roots is

$$y(t) = c_1e^{-bt/2a} + c_2te^{-bt/2a}$$

We can check the Wronskian to determine the independence of the solutions

$$\begin{aligned}
W(y_1, y_2)(t) &= \begin{vmatrix} e^{-bt/2a} & te^{-bt/2a} \\ -\frac{b}{2a}e^{-bt/2a} & \left(1 - \frac{bt}{2a}\right)e^{-bt/2a} \end{vmatrix} \\
&= e^{-bt/a}\left(1 - \frac{bt}{2a}\right) + e^{-bt/a}\left(\frac{bt}{2a}\right) \\
&= e^{-bt/a} \neq 0 \text{ for all } t
\end{aligned}$$

Example Problems

Example 1

Consider the differential equation

$$y'' - y' - 2y = 0$$

The characteristic equation is

$$r^2 - r - 2 = 0$$
$$(r-2)(r+1) = 0$$

In which case $r_1 = 2$ and $r_2 = -1$. This falls into the $r_1 \neq r_2$ (real) case.

$$y_1 = c_1 e^{2x}$$
$$y_2 = c_2 e^{-x}$$

The general solution is

$$y = c_1 e^{2x} + c_2 e^{-x}$$

Example 2

Consider the differential equation

$$y'' - 5y = 0$$

The characteristic equation is

$$r^2 - 5 = 0$$
$$r^2 = 5$$
$$r_1 = \sqrt{5}, r_2 = -\sqrt{5}$$

Giving us the general solution

$$y = c_1 e^{\sqrt{5}x} + c_2 e^{-\sqrt{5}x}$$

Example 3

Consider the differential equation

$$\ddot{y} + 10\dot{y} + 21y = 0$$

The characteristic equation is

$$r^2 + 10r + 21 = 0$$
$$(r+7)(r+3) = 0$$

So $r_1 = -7$ and $r_2 = -3$, giving us the general solution

$$y = c_1 e^{-7x} + c_2 e^{-3x}$$

Example 4

Consider the differential equation

$$y'' + 4y' + 5y = 0$$

The characteristic equation is

$$r_2 + 4r + 5 = 0$$

Which cannot be factored. We will have to whip out our trusty quadratic equation

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-4 \pm \sqrt{16 - 20}}{2} \\ x &= \frac{-4 \pm \sqrt{-4}}{2} \\ x &= \frac{-4 \pm 2i}{2} \\ x &= -2 \pm i \end{aligned}$$

Recall that for complex roots

$$\begin{aligned} r_1 &= \lambda + i\mu, \quad \mu, \lambda \in \mathbb{R} \\ r_2 &= \lambda - i\mu, \quad \mu, \lambda \in \mathbb{R} \end{aligned}$$

In this instance $\lambda = -2$ and $\mu = 1$, giving us the solutions

$$\begin{aligned} y_1 &= c_1 e^{-2x} \cos(1 \cdot x) \\ y_2 &= c_2 e^{-2x} \sin(1 \cdot x) \end{aligned}$$

And the general solution

$$y = c_1 e^{-2x} \cos x + c_2 e^{-2x} \sin x$$

Example 5

Consider the differential equation

$$\frac{d^2 I}{dt^2} + 20\frac{dI}{dt} + 200I = 0$$

The characteristic equation is

$$r^2 + 20r + 200 = 0$$

Which looks like it cannot be factored, so we will use the quadratic equation

$$\begin{aligned} r &= \frac{-20 \pm \sqrt{20^2 - 4(1)(200)}}{2(1)} \\ &= \frac{-20 \pm \sqrt{400 - 800}}{2} \\ &= \frac{-20 \pm 20i}{2} \\ r_1, r_2 &= \underbrace{-10}_{\lambda} \pm \underbrace{10i}_{\mu} \end{aligned}$$

The general solution is

$$y = c_1 e^{-10x} \cos(10x) + c_2 e^{-10x} \sin(10x)$$

Example 6

Consider the differential equation (in terms of $x(t)$)

$$\ddot{x} + 4\dot{x} + 4x = 0$$

The characteristic equation is

$$r^2 + 4r + 4 = 0(r+2)(r+2) = 0$$

So we have repeated roots

$$r_1, r_2 = -2$$

We still have two solutions

$$x_1 = c_1 e^{-2t} x_2 = c_2 x e^{-2t}$$

And the general solution

$$y = c_1 e^{-2t} + c_2 t e^{-2t}$$

Example 7

Consider the differential equation

$$y'' + y' + y = 0$$

For an exponential solution, the characteristic equation is

$$y(t) = e^{rt} \Rightarrow r^2 + r + 1 = 0 \Leftrightarrow r = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

We separate the real and imaginary components

$$\lambda = -\frac{1}{2}, \mu = \frac{\sqrt{3}}{2}$$

And obtain the general solution

$$y(t) = c_1 e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2} t\right) + c_2 e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2} t\right)$$

Suppose we wanted to determine a particular solution which satisfied the initial conditions $y(0) = 2$ and $y'(0) = 8$

$$y(0) = c_1 = 2$$
$$y'(0) = -1/2c_1 + 3c_2 = 8$$

Therefore, we obtain

$$c_1 = 2$$
$$c_2 = 3$$

Thus, the solution of the initial value problem is

$$y(t) = e^{-t/2}\left(2\cos\left(\frac{\sqrt{3}}{2} t\right) + 3\sin\left(\frac{\sqrt{3}}{2} t\right)\right)$$

Thus, y_1 and y_2 form a fundamental solution set of the equation.

Reduction of Order

The method that we have used for repeated roots also works for equations with non-constant coefficients

$$y'' + p(t)y' + q(t)y = 0$$

That is, given that y_1 is a solution, we can try $y_2 = v(t)y_1$

$$\begin{aligned} y_2 &= v(t)y_1(t) \\ y_2' &= v'(t)y_1(t) + v(t)y_1'(t) \\ y_2'' &= v''(t)y_1(t) + 2v'(t)y_1'(t) + v(t)y_1''(t) \end{aligned}$$

Substituting these into our ODE and collecting terms, we are given

$$y_1v'' + (2y_1' + py_1)v' + (y_1'' + py_1' + qy_1)v = 0$$

Since y_1 is a solution to the differential equation, the last equation reduces to a first order equation in v'

$$y_1v'' + (2y_1' + py_1)v' = 0$$

This is actually a first order differential equation for the function v' and can be solved either as a first-order linear equation or a separable equation. After v' has been found, then v can be obtained through integration. Then y_2 can be determined from the equation

$$y_2 = v(t)y_1(t)$$

This method is known as **reduction of order**, because the most crucial step of the solution is a first-order differential equation for v' rather than the given second-order differential equation.

Reduction of Order Example

Let us now apply reduction of order on the following equation

$$2t^2y'' + 3ty' - y = 0, \quad t > 0$$

given that $y_1(t) = t^{-1}$ is a solution.

We can begin by differentiating

$$\begin{aligned} y_1(t) &= t^{-1} \\ y_1'(t) &= -t^2 \\ y_1''(t) &= 2t^{-3} \end{aligned}$$

So we obtain the following for $y_2(t)$

$$\begin{aligned} y_2(t) &= v(t)t^{-1} \\ y_2'(t) &= v'(t)t^{-1} - v(t)t^{-2} \\ y_2''(t) &= v''(t)t^{-1} - 2v'(t)t^{-2} + 2v(t)t^{-3} \end{aligned}$$

Substituting these for y, y' and y'' in the first equation, and collecting terms, we obtain

$$\begin{aligned} 2t^2(v''(t)t^{-1} - 2v'(t)t^{-2} + 2v(t)t^{-3}) + 3t(v'(t)t^{-1} - v(t)t^{-2}) - vt^{-1} &= 0 \\ 2v''t - 4v' + 4vt^{-1} + 3v' - 3vt^{-1} - vt^{-1} &= 0 \\ 2tv' - v' &= 0 \\ 2tu' - u &= 0, \quad \text{where } \mu(t) = v't \end{aligned}$$

To solve, we can use the separation of variables method

$$2t\frac{du}{dt} - u = 0 \Leftrightarrow \int \frac{du}{u} = \int \frac{1}{2t}\,dt \Leftrightarrow \ln|u| = \frac{1}{2}\ln|t| + C$$

Thus

$$|u| = |t|^{1/2}e^C \Leftrightarrow u = ct^{1/2}, \quad \text{since } t > 0$$

And

$$v' = ct^{1/2}$$

So

$$v(t) = 2/3ct^{3/2} + k$$

Giving us the solution

$$y_2(t) = (2/3ct^{3/2} + k)t^{-1} = 2/3ct^{1/2} + kt^{-1}$$

The second term on the RHS is a multiple of $y_1(t)$, and can be dropped, giving us the new solution

$$y_2(t) = t^{1/2}$$

The Wronskian of y_1 and y_2 can be calculated to verify the linear independence. The general solution is

$$y(t) = c_1t^{-1} + c_2t^{1/2}$$

Nonhomogeneous Equations and the Method of Undetermined Coefficients

We now shift our attention to the nonhomogeneous second-order linear differential equation

$$y'' + p(t)y' + q(t)y = g(t)$$

where p, g, q are continuous functions on an open interval I. The associated homogeneous equation is

$$y'' + p(t)y' + q(t)y = 0$$

Now we will learn how to solve nonhomogeneous equations with the method of undetermined coefficients. The following two theorems provide a foundation for the construction of a general solution.

Theorem

If Y_1 and Y_2 are solutions to the nonhomogeneous linear differential equation

$$y'' + p(t)y' + q(t)y = g(t)$$

Then their difference $Y_1 - Y_2$ is a solution to the associated homogeneous equation

$$y'' + p(t)y' + q(t)y = 0$$

If, in addition, $\{y_1, y_2\}$ forms a fundamental solution set of the homogeneous equation, then there exist constants c_1 and c_2 such that

$$Y_1(t) - Y_2(t) = c_1 y_1(t) + c_2 y_2(t)$$

Theorem

The general solution of the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t)$$

can be written in the form

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + Y(t)$$

where y_1 and y_2 form a fundamental solution set for the homogeneous equation, c_1 and c_2 are arbitrary constants, and $Y(t)$ is a specific solution to the nonhomogeneous equation.

The second theorem states that essentially, we must do three things to solve the nonhomogeneous equation described above

1. Find the general solution $c_1 y_1(t) + c_2 y_2(t)$ of the corresponding homogeneous equation. This solution is frequently called the **complementary solution** and may be denoted $y_c(t)$.
2. Find any solution $Y(t)$ of the nonhomogeneous equation. Often, this solution is referred to as a **particular solution**.
3. Form the sum of the functions found in steps 1 and 2.

We have previously discussed how to find $y_c(t)$, at least when the homogeneous equation has constant coefficients. We will now focus on finding a particular solution $Y(t)$. There are two techniques that we can use – the **method of undetermined coefficients** and the **method of variation of parameters**. Both have their own advantages and disadvantages.

The Method of Undetermined Coefficients

The method of undetermined coefficients requires an educated guess about the form of the particular solution $Y(t)$, but with the coefficients left unspecified. We then substitute the assumed expression into the nonhomogeneous differential equation

$$y'' + p(t)y' + q(t)y = g(t)$$

and attempt to find coefficients to satisfy the equation. If we succeed, we have found a solution of the differential equation and can use it for the particular solution $Y(t)$. If we cannot determine coefficients, then it means there is no solution of the form we assumed. So, we will have to modify the initial assumption and try again.

The main advantage of this method is that it is straightforward to execute once the assumption is made about the form of $Y(t)$. Its major limitation is that it is useful mainly for equations for which we can easily write down the correct form of the particular solution in advance (in particular, polynomials, exponential functions, sines and cosines).

We will attempt to illustrate how the method of undetermined coefficients is used through a few examples.

Exponential $g(t)$

Consider the nonhomogeneous equation

$$y'' - 3y' - 4y = 3e^{2t}$$

We seek a function Y satisfying this equation, that is a Y such that the combination

$$Y''(t) - 3Y'(t) - 4Y(t) = 3e^{2t}$$

Since the exponential function replicates through differentation, the most plausible way to achieve the desired result is to assume that $Y(t)$ is a multiple of e^{2t}

$$Y(t) = Ae^{2t}$$

where A is a coefficient that has yet to be determined. To find A, we calculate the first two derivatives of Y

$$\begin{aligned} Y'(t) &= 2Ae^{2t} \\ Y''(t) &= 4Ae^{2t} \end{aligned}$$

We then substitute for y, y' and y'' in the nonhomogeneous differential equation at the top, obtaining

$$\begin{aligned} Y'' - 3Y' - 4Y &= 3e^{2t} \\ 4Ae^{2t} - 6Ae^{2t} - 4Ae^{2t} &= 3e^{2t} \\ -6Ae^{2t} &= 3e^{2t} \\ A &= -\frac{1}{2} \end{aligned}$$

Thus, a particular solution is

$$Y(t) = -\frac{1}{2}e^{2t}$$

Sine $g(t)$

Consider the nonhomogeneous equation

$$y'' - 3y' - 4y = 2\sin t$$

We seek Y satisfying this equation. Since sines replicate through differentiation, a good start for Y is $A\sin t$, where A is a constant that has yet to be determined. After calculating the derivatives

and substituting into the equation, we obtain

$$\begin{aligned} Y'' - 3Y' - 4Y &= 2\sin t \\ -A\sin t - 3A\cos t - 4A\sin t &= 2\sin t \\ (2+5A)\sin t + 3A\cos t &= 0 \\ c_1 \sin t + c_2 \cos t &= 0 \end{aligned}$$

However, since sin and cos are not multiples each other, we must have $c_1 = c_2 = 0$, and hence $2+5A = 3A = 0$, which is not possible. We then conclude that our guess for $Y(t)$ was inadequate.

The appearance of a cosine term suggests that we modify our original assumption to include a cosine term in $Y(t)$, that is a linear combination of sin and cos

$$\begin{aligned} Y(t) &= A\sin t + B\cos t \\ Y'(t) &= A\cos t - B\sin t \\ Y''(t) &= -A\sin t - B\cos t \end{aligned}$$

Substituting this into our equation, we obtain

$$\begin{aligned} (-A\sin t - B\cos t) - 3(A\cos t - B\sin t) - 4(A\sin t + B\cos t) &= 2\sin t \\ (-5A+3B)\sin t + (-3A-5B)\cos t - 2\sin t &= 0 \end{aligned}$$

Therefore, we must now find A and B to safisfy the equations

$$-5A + 3B = 2, \quad -3A - 5B = 0$$

Solving these algebraic equations, we obtain

$$A = -\frac{5}{17}, \quad B = \frac{3}{17}$$

giving us a particular solution

$$Y(t) = -\frac{5}{17}\sin t + \frac{3}{17}\cos t$$

Product $g(t)$

Consider the nonhomogeneous linear equation

$$y'' - 3y' - 4y = -8e^t \cos 2t$$

In this case, our guess is fairly complicated. It will involve the product of e^t and a linear combination of $\cos(2t)$ and $\sin(2t)$

$$Y(t) = Ae^t \cos 2t + Be^t \sin 2t$$

The algebra is a bit more to work out, but it follows that

$$Y'(t) = (A+2B)e^t \cos(2t) + (-2A+B)e^t \sin(2t)$$

and that

$$Y''(t) = (-3A + 4B)e^t \cos(2t) + (-4A - 3B)e^t \sin(2t)$$

By substituting these expressions into our equation, we find that A and B must satisfy

$$10A + 2B = 8, \quad 2A - 10B = 0$$

Hence

$$A = \frac{10}{13}, \quad B = \frac{2}{13}$$

And a particular solution is

$$Y(t) = \frac{10}{13}e^t \cos(2t) + \frac{2}{13}e^t \sin(2t)$$

Sum $g(t)$

Suppose that $g(t)$ is the sum of two terms

$$g(t) = g_1(t) + g_2(t)$$

and suppose that Y_1 and Y_2 are solutions of the equations

$$ay'' + by' + cy = g_1(t)$$

and

$$ay'' + by' + cy = g_2(t)$$

respectively. It is also true that $Y_1 + Y_2$ is a solution of the equation

$$ay'' + by' + cy = g(t)$$

For example, to find the particular solution of

$$y'' - 3y' - 4y = 3e^{2t} + 2\sin t - 8e^t \cos(2t)$$

By splitting up the right hand side of the equation, we obtain three equations

$$\begin{aligned} y'' - 3y' - 4y &= 3e^{2t} \\ y'' - 3y' - 4y &= 2\sin t \\ y'' - 3y' - 4y &= -8e^t \cos(2t) \end{aligned}$$

As we have already found the solutions of these three equations, a particular solution of this equation is their sum, namely

$$Y(t) = -\frac{1}{2}e^{2t} + \frac{3}{17}\cos t - \frac{5}{17}\sin t + \frac{10}{13}e^t \cos(2t) + \frac{2}{13}e^t \sin(2t)$$

To summarize our conclusions up to this point, to guess the function Y

1. If the nonhomogeneous term $g(t)$ in the differential equation is an exponential function $e^{\alpha t}$, then assume $Y(t)$ is proportional to the same exponential function.

2. If $g(t)$ is $\sin(\beta t)$ or $\cos(\beta t)$, then assume that $Y(t)$ is a linear combination of $\sin(\beta t)$ and $\cos(\beta t)$
3. If $g(t)$ is a polynomial of degree of n, then assume that $Y(t)$ is a polynomial of degree n. For example, if $g(t) = 4t^2 - 1$, we would assume $Y(t) = At^2 + Bt + C$.
4. If $g(t)$ is a sum of n terms, then separate each term into its only subproblem, in which $g(t) = g_i(t)$ from $i = 1..n$

Method of Undetermined Coefficients Summary

To summarize the steps involved in finding the solution of an initial value problem consisting of a nonhomogeneous linear differential equation of the form

$$ay'' + by' + cy = g(t)$$

where the coefficients a, b, c are constants, together with a set of initial conditions:

1. Find the general solution of the associated homogeneous equation
2. Make sure that the function $g(t)$ belongs to a class of functions discussed in this section – exponentials, sines, cosines, polynomials, or sums or products of such functions.
3. If $g(t) = g_1(t) + \cdots + g_n(t)$, then form n subproblems, each of which contains only one of the terms $g_1(t), \ldots, g_n(t)$, with the ith subproblem consisting of the equation

$$ay'' + by' + cy = g_i(t), \quad i = 1..n$$

4. For the ith subproblem, assume a particular solution $Y_i(t)$ consisting of the appropriate exponential function, sine, cosine, polynomial, or combination of each. If there is any duplication in the assumed form of $Y_i(t)$ with the solutions of the homogeneous equation (found in step 1), then $Y_i(t)$ must be multiplied by t, or (if required) by t^2, as to remove the duplication. That is to say, if the assumed function $Y(t)$ gives a particular solution that is equal to a solution of the associated homogeneous equation, we must multiply by t or t^2 accordingly.
5. Find a particular solution $Y_i(t)$ for each of the subproblems. Then $Y_1(t) + \cdots + Y_n(t)$ is a particular solution of the full nonhomogeneous equation
6. Form the sum of the general solution of the homogeneous equation in step 1 and the particular solution of the nonhomogeneous equation in step 5. This is the general solution of the nonhomogeneous equation
7. When initial conditions are provided, use them to determine the values of the arbitrary constants remaining in the general solution

Variation of Parameters

A second method of finding a particular solution of a nonhomogeneous equation is known as the **variation of parameters method**. The main advantage of this method is that it is a *general method*, and can be applied to any equation without a detailed assumption about the form of its solution. However, this method also involves evaluating various integrals involving the nonhomogeneous term in the differential equation, which may present difficulties.

Let us begin with an example to illustrate its usage

Varation of Parameters Example

We seek a particular solution to the equation below

$$y'' + 4y = 8\tan t \tag{1}$$

We cannot use the undetermined method of coefficients, as $g(t)$ is a quotient of $\sin t$ and $\cos t$ (rather than a sum or product). So, we need a different approach.

Observe that the homogeneous equation corresponding to the equation is

$$y'' + 4y = 0 \tag{2}$$

and that the general solution of the homogeneous equation is

$$y_c(t) = c_1\cos(2t) = c_2\sin(2t) \tag{3}$$

This basic idea in the method of variation of parameters is that we would like to replace the constants c_1 and c_2 in equation (3) by functions $u_1(t)$ and $u_2(t)$ respectively, and then determine these functions such that the resulting expression

$$y = u_1(t)\cos(2t) + u_2(t)\sin(2t) \tag{4}$$

is a solution of the nonhomogeneous equation (1). To determine these functions, we need to substitute y from equation (4) in differential equation (1). We can anticipate that the result will be a single equation involving a combination of u_1, u_2 and their first two derivatives. Since there is only one equation and two unknown functions, there will be many possible choices of u_1 and u_2 that meet our requirements, so it would be advantageous to us to impose a second condition (of our own choosing), which would give us two equations for the two unknown functions. It is possible to choose this second condition in a way that makes the computation much more efficient. Back to equation (4), if we differentiate and rearrange the terms, we obtain

$$y' = -2u_1(t)\sin(2t) + 2u_2(t)\cos(2t) + u_1'(t)\cos(2t) + u_2'(t)\sin(2t) \tag{5}$$

Since we are on the lookout for a second condition, let us require that the sum of the last two terms on the right hand side of equation (5) to be zero. That is

$$u_1'(t)\cos(2t) + u_2'(t)\sin(2t) = 0 \tag{6}$$

It then follows from equation (5) that

$$y' = -2u_1(t)\sin(2t) + 2u_2(t)\cos(2t) \tag{7}$$

The ultimate effect of the extra condition we imposed is not yet clear, but the removal of u_1' and u_2' has simplified the expression for y' quite a bit. Continuing onward, by differentiating (7) we obtain

$$y'' = -4u_1(t)\cos(2t) - 4u_2(t)\sin(2t) - 2u_1'(t)\sin(2t) + 2u_2'(t)\cos(2t) \tag{8}$$

Then, by substituting for y and y'' in equation (1) from equations (4) and (8), respectively, we find that

$$\begin{aligned} y'' + 4y = &- 4u_1(t)\cos(2t) - 4u_2(t)\sin(2t) - 2u_1'(t)\sin(2t) + 2u_2'(t)\cos(2t) \\ &+ 4u_1(t)\cos(2t) + 4u_2(t)\sin(2t) = 8\tan t \end{aligned}$$

Hence, u_1 and u_2 must satisfy

$$-2u_1'(t)\sin(2t) + 2u_2'(t)\cos(2t) = 8\tan t \tag{9}$$

So, we need to find a u_1 and u_2 that satisfy equations (6) and (9)[a]. These equations can be viewed as a pair of linear algebraic equations. To solve equation (6) for $u_2'(t)$ we have

$$u_2'(t) = -u_1'(t)\frac{\cos(2t)}{\sin(2t)} \tag{10}$$

Then substituting for $u_2'(t)$ in equation (9) and simplifying, we obtain

$$u_1'(t) = -\frac{8\tan t\sin(2t)}{2} = -8\sin^2 t \tag{11}$$

By putting this expression for $u_1'(t)$ back into equation (10) and using the double angle formulas, we find that

$$u_2'(t) = \frac{8\sin^2 t\cos(2t)}{\sin(2t)} = 4\frac{\sin t\left(2\cos^2 t - 1\right)}{\cos t} = 4\sin t\left(2\cos t - \frac{1}{\cos t}\right) \tag{12}$$

Having obtained $u_1'(t)$ and $u_2'(t)$, we integrate so as to find $u_1(t)$ and $u_2(t)$. The result is

$$u_1(t) = 4\sin t\cos t + 4t + c_1 \tag{13}$$

and

$$u_2(t) = 4\ln(\cos t) - 4\cos^2 t + c_2 \tag{14}$$

Substituting these into equation (4), we get

$$y = (4\sin t\cos t)\cos(2t) + (4\ln(\cos t) - 4\cos^2 t)\sin(2t) + c_1\cos(2t) + c_2\sin(2t) \tag{3.1}$$

Using the double angle formulas once more, we obtain

$$y = -2\sin(2t) - 4t\cos(2t) + 4\ln(\cos t)\sin(2t) + c_1\cos(2t) + c_2\sin(2t) \tag{15}$$

The terms involving the arbitrary constants c_1 and c_2 are the general solution of the corresponding homogeneous equation, while the other three are a particular solution of the nonhomogeneous equation (1). Therefore, equation (15) is the general solution of equation (1).

[a] Nice

In the example above, by choosing the constants $c_1 = 0$ and $c_2 = 2$, we obtain a particular soluition with only two terms

$$-4t\cos(2t) + 4\ln(\cos t)\sin(2t)$$

This explains why the method of undetermined coefficients is not a good candidate for this problem, and why the variation of parameters method is needed. Our next question is how this method can be generalized to any equation. Consider

$$y'' + p(t)y' + q(t)y = g(t) \tag{1}$$

where p, g, q are continuous functions. As a starting point, we assume we already know the general solution

$$y_c(t) = c_1y_1(t) + c_2y_2(t) \tag{2}$$

of the corresponding homogeneous equation

$$y'' + p(t)y' + q(t)y = 0 \tag{3}$$

This is quite the assumption to make, as so far we have only shown how to solve equation (3) only if it has constant coefficients. The crucial idea, as shown in the example above, is to replace the constants by functions $u_1(t)$ and $u_2(t)$, giving us

$$y = u_1(t)y_1(t) + u_2(t)y_2(t) \tag{4}$$

We will then try to determine $u_1(t)$ and $u_2(t)$ such that the expression in equation (4) is a solution of the nonhomogeneous equation (1) rather than the homogeneous equation (3). To do this, we differentiate equation (4), obtaining

$$y' = u_1'(t)y_1(t) + u_1(t)y_1'(t) + u_2'(t)y_2(t) + u_2(t)y_2'(t) \tag{5}$$

Just like in the example, we set the terms involving the first derivatives of our unknown functions in equation (5) to zero. That is, we require that

$$u_1'(t)y_1(t) + u_2'(t)y_2(t) = 0 \tag{6}$$

Thus, from equation (5), we have

$$y' = u_1(t)y_1'(t) + u_2(t)y_2'(t) \tag{7}$$

By differentiating a second time, we obtain

$$y'' = u_1'(t)y_1'(t) + u_1(t)y_1''(t) + u_2'(t)y_2'(t) + u_2(t)y_2''(t) \tag{8}$$

We then substitute for y, y' and y'' in equation (1) from equations (4), (7) and (8), respectively, obtaining the following

$$\begin{aligned} u_1(t)\,(y_1''(t) \;&+p(t)y_1'(t) + q(t)y_1(t)) \\ +u_2(t)\,(y_2''(t) \;&+p(t)y_2'(t) + q(t)y_2(t)) \\ &+ u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t) \end{aligned} \tag{9}$$

Each of the expressions in parentheses in the first two lines of equation (9) are both zero, because both y_1 and y_2 are solutions of the homogeneous equation (3). Therefore, equation (9) reduces to

$$u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t) \tag{10}$$

Equations (6) and (10) form a system of two linear algebraic equations for the derivatives $u_1'(t)$ and $u_2'(t)$. The correspond exactly to the equations from the example above. Solving the systems of equations, we obtain

$$u_1(t) = -\frac{y_2(t)g(t)}{W[y_1, y_2](t)}, \quad u_2(t) = \frac{y_1(t)g(t)}{W[y_1, y_2](t)} \tag{11}$$

where $W[y_1, y_2]$ is the Wronskian of y_1 and y_2. Note that division by the Wronskian is permissible since y_1 and y_2 are a fundamental set of solutions, and thus their Wronskian cannot be zero. By integrating equations (11), we find the desired functions $u_1(t)$ and $u_2(t)$, namely

$$u_1(t) = -\int \frac{y_2(t)g(t)}{W[y_1, y_2](t)}\,dt + c_1, \quad u_2(t) = \int \frac{y_1(t)g(t)}{W[y_1, y_2](t)}\,dt + c_2 \tag{12}$$

If the integrals can be evaluated in terms of elementary functions, then we substitute the result in equation (4), thereby obtaining the general solution of equation (1). More generally, the solution can always be expressed in terms of integrals, in the form of the following particular solution

$$Y(t) = -y_1(t)\int_{t_0}^{t} \frac{y_2(s)g(s)}{W\left[y_1, y_2\right](s)}ds + y_2(t)\int_{t_0}^{t} \frac{y_1(s)g(s)}{W\left[y_1, y_2\right](s)}ds$$

Mechanical and Electrical Vibrations

Two important areas of application for second order linear equations with constant coefficients are in modeling mechanical and electrical oscillations. The motion of a mass on a vibrating spring or the flow of an electric current in a simple circuit can be described by the solution of an initial value problem of the form

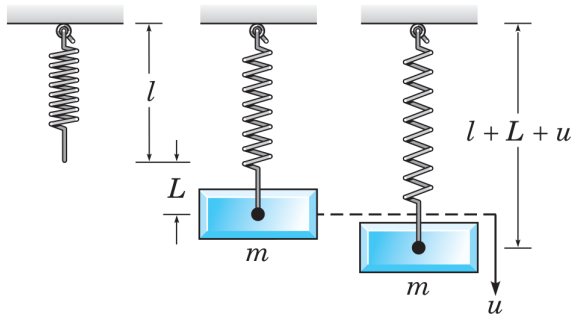
$$ay'' + by' + cy = g(t), \quad y(0) = y_0, y'(0) = y_0'$$

Many physical models have mathematically equivalent models. Once we know how to solve the initial value problem above, it is only necessary to make appropriate interpretations of the constants $a, b,$ and c and of the functions y and g to obtain solutions of different physical problems.

We will first study of a mass on a spring, as understanding this simple system will lay the foundation for investigation of more complex vibrating systems. The principles involved are also present in many problems.

Spring Model

Consider a mass m hanging at rest on the end of a vertical spring of original length l. The mass causes an elongation, L, of the spring in the downward direction (which we denote as the positive direction).



In this situation, there are two forces acting at the point, where the mass is attached to the spring

1. The gravitational force, or the weight of the mass, acts downward and has magnitude $w = mg$, where g is the acceleration due to gravity
2. The force F_s due to the spring, which acts upward. If we assume that the elongation L of the spring is small, the spring force is nearly proportional to L (Hooke's Law). Thus, we write $F_s = -kL$, where k is the spring constant (negative because the spring force acts in the upward direction)

As the mass is in equilibrium, the two forces balance each other, which means that

$$w + F_s = mg - kL = 0 \tag{1}$$

We are interested in studying the motion of the mass when it is acted upon by an external force, or is initially displaced. Let $u(t)$, measured positive in the downward direction, denote the displacement of the mass from its equilibrium position from time t. Then $u(t)$ is related to the forces acting on the mass through Newton's law of motion

$$mu''(t) = f(t) \tag{2}$$

where u'' is the acceleration of the mass and f is the net force acting on the mass (note that they are both functions of time). In this now dynamic problem, we must consider four separate forces

1. The weight $w = mg$ of the mass, which always acts downward
2. The spring force F_s, which is assumed to be proportional to the total elongation $L + u$ of the spring and always acts to restore the spring to its natural position. If the spring is extended ($L + u > 0$), the spring force is directed upward, and

$$F_s = -k(L + u) \tag{3}$$

   Alternatively, if the spring is compressed ($L + u < 0$), then the spring is compressed a distance $|L + u|$, and the spring force, which is now directed downward, is given by

$$F_s = k|L + u| = -k(L + u)$$

   as $L + u < 0$. Thus, the force exerted by the spring is always expressed by equation (3), regardless of the position of the mass.
3. The damping/resistive force F_d, which always acts in the direction opposite to the direction of the motion of the mass. This may arise from resistance to air (or other medium in which the mass moves), energy loss due to the extension or compression of the spring, friction, or more. In any case, we assume that the resistive force is proportional to the speed $|du/dt|$ of the mass (usually referred to as **viscous damping**).

   We find that if $du/dt > 0$, then u is increasing, so the mass is moving downwards and F_d is directed upward. This is given by

$$F_d(t) = -\gamma u'(t) \tag{4}$$

   where γ is a positive constant of proportionality known as the damping constant. However, if $du/dt < 0$, then u is decreasing, the mass is moving upward and F_d is directed downward. In this case

$$F_d = \gamma |u'(t)|$$

   but just like with the spring force, it turns out the damping force always works out to be expressed by equation (4).
4. An applied external force $F(t)$ is directed downward or upward as $F(t)$ is positive or negative. This force is often periodic.

Taking into account these forces, we can rewrite Newton's law (2) as

$$\begin{aligned} mu''(t) &= w + F_s(t) + F_d(t) + F(t) \\ &= mg - k(L + u(t)) - \gamma u'(t) + F(t) \end{aligned} \tag{5}$$

Since $mg - kL = 0$ by equation (1), it follows that the equation of the motion of the mass is

$$mu''(t) + \gamma u'(t) + ku(t) = F(t), \quad m, \gamma, k > 0 \tag{6}$$

Note that this has the same form as the equation mentioned at the beginning of the section – it is a nonhomogeneous second-order linear differential equation with constant coefficients. We must also note that the equation (6) is only an approximate equation for the displacement $u(t)$, and that the equations for spring force/damping force are approximates as well.

The complete vibration problem requires that we also specify two initial conditions, namely, the initial position u_0 and the initial velocity v_0 of the mass

$$u(0) = u_0, u'(0) = v_0 \tag{7}$$

Example

A mass weighing 4 lb stretches a spring 2 in. Suppose that the mass is displaced an additional 6 in in the positive direction and subsequently released. The mass is in a medium that exerts a viscous resistance of 6 lb when the mass has a velocity of 3 ft/s. Formulate the initial value problem that governs the motion of the mass.

Let us measure the displacement u in feet. Since nothing is said about an external force, assume that $F(t) = 0$. To determine m:

$$m = \frac{w}{g} = \frac{4 \text{ lb}}{32 \text{ ft/s}^2} = \frac{1}{8}\frac{\text{lb} \cdot \text{s}^2}{\text{ft}}$$

The damping coefficient γ is determined from the statement that $\gamma u'$ is equal to 6 lb when u' is 3 ft/s.

$$\gamma = \frac{6 \text{ lb}}{3\text{ft/s}} = 2\frac{\text{lb} \cdot \text{s}}{\text{ft}}$$

The spring constant k is found from the statement that the mass stretches the spring by 2 inches, or $\frac{1}{6}$ feet. Thus

$$k = \frac{4 \text{ lb}}{1/6 \text{ ft}} = 24\frac{\text{lb}}{\text{ft}}$$

Our differential equation

$$mu''(t) + \gamma\mu'(t) + ku(t) = F(t), \quad u(0) = u_0, \quad u'(0) = v_0$$

becomes

$$\frac{1}{8}u'' + 2u' + 24u = 0$$

which can be simplified to

$$u'' + 16u' + 192u = 0$$

The initial conditions are

$$u(0) = \frac{1}{2}, \quad u'(0) = 0$$

For the first initial condition, we have that the mass is displaced an additional 6 inches, which converts to $\frac{1}{2}$ feet. The second initial condition is implied by the word 'released' in the problem statement, which implies that the mass started with no initial velocity.

To solve this equation, we obtain the characteristic equation

$$r^2 + 16r + 192 = 0$$

We use the quadratic equation to solve for r

$$\begin{aligned} r &= \frac{-16 \pm \sqrt{16^2 - 4 \times 192}}{2} \\ &= \frac{-16 \pm \sqrt{-512}}{2} \\ &= \frac{-16 \pm 2^4\sqrt{-2}}{2} \\ &= -8 \pm 8\sqrt{2}i \end{aligned}$$

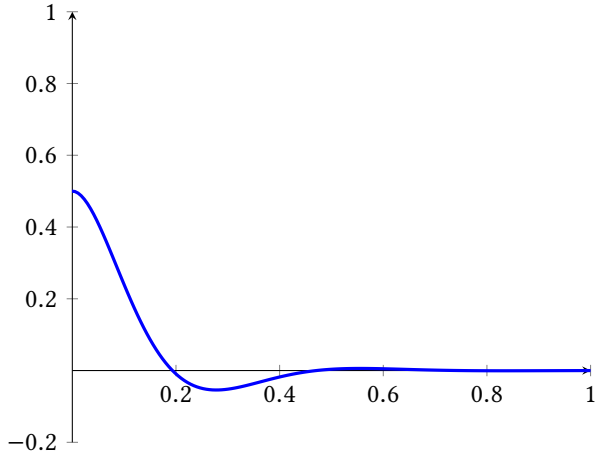
So $\lambda = -8$ and $\mu = 8\sqrt{2}$, giving us the general solution

$$y(t) = c_1 e^{-8t} \cos\left(8\sqrt{2}t\right) + c_2 e^{-8t} \sin\left(8\sqrt{2}t\right)$$

If we solve for constants using initial conditions, we obtain the solution

$$u(t) = \frac{1}{4} e^{-8t} (2\cos\left(8\sqrt{2}t\right) + \sqrt{2}\sin\left(8\sqrt{2}t\right))$$

Examining the graph, we see that this indeed follows the pattern of a spring vibrating up and down and eventually slowing down.



Undamped Free Vibrations

If there is no external force, then $F(t) = 0$ and $\gamma = 0$ in the equation

$$mu''(t) + \gamma u'(t) + ku(t) = F(t)$$

This should not happen in real life, as any spring will have some kind of force acting on it. However, if the actual damping is very small, then the assumption of no damping at all may yield satisfactory results over short to moderate time intervals. Nevertheless, our equation then becomes

$$mu''(t) + ku(t) = 0$$

The characteristic equation is

$$mr^2 + k = 0$$

and its roots are $r = \pm i\sqrt{k/m}$. Thus, the general solution of the equation is

$$u = A\cos(\omega_0 t) + B\sin(\omega_0 t)$$

where

$$\omega_0^2 = \frac{k}{m}$$

Using trigonometric identities, the solution can be rewritten as

$$u = R\cos(\omega_0 t - \delta)$$

or

$$u = R\cos\delta\cos(\omega_0 t) + R\sin\delta\sin(\omega_0 t)$$

By comparing these equations, we find that A, B, R and δ are related by the equations

$$A = R\cos\delta, \quad B = R\sin\delta$$

And thus

$$R = \sqrt{A^2 + B^2}, \quad \tan\delta = \frac{B}{A}$$

In calculating δ, we must be careful to choose the correct quadrant (can be done by checking the signs of $\cos\delta$ and $\sin\delta$. Thus, our solution is

$$u(t) = A\cos\omega_0 t + B\sin\omega_0 t = R\cos(\omega_0 t - \delta)$$

where

$$\omega_0 = \sqrt{k/m}$$

The solution is a shifted cosine (or sine) wave that describes simple harmonic motion. The period of the motion is

$$T = \frac{2\pi}{\omega_0} = 2\pi\left(\frac{m}{k}\right)^{1/2}$$

The circular frequency $\omega_0 = \sqrt{k/m}$, measured in radians per unit time, is called the **natural frequency** of the vibration. The maximum displacement R is the **amplitude** of the motion. The dimensionless parameter δ is called the **phase**, and measures the displacement of the wave from its normal position corresponding to $\delta = 0$.

Note that the motion described by the equation has a constant amplitude that does not diminish with time. This is because, in the absence of damping, there is no way for the system to dissipate the energy given to it by the initial displacement and velocity.

4 Higher-Order Linear Differential Equations

Handling higher-order linear differential equations involves methods which are very similar to those of first- and second-order linear differential equations.

For the Purposes of the Class

Essentially, dealing with higher-order linear differential equations boils down to being able to reduce them to first/second order linear differential equations using algebraic methods and solving them using the techniques we have already covered.

Let us begin with an attempt to solve the third-order linear differential equation

$$y''' - 6y'' + 11y' - 6y = 0$$

We can still use the characteristic equation

$$r^3 - 6r^2 + 11r - 6 = 0$$

We can simplify this equation using algebra. Suppose we wanted to divide both sides by $r - 1$ to gain it as a factor.

$$\begin{array}{r|l} & r^2 - 5r + 6 \\ \hline r - 1 & r^3 - 6r^2 + 11r - 6 \\ & \underline{r^3 - r^2} \\ & -5r^2 + 11r - 6 \\ & \underline{-5r^2 + 5r} \\ & 6r - 6 \\ & \underline{6r - r} \\ & 0 \end{array}$$

So we get

$$\begin{aligned} r^3 - 6r^2 + 11r - 6 &= 0 \\ (r-1)(r^2 - 5r + 6) &= 0 \\ (r-1)(r-2)(r-3) &= 0 \\ r &= 1, 2, 3 \end{aligned}$$

Giving us the general solution

$$y = c_1 e^t + c_2 e^{2t} + c_3 e^{3t}$$

We can do the same for a fourth-order linear differential equation

$$y^{(4)} - 9y'' + 20y = 0$$

If we substitute $r^2 = S$, the equation becomes

$$\begin{aligned} S^2 - 9S + 20 &= 0 \\ (S-4)(S-5) &= 0 \\ (r^2-4)(r^2-5) &= 0 \\ r &= \pm 2, \pm\sqrt{5} \end{aligned}$$

And we obtain the general solution

$$y = c_1 e^{-2t} + c_2 e^{2t} + c_3 e^{-\sqrt{5}t} + c_4 e^{\sqrt{5}t}$$

5 Series Solutions of Second-Order Linear Equations

Finding the general solution of a linear differential equation depends on determining a fundamental set of solutions of the homogeneous equation. So far, we have a systematic procedure for constructing fundamental solutions if an equation has constant coefficients, however, for a much larger class of equations with variable coefficients, we must search for solutions beyond the familiar elementary functions of calculus

The principle tool we need is the presentation of a given function by a power series. The idea is similar to that of the method of undetermined coefficients – we assume the solutions have power series representations, and then determine the coefficients so as to satisfy the equation.

Review of Power Series

Convergent Power Series

A power series about the point x_0 has the form

$$\sum_{n=0}^{\infty} a_n(x-x_0)^n = a_0 + a_1(x-x_0)^1 + a_2(x-x_0)^2 + \dots$$

and is said to *converge at a point* x if

$$\lim_{m\to\infty} \sum_{n=0}^{m} a_n(x-x_0)^n$$

exists for that x. Note that the series converges for $x = x_0$; however, it may not converge for all x, or may converge for some values of x and not others.

Absolute Convergence

A power series $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ is said to *converge absolutely* at a point x if the associated power series

$$\sum_{n=0}^{\infty} |a_n(x-x_0)^n| = \sum_{n=0}^{\infty} |a_n||x-x_0|^n$$

converges. If a power series converges absolutely, then the power series also converges, however, the converse is not necessarily true.

Ratio Test

One of the most useful tests for the absolute convergence of a power series is the ratio test. If $a_n \neq 0$, and if, for a fixed value of x

$$\lim_{n\to\infty}\left|\frac{a_{n+1}(x-x_0)^{n+1}}{a_n(x-x_0)^n}\right| = |x-x_0|\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = |x-x_0|L$$

then the power series converges absolutely at that value of x if $|x-x_0|L < 1$ and diverges if $|x-x_0|L > 1$. The test is inconclusive if $|x-x_0|L = 1$.

Ratio Test Example

For which values of x does the power series

$$\sum_{n=1}^{\infty}(-1)^{n+1}n(x-2)^n$$

converge? Using the ratio test, we obtain

$$\lim_{n\to\infty}\left|\frac{(-1)^{n+2}(n+1)(x-2)^{n+1}}{(-1)^{n+1}n(x-2)^n}\right| = |x-2|\lim_{n\to\infty}\frac{n+1}{n} = |x-2|$$

We see that the series converges absolutely for $|x-2| < 1$, or in other words $1 < x < 3$, and diverges for $|x-2| > 1$. As for each individual value of x corresponding to $|x-2| = 1$, we have $x = 1$ and $x = 3$. The series will diverge for each of these values. In conclusion, the series converges in the radius $1 < x < 3$ and diverges completely for $x \geq 3$ and $x \leq 1$.

Radius of Convergence

For a typical power series, there is a positive number ρ called the **radius of convergence**, such that $\sum_{n=0}^{\infty} a_n(x-x_0^n)$ converges absolutely for $|x-x_0| < \rho$ and diverges for $|x-x_0| > \rho$. The interval $|x-x_0| < \rho$ is called the **interval of convergence** $(x_0-\rho, x_0+\rho)$. The series may either converge or diverge when $|x-x_0| = \rho$.

If a power series converges for all values of x, it is customary to say that ρ is infinite and the interval of convergence is the entire real line. Conversely, when a series converges only at x_0, we say $\rho = 0$ and the series has no interval of convergence.

Operations on Series

Suppose that $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ and $\sum_{n=0}^{\infty} b_n(x-x_0)^n$ converge to $f(x)$ and $g(x)$, respectively, for $|x-x_0| < \rho, \rho > 0$.

Series Addition/Subtraction

These series can be added or subtracted termwise, and the resulting series converges at least for $|x-x_0| < \rho, \rho > 0$.

$$f(x) \pm g(x) = \sum_{n=0}^{\infty}(a_n \pm b_n)(x-x_0)^n$$

Series Multiplication

The two series can also be multiplied

$$f(x)g(x) = \left(\sum_{n=0}^{\infty} a_n(x-x_0)^n\right)\left(\sum_{n=0}^{\infty} b_n(x-x_0)^n\right) = \sum_{n=0}^{\infty} c_n(x-x_0)^n$$

where each $c_n = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0$. Again, the resulting series converges at least for $|x-x_0| < \rho, \rho > 0$.

Series Division

Furthermore, if $b_0 \neq 0$, then $g(x_0) \neq 0$, and the series $f(x)$ can be divided by the series $g(x)$

$$\frac{f(x)}{g(x)} = \sum_{n=0}^{\infty} d_n(x-x_0)^n$$

where each d_n can be obtained by equating coefficients in the equivalent relation

$$\begin{aligned}\sum_{n=0}^{\infty} a_n\,(x-x_0)^n &= \left[\sum_{n=0}^{\infty} d_n\,(x-x_0)^n\right]\left[\sum_{n=0}^{\infty} b_n\,(x-x_0)^n\right] \\ &= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} d_kb_{n-k}\right)(x-x_0)^n\end{aligned}$$

Derivatives of the Series

The function f is continuous and has derivatives of all orders for $|x-x_0| < \rho$. f', f'' and so on can be computed by taking the derivative of the series termwise

$$\begin{aligned}f'(x) &= a_1 + 2a_2(x-x_0) + \cdots + na_n(x-x_0)^{n-1} + \ldots \\ &= \sum_{n=1}^{\infty} na_n(x-x_0)^{n-1} \\ f''(x) &= 2a_2 + 6a_3(x-x_0) + \cdots + n(n-1)a_n(x-x_0)^{n-2} \\ &= \sum_{n=2}^{\infty} n(n-1)a_n(x-x_0)^{n-2}\end{aligned}$$

Taylor Series

Suppose that $\sum a_n(x-x_0)^n$ converges to $f(x)$ for $|x-x_0| < \rho$. Then the value of a_n is given by

$$a_n = \frac{f^{(n)}(x)}{n!}$$

and the series is called a **Taylor Series** for f about $x = x_0$. If it is also the case that

$$f(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

then f is continuous and has derivatives of all orders on the interval of convergence. Further, the derivatives of f can be computed by differentiating the relevant series term by term.

Analytic Functions

A function f that has a Taylor series expansion about $x = x_0$

$$f(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

with a radius of convergence $\rho > 0$, is said to be *analytic* at x_0. All of the familiar functions of calculus are analytic[1]. For example, $\sin x$ and e^x are analytic everywhere, $\frac{1}{x}$ is analytic except at $x = 0$ and $\tan x$ is analytic except at odd multiples of $\frac{\pi}{2}$.

It is also the case that if f and g are analytic at x_0, then so are $f \pm g$, $f \cdot g$ and f/g.

Series Equality

If two power series are equal, that is

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n = \sum_{n=0}^{\infty} b_n(x - x_0)^n$$

for each x in some open interval with center x_0, then $a_n = b_n$ for $n = 0, 1, 2, 3, \ldots$. In particular, if

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n = 0$$

then $a_0 = a_1 = \cdots = a_n = \cdots = 0$.

Shifting Index of Summation

The index of a summation in an infinite series is a dummy parameter, just as the integration variable in a definite integral is a dummy variable. Thus, the letter used for the index of summation is not that important. Just as we make changes in the variable of integration in a definite integral, we find it convenient to make changes of the index of summation in calculating series solutions of differential equations. Suppose we are asked to rewrite the series below as one starting with the index $n = 0$

$$\sum_{n=2}^{\infty} a_n(x)^n$$

By letting $m = n - 2$ in this series, $n = 2$ corresponds to $m = 0$ and hence

$$\sum_{n=2}^{\infty} a_n(x)^n = \sum_{m=0}^{\infty} a_{m+2}(x)^{m+2}$$

Replacing the dummy index m with n, we obtain

$$\sum_{n=2}^{\infty} a_n(x)^n = \sum_{n=0}^{\infty} a_{n+2}(x)^{n+2}$$

as desired.

[1] Except perhaps at certain points

Shifting Index of Summation Example

Suppose we wanted to rewrite the series

$$\sum_{n=2}^{\infty}(n+2)(n+1)a_n\,(x-x_0)^{n-2}$$

as a series whose generic term is $(x-x_0)^n$ rather than $(x-x_0)^{n-2}$. To do so, we can again shift the index by two (skipping the step in which we substitute in another variable name, as it will be reverted to n regardless)

$$\sum_{n=0}(n+4)(n+3)a_{n+2}(x-x_0)^n$$

Determining Coefficients

Assume that

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} a_n x^n$$

Determine what this implies about the coefficients.

We can begin by writing both series with the same powers of x. As before, for the series on the left, let $m = n-1$, then replace m by as we have been doing. The above equality becomes

$$\sum_{n=0}^{\infty}(n+1)a_{n+1}x^n = \sum_{n=0}^{\infty} a_n x^n \Rightarrow (n+1)a_{n+1} = a_n \Rightarrow a_{n+1} = \frac{a_n}{n+1}$$

for $n = 0, 1, 2, 3, \ldots$ Using this recurrence relation just derived, we can solve for the coefficients successively by letting $n = 0, 1, 2, \ldots, n$

$$a_1 = \frac{a_0}{2}, a_2 = \frac{a_1}{3} = \frac{a_0}{6}, a_3 = \frac{a_2}{4} = \frac{a_0}{24}, \ldots, a_n = \frac{a_0}{n!}$$

Using these coefficients in the original series, we get a recognizable Taylor Series

$$a_0 \sum_{n=0}^{\infty} \frac{x^n}{n!} = a_0 e^x$$

Series Solutions Near an Ordinary Point

In chapter 3, we examined methods of solving second order linear differential equations with constant coefficients. We now consider the case where the coefficients are functions of the independent variable, which we will denote by x. It is sufficient to consider the homogeneous equation

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0 \tag{1}$$

since the method for the nonhomogeneous case is similar. We primarily consider the case when P, Q and R are polynomials, and hence also continuous.

Ordinary Points

Assuming that P, Q and R are polynomials with no common factors, we would like to solve the equation (1) in the neighborhood of a point of interest x_0. This point x_0 is called an **ordinary point** if $P(x_0) \neq 0$. Since P is continuous, it follows that there is an open interval containing x_0 in which $P(x)$ is never zero. In that interval, which we call I, we can divide (1) by $P(x)$ to obtain

$$y'' + p(x)y' + q(x)y = 0 \tag{2}$$

where $p(x) = \frac{Q(x)}{P(x)}$ and $q(x) = \frac{R(x)}{P(x)}$. Since p and q are continuous, there is a unique solution, given initial conditions $y(x_0) = y_0, y'(x_0) = y_0'$[2].

Singular Points

On the other hand, if $P(x_0) = 0$, then x_0 is known as a **singular point** of equation (1). Since P, Q and R are polynomials with no common factors, it follows that $Q(x_0) \neq 0$ or $R(x_0) \neq 0$, or both. Then, at least one of p or q becomes unbounded as $x \longrightarrow x_0$, and we can no longer rely on finding a solution using methods similar to chapter 3.

Series Solutions Near Ordinary Points

To solve the problem of solving equation (1) in the neighborhood of an ordinary point x_0, we will assume a series representation of an unknown solution function y

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

As long as we remain within the interval of convergence, this representation of y is continuous and has derivatives of all orders.

While it may seem like seeking a solution in the form of a power series is not the best method, it is actually a convenient and useful form for a solution. Power series behave very much like polynomials and are easy to manipulate, both analytically and numerically. We can begin with an example.

Series Solution Example

Find a series solution of the equation

$$y'' + y = 0, -\infty < x < \infty$$

Here $P(x) = 1, Q(x) = 0, R(x) = 1$. Thus, every point x is an ordinary point. We will take $x_0 = 0$. Assume a series solution of the form

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

[2] By Theorem 3.2.1 in chapter 3

We can differentiate term by term to obtain

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$
$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$
$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting these expressions into the equation, we obtain

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0$$

Shifting indices we obtain

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

Or

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} + a_n\right] x^n = 0$$

For this equation to be valid for all x, the coefficient of each power of x must be zero, and hence

$$(n+2)(n+1) a_{n+2} + a_n = 0, n = 0, 1, 2, \ldots$$

Or in other terms

$$a_{n+2} = \frac{-a_n}{(n+2)(n+1)}, \quad n = 0, 1, 2, \ldots$$

This type of equation is called a **recurrence relation**. Next, we find the individual coefficients $a_0, a_1, a_2, \ldots$. To find the even coefficients, $a_2, a_4, a_6, \ldots$, we proceed as follows

$$\begin{aligned}
a_2 &= -\frac{a_0}{2 \cdot 1} \\
a_4 &= -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4 \cdot 3 \cdot 2 \cdot 1} \\
a_6 &= -\frac{a_4}{6 \times 5} = -\frac{a_0}{6!} \\
&\vdots \\
a_{2k} &= \frac{(-1)^k a_0}{(2k)!}, k = 1, 2, 3 \ldots
\end{aligned}$$

The same can be done for the odd coefficients

$$\begin{aligned} a_3 &= -\frac{a_1}{3\cdot 2} \\ a_5 &= -\frac{a_3}{5\cdot 4} = \frac{a_1}{5\cdot 4\cdot 3\cdot 2\cdot 1} \\ a_7 &= -\frac{a_5}{7\cdot 6} = -\frac{a_1}{7\cdot 6\cdot 5\cdot 4\cdot 3\cdot 2\cdot 1} \\ &\vdots \\ a_{2k+1} &= \frac{(-1)^k a_1}{(2k+1)!}, k = 1, 2, 3 \ldots \end{aligned}$$

We now have the following information

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

where

$$a_{2k} = \frac{(-1)^k}{(2k)!} a_0, \quad a_{2k+1} = \frac{(-1)^k}{(2k+1)!} a_1$$

So we have two series solutions

$$\begin{aligned} y_1(x) &= a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\ y_2(x) &= a_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \end{aligned}$$

And the general solution

$$y(x) = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + a_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

From calculus, we know this solution is equivalent to

$$y(x) = a_0 \cos x + a_1 \sin x$$

Although even if we didn't, we would still have two solutions in series form.

Series Solution of Airy's Equation

Find a series solution of Airy's Equation about $x_0 = 0$:

$$y'' - xy = 0, \quad -\infty < x < \infty$$

Here, $P(x) = 1, Q(x) = 0, R(x) = -x$. Thus, every point x is an ordinary point. We will take

$x_0 = 0$. Assuming a series solution and differentiating, we obtain

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$
$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$
$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting these expressions into Airy's Equation, we obtain

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

We can simplify this by shifting the indices

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=1}^{\infty} a_{n-1} x^n = 0$$

Or

$$2 \cdot 1 \cdot a_2 + \sum_{n=1}^{\infty} \left[(n+2)(n+1) a_{n+2} - a_{n-1} \right] x^n = 0$$

For this equation to be valid for all x, the coefficient of each power of x must be zero, and hence $a_2 = 0$ and we obtain the recurrence relation

$$(n+2)(n+1) a_{n+2} - a_{n-1} = 0, \quad n = 1, 2, 3 \ldots$$

As a_{n+2} is given in terms of a_{n-1}, the a's are determined in steps of three. This means that a_0 determines a_3, which determines a_6, and so on. Since $a_2 = 0$, we can conclude that $a_5 = a_8 = a_{11} = 0$. To find $a_0, a_3, a_6, a_9, \ldots$, we set $n = 1, 4, 7, 10, \ldots$

$$a_3 = \frac{a_0}{2 \cdot 3}$$
$$a_6 = \frac{a_3}{5 \cdot 6} = \frac{a_0}{2 \cdot 3 \cdot 5 \cdot 6}$$
$$a_9 = \frac{a_6}{8 \cdot 9} = \frac{a_0}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9}$$

These suggest the general formula

$$a_{3n} = \frac{a_0}{2 \cdot 3 \cdot 5 \cdot 6 \cdot \cdots \cdot (3n-1)(3n)}, \quad n \geq 4$$

For the sequence $a_1, a_4, a_7, a_{10}, \ldots$, we set $n = 2, 5, 8, 11, \ldots$ in the recurrence relation and obtain

$$a_4 = \frac{a_1}{3 \cdot 4}$$
$$a_7 = \frac{a_4}{6 \cdot 7} = \frac{a_1}{3 \cdot 4 \cdot 6 \cdot 7}$$
$$a_{10} = \frac{a_7}{9 \cdot 10} = \frac{a_1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10}$$

In general, we have

$$a_{3n+1} = \frac{a_1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot \cdots \cdot (3n)(3n+1)}, \quad n \geq 4$$

Thus we have the general solution

$$\begin{aligned} y(x) = a_0 &\left[1 + \frac{x^3}{2 \cdot 3} + \frac{x^6}{2 \cdot 3 \cdot 5 \cdot 6} + \cdots + \frac{x^{3n}}{2 \cdot 3 \cdot \cdots \cdot (3n-1)(3n)} + \ldots\right] \\ + a_1 &\left[x + \frac{x^4}{3 \cdot 4} + \frac{x^7}{3 \cdot 4 \cdot 6 \cdot 7} + \cdots + \frac{x^{3n+1}}{3 \cdot 4 \cdot \cdots \cdot (3n)(3n+1)} + \ldots\right] \end{aligned}$$

Or in other terms

$$y(x) = a_0 \left[1 + \sum_{n=1}^{\infty} \frac{x^{3n}}{2 \cdot 3 \cdot \cdots \cdot (3n-1)(3n)}\right] + a_1 \left[x + \sum_{n=1}^{\infty} \frac{x^{3n+1}}{3 \cdot 4 \cdot \cdots \cdot (3n)(3n+1)}\right]$$

In this case a_0 and a_1 are arbitrary as they are determined by initial conditions. Consider the following two cases

1. $a_0 = 1, a_1 = 0 \Leftrightarrow y(0) = 1, y'(0) = 0$
2. $a_0 = 0, a_1 = 1 \Leftrightarrow y(0) = 0, y'(0) = 1$

The corresponding solutions $y_1(x), y_2(x)$ are linearly independent, since $W(y_1, y_2)(0) = 1 \neq 0$ where

$$W(y_1, y_2)(0) = \begin{vmatrix} y_1(0) & y_2(0) \\ y_1'(0) & y_2'(0) \end{vmatrix} = y_1(0)y_2'(0) - y_1'(0)y_2(0)$$

Therefore we obtain the general solution

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where $y_1(x)$ and $y_2(x)$ are the first and second bracketed expressions in the summation equation above.

Series Solutions Near an Ordinary Point II

Previously, we considered the problem of finding a solution

$$P(x)'' + Q(x)y' + R(x)y = 0 \tag{1}$$

where P, Q and R are polynomials in the neighborhood of an ordinary point x_0. We assume that the equation (1) does have a solution $y = \phi(x)$ and that has a ϕ has a Taylor series

$$\phi(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \tag{2}$$

which converges for $|x - x_0| < \rho$, where $\rho > 0$. We found that a_n can be determined by directly substituting the series (2) for y in equation (1).

Theorem 5.3.1

If x_0 is an ordinary point of the differential equation

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$$

Then the general solution for this equation is

$$y(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n = a_0y_1(x) + a_1y_2(x)$$

where a_0 and a_1 are arbitrary and y_1 and y_2 are linearly independent series solutions that are analytic at x_0. The solutions y_1 and y_2 form a fundamental set of solutions. Further, the radius of convergence for each of the series solutions y_1 and y_2 is as least as large as the minimum of the radii of convergence of the series for p and q.

To find the radius of convergence for each of the series solutions, we can use two methods

1. Find the series p and q and determine their radii of convergence using a convergence test. This is a bit troublesome, as we need to solve $y(x)$ to know the a_n. Next, we must also solve the limit for the convergence test
$$\lim_{n\to\infty}\left|\frac{a_{n+1}(x - x_0)^{n+1}}{a_n(x - x_0)^n}\right|$$
2. If P, Q, R are polynomials with no common factors, then it can be shown that Q/P and R/P are analytic at x_0, if $P(x_0) \neq 0$, and the radius of convergence of the power series for Q/P and R/P about x_0 is the distance to the nearest zero of P (including complex zeros)

Radius of Convergence Example

What is the radius of convergence for the Taylor series for $(1 + x^2)^{-1}$ about $x = 0$.

First, is there a singular point for this series? It is of the form

$$\frac{1}{1 + x^2}, \quad 1 + x^2 \geq 1$$

So there are no possible gaps (divide by zero) and we have real output. The Taylor series of f about $x_0 = 0$ is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n = 1 - x^2 + x^4 - x^6 + \cdots + (-1)^n x^{2n} + \ldots$$

Using the ratio test, we have

$$\lim_{n\to\infty}\left|\frac{a_{n+1}(x - x_0)^{n+1}}{a_n(x - x_0)^n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^{n+1}x^{2n+2}}{(-1)^n x^{2n}}\right| = \lim_{n\to\infty} x^2 < 1, \text{ for } |x| < 1$$

Thus, the radius of convergence is $\rho = 1$.

A series solution may converge for a wider range of x than indicated by Theorem 5.3.1, so the theorem only gives a lower bound on the radius of convergence of the series solution.

Legendre Equation

Determine a lower bound for the radius of convergence of the series solution about $x_0 = 0$ for the Legendre equation

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0, \quad \alpha \text{ is constant}$$

In this instance

- $P(x) = 1 - x^2$
- $Q(x) = -2x$
- $R(x) = \alpha(\alpha + 1)$

Thus $x_0 = 0$ is an ordinary point, since

$$p(x) = Q(x)/P(x) = -2x/(1 - x^2)$$

and

$$q(x) = R(x)/P(x) = \alpha(\alpha + 1)/(1 - x^2)$$

are analytic at $x_0 = 0$, because $P(0) \neq 0$. Also, p and q have singular points at $x = \pm 1$, as $P(1)$ and $P(-1)$ are both zero. Thus, the radius of convergence for the Taylor series expansions of p and q about $x_0 = 0$ is $\rho = 1$.

Solution Theory

Suppose that we have a similar equation

$$\left(1 + x^2\right) y'' + 2xy' + 4x^2 y = 0$$

and we determine that the radius of convergence for the series solution about $x_0 = 0$ is at least $\rho = 1$.

Suppose that initial conditions $y(0) = y_0$ and $y'(0) = y_0'$, are given. Since $1 + x^2 \neq 0$ for all x, there exists a unique solution on the initial value problem on $-\infty < x < \infty$ (by theorem 3.2.1)

- On the other hand, Theorem 5.3.1 only guarantees a solution of the form $\sum a_n x^n$ for $-1 < x < 1$ where $a_0 = y_0$ and $a_1 = y_0'$. If we plug in an x where $|x| \not< 1$, the series will grow to infinity
- So, the unique solution on $-\infty < x < \infty$ may not have a power series about $x_0 = 0$ that converges for all x

Radius of Convergence Example 2

Determine a lower bound for the radius of convergence of the series solution about $x_0 = 0$ for the equation

$$y'' + (\sin x)y' + (1 + x^2)y = 0$$

- $P(x) = 1$
- $Q(x) = \sin x$
- $R(x) = 1 + x^2$

Note that $p(x) = \sin x$ is not a polynomial, but recall that it does have a Taylor series about $x_0 = 0$

that converges $\forall\, x$.

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Similarly, $q(x) = 1 + x^2$ has a Taylor series about $x_0 = 0$ (namely $1 + x^2$, any polynomial has a power series which is itself) which converges $\forall\, x$.

$$a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

Where $a_0 = a_2 = 1$ and the remaining a_i are zero. Therefore, by Theorem 5.3.1, the radius of convergence for the series solution about $x_0 = 0$ is infinite.

Euler Equations

A relatively simple differential equation that has a regular singular point is the *Euler equation*:

$$L[y] = x^2 y'' + \alpha x y' + \beta y = 0, \quad \alpha, \beta \text{ are constants}$$

Note that x_0 is a regular point since

$$\begin{aligned} P(x) &= x^2 \\ Q(x) &= \alpha x \\ R(x) &= \beta \\ p(x) &= \alpha x / x^2 \\ q(x) &= \beta / x^2 \end{aligned}$$

The solution of the Euler equation is typical of the solutions of all differential equations with regular singular points, so we can examine Euler equations before discussing more general problems. In many interval not containing the origin, the general solution of the Euler equation has the form

$$y(x) = c_1 y_1 + c_2 y_2$$

Suppose that $x \in (0, \infty)$ and assume a solution of the form $y = x^r$, then

$$y = x^r, \quad y' = r x^{r-1}, \quad y'' = r(r-1) x^{r-2}$$

Substituting these into the differential equation, we obtain

$$\begin{aligned} L[x^r] &= r(r-1)x^r + \alpha r x^r + \beta x^r = 0 \\ L[x^r] &= x^r [r(r-1) + \alpha r + \beta] = 0 \end{aligned}$$

If r is a root of the quadratic equation

$$F(r) = r(r-1) + \alpha r + \beta = 0$$

then $L[x^r]$ is zero, and $y = x^r$ is a solution of the Euler equation. The roots of $F(r)$ are

$$r_1, r_2 = \frac{(-\alpha - 1) \pm \sqrt{(\alpha - 1)^2 - 4\beta}}{2}$$

Note that the quadratic polynomial $F(r)$ can also be written as

$$F(r) = (r - r_1)(r - r_2)$$

Similarly to chapter three, we again consider the cases in which roots are real and different, real and equal and complex conjugates.

Real and Distinct Roots

If $F(r) = 0$ has real roots r_1 and r_2, with $r_1 \neq r_2$, then

$$y_1(x) = x^{r_1}, y_2(x) = x^{r_2}$$

are solutions of the Euler equation. Since

$$\begin{aligned} W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} x^{r_1} & x^{r_2} \\ r_1 x^{r_1-1} & r_2 x^{r_2-1} \end{vmatrix} \\ &= r_2 x^{r_1+r_2-1} - r_1 x^{r_1+r_2-1} \\ &= (r_2 - r_1)\, x^{r_1+r_2-1} \neq 0 \text{ for all } x > 0 \end{aligned}$$

is nonzero for $r_1 \neq r_2$ and $x > 0$. Therefore, y_1 and y_2 are linearly independent, and the general solution to our differential equation is

$$y(x) = c_1 x^{r_1} + c_2 x^{r_2}, \quad x > 0$$

Real and Distinct Roots Example

Consider the equation

$$2x^2 y'' + 3xy' - y = 0, \quad x > 0$$

Substituting $y = x^r$ into this equation, we obtain

$$y = x^r, \quad y' = r x^{r-1}, \quad y'' = r(r-1)x^{r-2}$$

and

$$\begin{aligned} 2r(r-1)x^r + 3rx^r - x^r &= 0 \\ x^r[2r(r-1) + 3r - 1] &= 0 \\ x^r\left[2r^2 + r - 1\right] &= 0 \\ x^r(2r-1)(r+1) &= 0 \end{aligned}$$

Thus, $r_1 = \frac{1}{2}$ and $r_2 = -1$, and we obtain the general solution

$$y(x) = c_1 x^{1/2} + c_2 x^{-1}, \quad x > 0$$

Equal Roots

If $F(r)$ has equal roots $r_1 = r_2$, then we obtain only one solution

$$y_1(x) = x^{r_1}$$

A second solution could be obtained through reduction of order, but instead we will consider an alternate method. As $r_1 = r_2$, it follows that $F(r) = (r - r_1)^2$. Therefore, not only does $F(r_1) = 0$, but $F'(r_1) = 0$ as well. This suggests differentiating $L[x^r]$ with respect to r and then setting r equal to r_1

$$\begin{aligned} L[x^r] &= x^r[r^2 + (\alpha - 1)r + \beta] \\ &= x^r(r - r_1)^2 \\ \frac{\partial}{\partial r} L[x^r] &= \frac{\partial}{\partial r} x^r (r - r_1)^2 \\ &= x^r \ln x (r - r_1)^2 + 2(r - r_1)x^r \end{aligned}$$

By interchanging differentiation with respect to x and with respect to r, we also obtain

$$\frac{\partial}{\partial r} L[x^r] = L\left[\frac{\partial}{\partial r} x^r\right] = L\left[x^r \ln x\right]$$

As the right hand side of our first differentiation result is zero for $r = r_1$, it follows that $L\left[x^{r_1} \ln x\right] = 0$ as well. Therefore, we obtain the second solution

$$y_2(x) = x^{r_1} \ln x, \quad x > 0$$

To determine if our solutions are linearly independent, we again take the Wronskian

$$\begin{aligned} W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} x^{r_1} & x^{r_1} \ln x \\ r_1 x^{r_1 - 1} & x^{r_1 - 1}(r_1 \ln x + 1) \end{vmatrix} \\ &= x^{2r_1 - 1}(r_1 \ln x + 1) - r_1 x^{2r_1 - 1} \ln x \\ &= x^{2r_1 - 1} \neq 0 \, \forall x > 0 \end{aligned}$$

Therefore, the general solution to our Euler equation for repeated roots is

$$y(x) = c_1 x^{r_1} + c_2 x^{r_1} \ln x = (c_1 + c_2 \ln x) x^{r_1}, \quad x > 0$$

Repeated Roots Example

Consider the equation

$$x^2 y'' + 5xy' + 4y = 0, x > 0$$

In this instance $\alpha = 5$ and $\beta = 4$. Then

$$y = x^r, \quad y' = r x^{r-1}, \quad y'' = r(r-1) x^{r-2}$$

and

$$\begin{aligned} r(r-1)x^r + 5rx^r + 4x^r &= 0 \\ x^r[r(r-1) + 5r + 4] &= 0 \\ x^r\left[r^2 + 4r + 4\right] &= 0 \\ x^r(r+2)^2 &= 0 \end{aligned}$$

So we have repeated roots $r_1 = r_2 = -2$. Our general solution is

$$y(x) = (c_1 + c_2 \ln x) x^{r_1}, \quad x > 0$$

Complex Roots

Suppose that $F(r)$ has complex roots

$$\begin{aligned} r_1 &= \lambda + i\mu \\ r_2 &= \lambda - i\mu \end{aligned}$$

with $\mu \neq 0$. Remembering that

$$x^r = e^{r \ln x}$$

when $x > 0$ and r is real, we can use this equation to define x^r when r is complex. We already know Euler's formula from chapter three:

$$\begin{aligned} x^r &= e^{\ln x^r} = e^{r \ln x} = e^{(\lambda + i\mu) \ln x} = e^{\lambda \ln x} e^{i\mu \ln x} \\ &= e^{\ln x^\lambda} e^{i\mu \ln x} = x^\lambda \left[\cos(\mu \ln x) + i \sin(\mu \ln x)\right], \quad x > 0 \end{aligned}$$

Thus, x_r has been defined for complex r, and it can be shown that the general solution to the differential equation has the form

$$y(x) = c_1 x^{\lambda + i\mu} + c_2 x^{\lambda - i\mu}, \quad x > 0$$

However, these solutions are complex-valued. It is possible to show that the following real-valued functions are solutions as well (similar to what we did with the characteristic equation)

$$y_1(x) = x^\lambda \cos(\mu \ln x), \quad y_2(x) = x^\lambda \sin(\mu \ln x)$$

Taking the Wronskian, it can be shown that the solutions are linearly independent, so the general solution can be written as

$$y(x) = c_1 x^\lambda \cos(\mu \ln x) + c_2 x^\lambda \sin(\mu \ln x), \quad x > 0$$

Complex Roots Example

Solve

$$x^2 y'' + x y' + y = 0$$

Substituting $y = x^r$ in the equation gives

$$x^r(r(r-1) + r + 1) = x^r(r^2 + 1) = 0$$

Hence $r = \pm i$ and the general solution is

$$y = c_1 \cos(\ln x) + c_2 \sin(\ln x), \quad x > 0$$

Note that the factor x^λ does not appear as in this example $\lambda = 0$ and $x^\lambda = 1$.

Series Solutions Near a Regular Singular Point

We will now consider solving the general second order linear equation

$$P(x)y'' + Q(x)y' + R(x)y = 0 \tag{1}$$

in the neighborhood of a **regular singular point** x_0. A point $x_0 = 0$ is a regular single point of (1) if and only if

$$x\frac{Q(x)}{P(x)} = xp(x) \text{ and } x^2\frac{R(x)}{P(x)} = x^2 q(x)$$

are analytical at $x = 0$ and the series

$$xp(x) = \sum_{n=0}^{\infty} p_n x^n \text{ and } x^2 q(x) = \sum_{n=0}^{\infty} q_n x^n$$

are convergent on some interval $|x| < \rho$, where $\rho > 0$.

Transforming the Differential Equation

To make the quantities $xp(x)$ and $x^2 q(x)$ appear in equation (1), it's generally convenient to divide (1) by $P(x)$ and multiply by x^2

$$P(x)y'' + Q(x)y' + R(x)y = 0$$
$$x^2 y'' + x\left[xp(x)\right]y' + \left[x^2 q(x)\right]y = 0$$

Substituting in the power series representations of p and q (as both $xp(x)$ and $x^2q(x)$ no longer have singular points), we obtain the equation

$$x^2y'' + x\left(p_0 + p_1x + p_2x^2 + \dots\right)y' + \left(q_0 + q_1x + q_2x^2 + \dots\right)y = 0 \tag{2}$$

We generally write the equation in infinite series form when we do not care about convergence (for example, solving recurrence relations with generating functions).

Comparison with Euler Equations

Note that the first terms of $xp(x)$ and $x^2(x)$ are

$$p_0 = \lim_{x\to 0}\frac{xQ(x)}{P(x)} \text{ and } q_0 = \lim_{x\to 0}\frac{x^2R(x)}{P(x)}$$

If all other coefficients p_n and q_n for $n \geq 1$ are zero, then equation (2) reduces to the Euler equation which was previously discussed

$$x^2y''p_0xy' + q_0y = 0 \tag{3}$$

In general, as one would assume, some of the coefficients will not always be zero. However, essentially, for equation (2), solutions in the neighborhood of the singular point are practically identical to that of solutions of the Euler equation (3). The terms $p_1x + \dots + p_nx^n + \dots$ and $q_1x + \dots + q_nx^n + \dots$ just make the calculations increasingly complicated.

Therefore, our solution method is to assume that solutions have the form of "Euler solutions" times power series

$$y(x) = x^r(a_0 + a_1x + a_2x^2 + \dots) = x_r\sum_{n=0}^{\infty}a_nx^n\sum_{n=0}^{\infty}a_nx^{r+n} \tag{4}$$

where $a_0 \neq 0$ and $x > 0$. In other words, r is the exponent of the first nonzero term in the series, and a_0 is its coefficient. For the solution, we must determine

1. The values of r for which equation (1) has a solution of the form (4)
2. The recurrence relation for the coefficients a
3. The radius of convergence of the series $\sum_{n=0}^{\infty}a_nx^n$

This general method was constructed by Ferdinand Georg Frobenius, but is too complicated to analyze, so we will assume that any power series in an expression for a solution has a radius of convergence that is nonzero, and determinable coefficients. We can begin with an example.

Regular Single Points Example

Consider the differential equation

$$2x^2y'' - xy' + (1+x)y = 0$$

We can rewrite it as follows, dividing by $P(x) = 2x^2$ and then multiplying by x^2

$$x^2y'' - \frac{x}{2}y' + \frac{1+x}{2}y = 0$$

Since the coefficients are polynomial, if follows that $x = 0$ is a regular singular point, since both limits below are finite:

$$\lim_{x\to 0}x\left(-\frac{x}{2x^2}\right) = -\frac{1}{2} < \infty \text{ and } \lim_{x\to 0}x^2\left(\frac{1+x}{2x^2}\right) = \frac{1}{2} < \infty$$

Now $xp(x) = -\frac{1}{2}$ and $x^2q(x) = (1+x)/2$ and thus for

$$xp(x) = \sum_{n=0}^{\infty} p_n x^n, \quad x^2q(x) = \sum_{n=0}^{\infty} q_n x^n$$

it follows that

$$p_0 = -1/2, q_0 = 1/2, q_1 = 1/2, p_1 = p = \cdots = q_2 = q_3 = \cdots = 0$$

As we know that when $x = 0$, $\lim_{x\to 0} xp(x) = -\frac{1}{2}$, and all the remaining terms are zero. The same applies for $x^2q(x)$. Thus, the corresponding Euler Equation is

$$2x^2y'' - xy' + y = 0$$

To solve this, we assume that there is a solution of the form (4). Then, y' and y'' are given by

$$y' = \sum_{n=0}^{\infty} a_n(r+n)x^{r+n-2}$$

and

$$y'' = \sum_{n=0}^{\infty} a_n(r+n)(r+n-1)x^{r+n-2}$$

Substituting these into our original equation, we obtain

$$\begin{aligned} 2x^2y'' - xy' + (1+x)y = &\sum_{n=0}^{\infty} 2a_n(r+n)(r+n-1)x^{r+n} \\ &- \sum_{n=0}^{\infty} a_n(r+n)x^{r+n} + \sum_{n=0}^{\infty} a_n x^{r+n} + \sum_{n=0}^{\infty} a_n x^{r+n+1} \end{aligned}$$

The last term can be rewritten as $\sum_{n=1}^{\infty} a_{n-1}x^{r+n}$, so combining the terms, we obtain

$$\begin{aligned} 2x^2y'' - xy' + (1+x)y = &a_0[2r(r-1) - r + 1]x^r \\ &+ \sum_{n=1}^{\infty} \left((2(r+n)(r+n-1) - (r+n) + 1)a_n + a_{n-1}\right) x^{r+n} = 0 \end{aligned}$$

If this equation is to be satisfied for all x, the coefficient of each power of x must be zero. From the coefficient of x^r, we obtain (since $a_0 \neq 0$):

$$2r(r-1) - r + 1 = 2r^2 - 3r + 1 = (r-1)2r - 1) = 0$$

This equation is known as the **indicial equation** for our differential equation. It is exactly the polynomial equation we would obtain for the Euler equation associated with the differential equation. The roots are

$$r_1 = 1, \quad r_2 = \frac{1}{2}$$

These values of r are known as the **exponents at the singularity** for the regular singular point $x = 0$. They determine the qualitative behavior of the solution in the neighborhood of the singular point. Continuing, we can set the coefficient of x^{r+n} equal to zero, which gives the relation

$$(2(r+n)(r+n-1) - (r+n) + 1)a_n + a_{n-1} = 0, \quad n \geq 1$$

or in other terms

$$\begin{aligned} a_n &= -\frac{a_{n-1}}{2(r+n)^2 - 3(r+n) + 1} \\ &= \frac{a_{n-1}}{((r+n)-1)(2(r+n)-1)}, \quad n \geq 1 \end{aligned}$$

For each root of the indicial equation, we use the recurrence relation we determined above to determine a set of coefficients $a_1, a_2, \ldots$. For the root $r_1 = 1$, we obtain

$$a_n = -\frac{a_{n-1}}{(2n+1)n}, \quad n \geq 1$$

Thus, we try calculating some coefficients to determine a pattern

$$\begin{aligned} a_1 &= -\frac{a_0}{3 \cdot 1} \\ a_2 &= -\frac{a_1}{5 \cdot 2} = \frac{a_0}{(3 \cdot 5)(1 \cdot 2} \\ a_3 &= -\frac{a_2}{7 \cdot 3} = \frac{a_0}{(3 \cdot 5 \cdot 7)(1 \cdot 2 \cdot 3)} \end{aligned}$$

In general, it appears to be the case that we have the general relation

$$a_n = \frac{(-1)^n}{(3 \cdot 5 \cdot 7 \cdot \cdots \cdot (2n+1))n!} a_0, \quad n \geq 4$$

However, it we multiply both the numerator and denominator by $2 \cdot 4 \cdot 6 \cdot \cdots \cdot 2n = 2^n n!$, we can rewrite a_n as

$$a_n = \frac{(-1)^n 2^n}{(2n+1)!} a_0, \quad n \geq 1$$

Omitting the constant multiplier a_0, one solution to our differential equation is

$$y_1(x) = x \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{(2n+1)!} x^n \right), \quad x > 0$$

We can use the ratio test to find the radius of convergence

$$\lim_{n \to \infty} \left| \frac{a_{n+1} x^{n+1}}{a_n x^n} \right| = \lim_{n \to \infty} \frac{2|x|}{(2n+2)(2n+3)} = 0 \; \forall \, x$$

Thus, the series converges for all x.

We can do the same for the second root $r_2 = \frac{1}{2}$. From the recurrence relation, we obtain

$$a_n = -\frac{a_{n-1}}{2n\left(n - \frac{1}{2}\right)} = -\frac{a_{n-1}}{n(2n-1)}, \quad n \geq 1$$

Again, we can calculate some coefficient terms to see if there are any patterns:

$$\begin{aligned} a_1 &= -\frac{a_0}{1 \cdot 1} \\ a_2 &= -\frac{a_1}{2 \cdot 3} = \frac{a_0}{(1 \cdot 2)(1 \cdot 3)} \\ a_3 &= -\frac{a_2}{3 \cdot 5} = -\frac{a_0}{(1 \cdot 2 \cdot 3)(1 \cdot 3 \cdot 5)} \end{aligned}$$

So we see that in general:

$$a_n = \frac{(-1)^n}{n!(1 \cdot 3 \cdot 5 \cdots (2n-1))} a_0, \quad n \geq 4$$

Similar to the first root, we can multiply the numerator and denominator by $2^n n!$ and obtain

$$a_n = \frac{(-1)^n 2^n}{(2n)!} a_0, \quad n \geq 1$$

Again omitting the constant multiplier a_0, we obtain a second solution

$$y_2(x) = x^{1/2} \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{(2n)!} x^n \right), \quad x > 0$$

We can show that the series also converges for all x. Since y_1 and y_2 behave essentially like x and $x^{1/2}$, respectively, near $x = 0$, they are linearly independent and form a fundamental set of solutions. Hence, we obtain the general solution

$$y = c_1 y_1(x) + c_2 y_2(x), \quad x > 0$$

7 Systems of First-Order Linear Equations

Physical problems often involve a number of separate, yet interconnected components. In these cases, the corresponding mathematical problem consists of a *system* of two or more differential equations, which can always be written as first-order differential equations. In this chapter, we will set our sights on solving systems of first-order *linear* differential equations.

Introduction to Systems of First-Order Linear Equations

Systems of ordinary differential equations occur naturally in problems involving multiple dependant variables, each of which is a function of the same single independent variable. We will denote this independent variable by t and let $x_1, x_2, x_3, \ldots$ denote variables that are functions of t. Differentiation with respect to t is denoted by $\frac{dx_i}{dt}$ or x_i'

$$\begin{aligned} x_1' &= F(t, x_1, x_2, \ldots, x_n) \\ x_2' &= F(t, x_1, x_2, \ldots, x_n) \\ &\vdots \\ x_n' &= F(t, x_1, x_2, \ldots, x_n) \end{aligned}$$

If each F_k is a linear function of $x_1, x_2, \ldots, x_n$, then the system of equations is said to be *linear*. Otherwise it is *nonlinear*. Systems of higher order differential equations can be similarly defined.

Spring Mass System

The motion of a certain spring-mass system from Chapter 3 was described by the second order linear differential equation

$$u''(t) + \frac{1}{8}u'(t) + u(t) = 0$$

Where u is the position of a mass on a spring. This second order equation can be converted into a *system of first-order equations* by letting $x_1 = u$ and $x_2 = u'$. It follows then that:

$$\begin{aligned} x_1' &= x_2 \\ x_2' + \frac{1}{8}x_2 + x_1 &= 0 \end{aligned}$$

Therefore, x_1 and x_2 satisfy the following system of two first-order differential equations

$$\begin{aligned} x_1' &= x_2 \\ x_2' &= -x_1 - \frac{1}{8}x_2 \end{aligned}$$

n^{th} Order Ordinary Differential Equations and Linear First-Order Systems

The general equation of motion of a spring-mass system

$$mu'' + \gamma u' + ku = F(t)$$

can be transformed into a system of first-order differential equations in the same manner, by letting $x_1 = u$ and $x_2 = u'$ to obtain the system

$$\begin{aligned} x_1' &= x_2 \\ x_2' &= -\frac{k}{m}x_1 - \frac{\gamma}{m}x_2 + \frac{1}{m}F(t) \end{aligned}$$

This method can also be used to transform an arbitrary n^{th} order equation

$$y^{(n)} = F(t, y, y', y'', \ldots, y^{(n-1)}) \tag{1}$$

into a system of n first order equations, by defining

$$x_1 = y,\ x_2 = y',\ x_3 = y'',\ \ldots,\ x_n = y^{(n-1)}$$

It follows that

$$\begin{aligned} x_1' &= x_2 \\ x_2' &= x_3 \\ x_3' &= x_4 \\ &\vdots \\ x_{n-1}' &= x_n \end{aligned}$$

and from (1)

$$x_n' = F_n(t, x_1, x_2, \ldots, x_n)$$

This is a special case of the more general system

$$\begin{aligned} x_1' &= F(t, x_1, x_2, \ldots, x_n) \\ x_2' &= F(t, x_1, x_2, \ldots, x_n) \\ &\vdots \\ x_n' &= F(t, x_1, x_2, \ldots, x_n) \end{aligned} \tag{2}$$

Solutions of First Order Systems

A *solution* of the system (2) on the interval $I : \alpha < t < \beta$ consists of n functions

$$x_1 = \phi_1(t),\ x_2 = \phi_2(t),\ \ldots,\ x_n = \phi_n(t) \tag{3}$$

where each function is differentiable at all points in interval I and the system of equations (2) is satisfied at all points in interval I. Initial conditions may also be prescribed to form an initial value problem

$$x_1(t_0) = x_1^0,\ x_2(t_0) = x_2^0,\ \ldots,\ x_n(t_0) = x_n^0 \tag{4}$$

where t_0 is a specified value of t in I and $x_1^0, x_2^0, \ldots, x_n^0$ are prescribed numbers.

The following conditions on $F_1, F_2, \ldots, F_n$ are sufficient to ensure that the initial value problem has a unique solution.

Theorem 7.1.1

Let each of the n functions $F_1, \ldots, F_n$ and the n^2 first partial derivatives $\frac{\partial}{\partial x_1}F_1, \ldots, \frac{\partial}{\partial x_n}F_1, \ldots, \frac{\partial}{\partial x_1}F_n, \ldots, \frac{\partial}{\partial x_n}F_n$ be continuous in the region R of $tx_1x_2\ldots x_n$-space defined by $\alpha < t < \beta, a_1 < x_1 < \beta_1, \ldots, \alpha_n < x_n < \beta_n$, and let the point $\left(t_0, x_1^0, x_2^0, \ldots, x_n^0\right)$ be in R. Then, there is an interval $(t_0 - h, t_0 + h)$ in which there exists a unique solution

$$x_1 = \phi_1(t),\ x_2 = \phi_2(t),\ \ldots,\ x_n = \phi_n(t)$$

of the system of differential equations (2) which also satisfies the initial conditions (3).

Linear Systems of Equations

If each of the functions $F_1, \ldots, F_n$ in equations (2) are linear functions of $x_1, x_2, \ldots, x_n$, then the system of equations has the general form

$$\begin{aligned}
x_1' &= p_{11}(t)x_1 + p_{12}(t)x_2 + \cdots + p_{1n}(t)x_n + g_1(t)\\
x_2' &= p_{21}(t)x_1 + p_{22}(t)x_2 + \cdots + p_{2n}(t)x_n + g_2(t)\\
&\vdots\\
x_n' &= p_{n1}(t)x_1 + p_{n2}(t)x_2 + \cdots + p_{nn}(t)x_n + g_n(t)
\end{aligned}$$

If each of the functions $g_1(t)$ is zero for all t in the interval I, then the system is said to be *homogeneous*, otherwise it is *nonhomogeneous*. For the linear system above, the existence and uniqueness theorem is simpler and also has a strong conclusion:

Theorem 7.1.2

If the functions $p_{11}, p_{12}, \ldots, p_{nn}$ and $g_1, \ldots, g_n$ are continuous on an open interval $I : \alpha < t < \beta$, then there exists a unique solution

$$x_1 = \phi_1(t),\ x_2 = \phi_2(t),\ \ldots,\ x_n = \phi_n(t)$$

of the linear system of equations that also satisfies the initial conditions (3), where t_0 is any point in I and $x_1^0, x_2^0, \ldots, x_n^0$ are any prescribed numbers. The solution exists throughout the interval I.

Note that unlike the situation for a nonlinear system, the existence an uniqueness of the solution of a linear system are guaranteed throughout the interval in which the hypotheses are satisfied.

Matrices

For both theoretical and computational reasons, we will be using matrices to bring some of the results of matrix algebra to apply to the initial value problem for a system of linear differential equations.

A *matrix* is an $m \times n$ rectangular array of elements arranged in m rows and n columns. The element located in the ith row and j^{th} column is designated as a_{ij}

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n}\\ a_{21} & a_{22} & \ldots & a_{2n}\\ \vdots & \vdots & \ddots & \vdots\\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

Each matrix $\mathbf{A}$ has an associated matrix $\mathbf{A}^T$, which is known as the *transpose* $\mathbf{A}$ as is obtained by interchanging the rows and columns of $\mathbf{A}$. Thus, if $\mathbf{A} = (a_{ij})$, $\mathbf{A}^T = (a_{ji})$:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \Rightarrow \mathbf{A}^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}$$

Furthermore, if we replace each a_{ij} in the matrix by its complex conjugate $\overline{a_{ij}}$, we obtain the *conjugate* of $\mathbf{A}$, denoted by $\overline{\mathbf{A}}$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \Rightarrow \overline{\mathbf{A}} = \begin{pmatrix} \overline{a_{11}} & \overline{a_{12}} & \dots & \overline{a_{1n}} \\ \overline{a_{21}} & \overline{a_{22}} & \dots & \overline{a_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a_{m1}} & \overline{a_{m2}} & \dots & \overline{a_{mn}} \end{pmatrix}$$

The transpose of the conjugate matrix $\overline{\mathbf{A}}^T$ is known as the *adjoint* of $\mathbf{A}$ and is denoted as $\mathbf{A}^*$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \Rightarrow \mathbf{A}^* = \begin{pmatrix} \overline{a_{11}} & \overline{a_{21}} & \dots & \overline{a_{m1}} \\ \overline{a_{12}} & \overline{a_{22}} & \dots & \overline{a_{m2}} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a_{1n}} & \overline{a_{2n}} & \dots & \overline{a_{mn}} \end{pmatrix}$$

We are particular interested in two special kinds of matrices: *square matrices*, which have the same number of rows and columns

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

and *vectors* (or *column vectors*), which can be though of as $n \times 1$ matrices

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Matrix Properties

1. **Equality** – two $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ are said to be equal if all corresponding elements are equal – that is, if $a_{ij} = b_{ij}$ for each i and j.
2. **Zero** – the symbol $\mathbf{0}$ is used to denote a matrix (or vector), each of whose elements is zero. Dimensions are contextual.
3. **Addition** – the sum of two $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ is defined as the matrix obtained by adding corresponding elements

   $$\mathbf{A} + \mathbf{B} = (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$$

   Using this definition, it follows that matrix addition is commutative and associative:

   $$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \ \mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$$

4. **Multiplication by a Number** – the product of a matrix $\mathbf{A}$ by a real or complex number α is defined as follows
$$\alpha\mathbf{A} = \alpha(a_{ij}) = (\alpha a_{ij})$$
that is, each element of $\mathbf{A}$ is multiplied by α. Distributed laws are also satisfied for this type of multiplication
$$\alpha(\mathbf{A}+\mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}, \quad (\alpha+\beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{A}$$
This is also used for the negative of $\mathbf{A}$
$$-\mathbf{A} = (-1)\mathbf{A}$$

5. **Subtraction** – the difference $\mathbf{A}-\mathbf{B}$ of two $m \times n$ matrices is defined by
$$\mathbf{A}-\mathbf{B} = \mathbf{A} + (-\mathbf{B}$$
Thus
$$\mathbf{A}-\mathbf{B} = (a_{ij}) - (b_{ij}) = (a_{ij} - b_{ij})$$

6. **Multiplication** – the product $\mathbf{AB}$ of two matrices can occur whenever the number of columns in the first factor is the same as the number of rows in the second. If $\mathbf{A}$ and $\mathbf{B}$ are $m \times n$ and $n \times r$ matrices, respectively, then the product $\mathbf{C} = \mathbf{AB}$ is an $m \times r$ matrix. The element in the ith and j^{th} column of $\mathbf{C}$ is found by multiplying each element of the ith row of $\mathbf{A}$ by the corresponding element of the jth column of $\mathbf{B}$ and adding the resulting products. In other terms
$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$
Matrix multiplication satisfies the associative law and distributive law
$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}), \ \mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$
However, *matrix multiplication is not commutative*
$$\mathbf{AB} \neq \mathbf{BA}$$

7. **Vector Multiplication** – there are several ways to form a product of two vectors $\mathbf{x}$ and $\mathbf{y}$, each with n components. One way is to take the *dot product*
$$\mathbf{x}^T\mathbf{y} = \sum_{i=1}^{n} x_i y_i$$
Another vector product method is known as the *scalar* or *inner product* and is defined by
$$(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\overline{\mathbf{y}}$$
This is generally used on vectors containing complex numbers.

8. **Vector Length** – the *length* or *magnitude* of a $n \times 1$ vector $\mathbf{x}$ is defined as
$$\|\mathbf{x}\| = (\mathbf{x}, \mathbf{x})^{1/2} = \left[\sum_{k=1}^{n} x_k \bar{x}_k\right]^{1/2} = \left[\sum_{k=1}^{n} |x_k|^2\right]^{1/2}$$
Note that we have used the fact that if $x = a + bi$, then
$$x \cdot \bar{x} = \underbrace{(a+bi)}_{x}\underbrace{(a-bi)}_{\bar{x}} = a^2 + b^2 = |x|^2$$

9. **Orthogonality** – two $n \times 1$ vectors $\mathbf{x}$ and $\mathbf{y}$ are *orthogonal* if $(\mathbf{x}, \mathbf{y}) = 0$

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \ \mathbf{y} = \begin{pmatrix} 11 \\ -4 \\ -1 \end{pmatrix}$$
$$(\mathbf{x}, \mathbf{y}) = (1)(11)+(2)(-4) + (3)(-1) = 0$$

10. **The Identity Matrix** – the multiplicative *identity matrix* $\mathbf{I}$ is an $n \times n$ matrix given by

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

For any square matrix $\mathbf{A}$, we have

$$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$$

Note that the dimensions of $\mathbf{I}$ depend on context as well.

11. **Inverse Matrices** – an $n \times n$ matrix $\mathbf{A}$ is said to be *nonsingular* or *invertible* if there is another matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}$$

where $\mathbf{I}$ is the identity matrix. If there is such a $\mathbf{B}$, it is unique and is denoted by $\mathbf{A}^{-1}$, or alternatively, the *multiplicative inverse* of $\mathbf{A}$. Writing $\mathbf{B} = \mathbf{A}^{-1}$, we obtain

$$\mathbf{AA^{-1}} = \mathbf{A^{-1}A} = \mathbf{I}$$

Matrices that do not have an inverse are called *singular* or *noninvertible*. It turns out that $\mathbf{A}^{-1}$ exists if and only if $\det \mathbf{A} \neq 0$. The inverse matrix can be found using *row reduction* (or *Gaussian elimination* on the augmented matrix $(\mathbf{A} \mid \mathbf{I})$. For example, if

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & -1 \\ 3 & -1 & -2 \\ 2 & 2 & 3 \end{pmatrix}$$

We would augment the matrix by the identity matrix and perform row reduction

$$(\mathbf{A} \mid \mathbf{I}) = \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{-3R_1+R_2, -2R_1+R_3} \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & -3 & 1 & 0 \\ 0 & 4 & 5 & -2 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{\frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 5/2 & -3/2 & 1/2 & 0 \\ 0 & 4 & 5 & -2 & 0 & 1 \end{array}\right)$$
$$\xrightarrow{-4R_2+R_3, R_2+R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 3/2 & -1/2 & 1/2 & 0 \\ 0 & 1 & 5/2 & -3/2 & 1/2 & 0 \\ 0 & 0 & -5 & 4 & -2 & 1 \end{array}\right)$$
$$\xrightarrow{-\frac{1}{5}R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 7/10 & -1/10 & 3/10 \\ 0 & 1 & 0 & 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 & -4/5 & 2/5 & -1/5 \end{array}\right)$$

Thus we obtain the inverse

$$\mathbf{A}^{-1} = \begin{pmatrix} 7/10 & -1/10 & 3/10 \\ 1/2 & -1/2 & 1/2 \\ -4/5 & 2/5 & -1/5 \end{pmatrix}$$

Matrix Functions

There are times when we need to consider vectors or matrices whose elements are functions of a real variable t.

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_m(t) \end{pmatrix}, \mathbf{A}(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}(t) & a_{m2}(t) & \dots & a_{mn}(t) \end{pmatrix}$$

The matrix $\mathbf{A}(t)$ is said to be continuous at $t = t_0$, or on the interval $\alpha < t < \beta$ if each element of $\mathbf{A}$ is a continuous function at the given point or on the given interval. $\mathbf{A}(t)$ is said to be differentiable if each of its elements is differentiable. Its derivative is defined as

$$\frac{d\mathbf{A}}{dt} = \left(\frac{da_{ij}}{dt}\right)$$

In similar fashion, the integral of a matrix function is defined as

$$\int_a^b \mathbf{A}(t)\, dt = \left(\int_a^b a_{ij}(t)\, dt\right)$$

For example, if we had the following matrix

$$\mathbf{A}(t) = \begin{pmatrix} 3t^2 & \sin t \\ \cos t & 4 \end{pmatrix}$$

The derivative would be integral would be

$$\frac{d\mathbf{A}}{dt} = \begin{pmatrix} 6t & \cos t \\ -\sin t & 0 \end{pmatrix}, \int_0^\pi \mathbf{A}(t)\, dt = \begin{pmatrix} 2 & \pi^2/2 \\ \pi & 0 \end{pmatrix}$$

Many rules from Calculus apply in this setting

$$\begin{aligned} \frac{d(\mathbf{CA})}{dt} &= \mathbf{C}\frac{d\mathbf{A}}{dt}, \text{where } \mathbf{C} \text{ is a constant matrix} \\ \frac{d(\mathbf{A}+\mathbf{B})}{dt} &= \frac{d\mathbf{A}}{dt} + \frac{d\mathbf{B}}{dt} \\ \frac{d(\mathbf{AB})}{dt} &= \left(\frac{d\mathbf{A}}{dt}\right)\mathbf{B} + \mathbf{A}\left(\frac{d\mathbf{B}}{dt}\right) \end{aligned}$$

Systems of Linear Equations

A set of n simultaneous linear algebraic equations in n variables

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Can be expressed as the matrix equation $\mathbf{Ax} + \mathbf{b}$

$$\begin{pmatrix} a_{11} & a_{22} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

If $\mathbf{b} = \mathbf{0}$, then the system of equations is *homogeneous*, otherwise it is *heterogeneous*.

Nonsingular Solutions

In the case that the coefficient matrix $\mathbf{A}$ is nonsingular (i.e. invertible, the determinant is nonzero), then there exists a unique solution of the system for any vector $\mathbf{b}$. We can solve $\mathbf{Ax} = \mathbf{b}$ as follows:

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ \mathbf{A}^{-1}\mathbf{Ax} &= \mathbf{A}^{-1}\mathbf{b} \\ \mathbf{Ix} &= \mathbf{A}^{-1}\mathbf{b} \\ \mathbf{x} &= \mathbf{A}^{-1}\mathbf{b} \end{aligned}$$

It also follows that if our equation is homogeneous, i.e. $\mathbf{b} = \mathbf{0}$, then the only solution to $\mathbf{Ax} = \mathbf{0}$ is

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{0} = \mathbf{0}$$

Singular Solutions

If $\mathbf{A}$ is singular – i.e. $\det \mathbf{A}$ is zero – the depending on the values of $\mathbf{b}$, solutions either do not exist, or do exist but are not unique. Even the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has more than one solution besides the trivial solution $\mathbf{x} = \mathbf{0}$.

The nonhomogeneous case $\mathbf{Ax} = \mathbf{b}$ has no solution unless the vector $\mathbf{b}$ satisfies the condition

$$(\mathbf{b}, \mathbf{y}) = \mathbf{0}$$

for all vectors $\mathbf{y}$ satisfying $\mathbf{A}^*\mathbf{y} = \mathbf{0}$, where $\mathbf{A}^*$ is the adjoint of $\mathbf{A}$. In this case, $\mathbf{Ax} = \mathbf{b}$ has infinitely many solutions, each of the form $\mathbf{x} = \mathbf{x}^{(0)} + \xi$, where $\mathbf{x}^{(0)}$ is a particular solution of $\mathbf{Ax} = \mathbf{b}$ and ξ is any solution of $\mathbf{Ax} = \mathbf{0}$.

In order to solve particular systems, it is generally useful to transform the system of equations by forming the augmented matrix $(\mathbf{A} \mid \mathbf{b})$ and transforming $\mathbf{A}$ into an *upper triangular matrix* (that is, a matrix whose elements below the main diagonal are all zero).

System of Linear Equations

Solve the system of equations

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 7 \\ -x_1 + x_2 - 2x_3 &= -5 \\ 2x_1 - x_2 - x_3 &= 4 \end{aligned}$$

We obtain the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ -1 & 1 & -2 & -5 \\ 2 & -1 & -1 & 4 \end{array}\right)$$

We can then perform row reduction

a) $R_1 + R_2$ and $-2R_1 + R_3$

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & -1 & 1 & 2 \\ 0 & 3 & -7 & -10 \end{array}\right)$$

b) $-1R_2$

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 3 & -7 & -10 \end{array}\right)$$

c) $-3R_2 + R_3$

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & -4 & -4 \end{array}\right)$$

d) $-\frac{1}{4}R_3$

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

The matrix now corresponds to the following system of equations

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 7 \\ x_2 - x_3 &= -2 \\ x_3 &= 1 \end{aligned}$$

Thus we obtain

$$x_1 = 2,\ x_2 = -1,\ x_3 = 1$$

And write the solution of the system as

$$\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

As the solution is unique, we also conclude that the matrix is nonsingular.

System of Linear Equations #2

Solve the nonhomogeneous linear system of equations using row reduction

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= b_1 \\ -x_1 + x_2 - 2x_3 &= b_2 \\ 2x_1 - x_2 + 3x_3 &= b_3 \end{aligned}$$

We can form the the augmented matrix $(\mathbf{A} \mid \mathbf{b})$ and use some of the steps we did in the first example to transform the matrix more quickly

$$(\mathbf{A} \mid \mathbf{b}) = \left(\begin{array}{ccc|c} 1 & -2 & 3 & b_1 \\ -1 & 1 & -2 & b_2 \\ 2 & -1 & 3 & b_3 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & -2 & 3 & b_1 \\ 0 & 1 & -1 & -b_1 - b_2 \\ 0 & 0 & 0 & b_1 + 3b_2 + b_3 \end{array}\right)$$

We now have the system of equations

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= b_1 \\ x_2 - x_3 &= -b_1 - b_2 \\ 0 &= b_1 + 3b_2 + b_3 \end{aligned}$$

If $b_1 + 3b_2 + b_3 \neq 0$, there is no solution to the system of equations. If it is, then there are infinitely many solutions. Assume that

$$\begin{aligned} b_1 &= 2 \\ b_2 &= 1 \\ b_3 &= -5 \end{aligned}$$

The system of equations then becomes

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 2 \\ x_2 - x_3 &= -3 \\ 0 &= 0 \end{aligned}$$

So we get the solution

$$\mathbf{x} = \begin{pmatrix} -x_3 - 4 \\ x_3 - 3 \\ x_3 \end{pmatrix} \longrightarrow \mathbf{x} = x_3 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -4 \\ -3 \\ 0 \end{pmatrix}$$

Linear Independence and Independence

A set of vectors $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ is said to be *linearly dependent* if there exist scalars $c_1, c_2, \ldots, c_n$, one of which nonzero, such that

$$c_1\mathbf{x}^{(1)} + c_2\mathbf{x}^{(2)} + \cdots + c_n\mathbf{x}^{(n)} = \mathbf{0}$$

In other words, $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ are linearly dependent if there is a linear relation among them. On the other hand, if the only values of the coefficients for which the equation above is satisfied are

$$c_1 = c_2 = \cdots = c_n = 0$$

then $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ are said to be *linearly independent.*

Determining Linear Independence

Determine whether the following vectors are linearly dependent or linearly independent

$$\mathbf{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \ \mathbf{x}^{(2)} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \ \mathbf{x}^{(3)} = \begin{pmatrix} -4 \\ 1 \\ -11 \end{pmatrix}$$

In essence, we are solving the equation

$$c_1\mathbf{x}^{(1)} + c_2\mathbf{x}^{(2)} + c_3\mathbf{x}^{(3)} = \mathbf{0}$$

Plugging in our $\mathbf{x}$s, we obtain

$$c_1\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + c_2\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + c_3\begin{pmatrix} -4 \\ 1 \\ -11 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Or equivalently

$$\begin{pmatrix} 1 & 2 & -4 \\ 2 & 1 & 1 \\ -1 & 3 & -11 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This is a homogeneous system, so we know that if the array is invertible then

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

We can reduce the augmented matrix $(\mathbf{A} \mid \mathbf{b})$

$$\left(\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 2 & 1 & 1 & 0 \\ -1 & 3 & -11 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 0 & -3 & 9 & 0 \\ 0 & 5 & -15 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Now that we have a row of zeros on the bottom, it means that $\mathbf{A}$ is not invertible, or rather $\det(A) = 0$. We now have the following system

$$\begin{aligned} c_1 + 2c_2 - 4c_3 &= 0 \\ c_2 - 3c_3 &= 0 \\ 0 &= 0 \end{aligned}$$

To find a linear relation among $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}$ and $\mathbf{x}^{(3)}$, note that the second equation above can be rewritten as $c_2 = 3c_3$, so from the first we obtain $c_1 = 4c_3 - 2c_2 = -2c_3$. Therefore, we have solved for c_1 and c_2 in terms of c_3, the latter of which remaining arbitrary. If we choose $c_3 = -1$ (for convenience), then $c_1 = 2$ and $c_2 = -3$ and our linear relation becomes

$$2\mathbf{x}^{(1)} - 3\mathbf{x}^{(2)} - \mathbf{x}^{(3)} = \mathbf{0}$$

and the given vectors are linearly dependent. Alternatively, we could show that

$$\det(x_{ij}) = \begin{vmatrix} 1 & 2 & -4 \\ 2 & 1 & 1 \\ -1 & 3 & -11 \end{vmatrix} = 0$$

General Independence

In general, the columns (of rows) of $\mathbf{A}$ are linearly independent if and only if $\mathbf{A}$ is nonsingular (which is if and only if $\mathbf{A}^{(-1)}$ exists. Also, $\mathbf{A}$ is nonsingular if and only if $\det(\mathbf{A}) \neq 0$ (note that we can only take the determinant of the matrix is square, otherwise we will have use elementary row operations).

Further if $\mathbf{C} = \mathbf{AB}$, then $\det(\mathbf{C}) = \det(\mathbf{A})\det(\mathbf{B})$ — so if the columns/rows of $\mathbf{A}$ and $\mathbf{B}$ are linearly independent, then the columns/rows of $\mathbf{C}$ are as well.

Linear Dependence of Vector Functions

Now consider vector functions $\mathbf{x}^{(1)}(t), \mathbf{x}^{(2)}(t), \ldots, \mathbf{x}^{(n)}(t)$ where

$$\mathbf{x}^{(k)}(t) = \begin{pmatrix} x_1^{(k)}(t) \\ x_2^{(k)}(t) \\ \vdots \\ x_m^{(k)}(t) \end{pmatrix}, \quad k = 1, 2, \ldots, n, \quad t \in I = (\alpha, \beta)$$

As before, $\mathbf{x}^{(1)}(t), \mathbf{x}^{(2)}(t), \ldots, \mathbf{x}^{(n)}(t)$ is *linearly dependent* on I if there exists scalars $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1\mathbf{x}^{(1)}(t) + c_2\mathbf{x}^{(2)}(t) + \cdots + c_n\mathbf{x}^{(n)}(t) = \mathbf{0}, \ \forall\, t \in I$$

Otherwise $\mathbf{x}^{(1)}(t), \mathbf{x}^{(2)}(t), \ldots, \mathbf{x}^{(n)}(t)$ is *linearly independent* on I. Note that if $\mathbf{x}^{(1)}(t), \mathbf{x}^{(2)}(t), \ldots, \mathbf{x}^{(n)}(t)$ are linearly dependent on an interval, they are linearly dependent on every single point in the interval, however the inverse is not true if they are linearly independent. They may, in fact, be linearly dependent on each point, but with different sets of constants at different points.

Eigenvalues and Eigenvectors

The equation

$$\mathbf{Ax} = \mathbf{y}$$

can be viewed as a linear transformation that maps, or transforms, $\mathbf{x}$ into a new vector $\mathbf{y}$. Vectors that are transformed into multiples of themselves are important in many applications, including solving first order linear differential equations with constant coefficients.

To find such vectors, we will set

$$\mathbf{y} = \lambda\mathbf{x}$$

where λ is a scalar proportionality factor, and seek solutions of the equation

$$\mathbf{Ax} = \lambda\mathbf{x}$$

or

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$$

The equation has a nonzero solution of we choose λ such that

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \mathbf{0}$$

or in other words, if $\mathbf{A} - \lambda\mathbf{I}$ is singular. The equation above is a polynomial one of degree n in λ, and is known as the *characteristic equation* of the matrix $\mathbf{A}$. Values of λ that satisfy the equation may either be real or complex valued, and are known as *eigenvalues* of $\mathbf{A}$. The nonzero solutions $\mathbf{x}$ that are obtained by using such a value of λ are called the *eigenvectors* corresponding to that eigenvalue.

Eigenvalues and Eigenvectors Example

Find the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}$$

The eigenvalues λ and eigenvectors $\mathbf{x}$ satisfy the equation $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$, or

$$\begin{pmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

These eigenvalues are the roots of the characteristic equation of

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{vmatrix} = \lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0$$

Therefore, we obtain the eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. To find the eigenvectors of the matrix $\mathbf{A}$, we must solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$.

For $\lambda = 2$, we have

$$\begin{pmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} 1 & -1 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Therefore, it seems to be the case that $x_1 = x_2$, as we have the condition $x_1 - x_2 = 0$. If $x_1 = c$, then $x_2 = c$ also, and the eigenvector $\mathbf{x}^{(1)}$ is

$$\mathbf{x}^{(1)} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad c \neq 0$$

So, we have an infinitely family of eigenvectors for the eigenvalue $\lambda_1 = 2$, indexed by the arbitrary constant c. We can choose a single member of the family to act as a representative of the rest, and in this example it seems simplest to let $c = 1$, so we obtain the eigenvector

$$\mathbf{x}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda = -1$, we obtain

$$\begin{pmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} 4 & -1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

So we obtain another condition $4x_1 - x_2 = 0$, therefore the corresponding eigenvector to $\lambda_2 = -1$ is

$$\mathbf{x}^{(2)} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

Note that any nonzero multiple of these eigenvectors are also eigenvectors.

When eigenvectors are determined up to an arbitrary nonzero multiplicative constant, and the constant is specified in some fashion, then the eigenvector is said to be *normalized*. Sometimes it is useful to normalize by choosing a constant so that the length of the eigenvector $\mathbf{x}$ is $\|\mathbf{x}\| = (\mathbf{x}, \mathbf{x})^{1/2} = 1$.

Algebraic and Geometric Multiplicity

Since the characteristic equation for an $n \times n$ matrix is a polynomial equation of degree n in λ, each $n \times n$ matrix has n eigenvalues $\lambda_1, \ldots, \lambda_n$ (some of which may be repeated). If an eigenvalue is repeated m times as a root for the characteristic equation, then we say its *algebraic multiplicity* is m.

Each eigenvalue has at least one associated eigenvector, although it can have other linearly independent ones. If an eigenvalue has q linearly independent associated eigenvectors, where $1 \leq q \leq m$, we say that the eigenvalue has *geometric multiplicity* q. If each eigenvalue has algebraic multiplicity 1, we say it is *simple*, and each eigenvalue has geometric multiplicity of 1 as well.

If each eigenvalue of an $n \times n$ matrix $\mathbf{A}$ is simple, then $\mathbf{A}$ has n distinct eigenvalues, and it can be shown that each corresponding eigenvector is linearly independent. However, if an eigenvalue has one or more repeated eigenvalues, then there may be fewer than n linearly independent eigenvectors, as for each repeated eigenvalue, we may have $q < m$. This can complicate things.

Eigenvalue and Eigenvector Example #2

Find the eigenvalues and eigenvectors of the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

We can choose λ such that $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$, as follows

$$\begin{aligned} \mathbf{A} &= \begin{pmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} \\ &= \lambda^3 + 3\lambda + 2 \\ &= (\lambda - 2)(\lambda + 1)^2 \end{aligned}$$

Therefore we obtain $\lambda_1 = 2, \lambda_2 = -1, \lambda_3 = -1$. Therefore 2 is a simple eigenvalue, and -1 is an eigenvalue of algebraic multiplicity 2, or a double eigenvalue.

To find the eigenvector for $\lambda_1 = 2$, we substitute it:

$$\begin{aligned} \left(\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array}\right) &\to \left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 1 & -2 & 1 & 0 \\ -2 & 1 & 1 & 0 \end{array}\right) \\ &\to \left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array}\right) \\ &\to \left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \\ &\to \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \end{aligned}$$

So now we have the system

$$\begin{aligned} 1x_1 - 1x_3 &= 0 \\ 1x_2 - 1x_3 &= 0 \\ 0x_3 &= 0 \end{aligned}$$

Solving this system yields the eigenvector

$$\mathbf{x}^{(1)} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

For $\lambda = -1$, the equation reduces immediately to the single equation

$$x_1 + x_2 + x_3 = 0$$

So we obtain the second eigenvector

$$\mathbf{x}^{(2)} = \begin{pmatrix} -x_2 - x_3 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

where x_2 and x_3 are arbitrary. In this case we can choose $\mathbf{x}^{(2)}$ and $\mathbf{x}^{(3)}$ as

$$\mathbf{x}^{(2)} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{x}^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

So the three eigenvectors of $\mathbf{A}$ are

$$\mathbf{x}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{x}^{(2)} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{x}^{(3)} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

where both $\mathbf{x}^{(2)}$ and $\mathbf{x}^{(3)}$ correspond to the double eigenvalue $\lambda = -1$. It can be shown that all $\mathbf{x}$s are linearly independent. Hence $\mathbf{A}$ is a 3×3 *symmetric matrix* ($\mathbf{A} = \mathbf{A}^{\mathbf{T}}$) with 3 real eigenvalues and 3 linearly independent eigenvectors.

Basic Theory of Systems of First Order Linear Equations

The general theory of a system of n first order linear equations

$$\begin{aligned} x_1' &= p_{11}(t)x_1 + p_{12}(t)x_2 + \cdots + p_{1n}(t)x_n + g_1(t) \\ x_2' &= p_{21}(t)x_1 + p_{22}(t)x_2 + \cdots + p_{2n}(t)x_n + g_2(t) \\ &\vdots \\ x_n' &= p_{n1}(t)x_1 + p_{n2}(t)x_2 + \cdots + p_{nn}(t)x_n + g_n(t) \end{aligned}$$

parallels that of a single n^{th} order linear equation. This system can be rewritten as $\mathbf{x}' = \mathbf{P}(t)\mathbf{x} + \mathbf{g}(t)$ where

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \ \mathbf{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix}, \ \mathbf{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \dots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \dots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \dots & p_{nn}(t) \end{pmatrix}$$

A vector

$$\mathbf{x}' = \mathbf{P}(t)\mathbf{x} + \mathbf{g}(t)$$

is said to be a solution of the equation if its components satisfy the system of equations. Assuming that $\mathbf{P}$ and $\mathbf{g}$ are continuous on the interval $I : \alpha < t < \beta$, such a solution exists by Theorem 7.1.2.

We can first consider the homogeneous equation where $\mathbf{g}(t) = \mathbf{0}$

$$\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$$

Regardless of order, once the homogeneous equation to a differential equation has been solved, there are several methods that can be used to solve the nonhomogeneous equation. We use the notation

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} x_{11}(t) \\ x_{21}(t) \\ \vdots \\ x_{n1}(t) \end{pmatrix}, \dots, \mathbf{x}^{(k)}(t) = \begin{pmatrix} x_{1k}(t) \\ x_{2k}(t) \\ \vdots \\ x_{nk}(t) \end{pmatrix}, \dots$$

to designate specific solutions of the system.

Theorem 7.4.1

If the vector functions $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are solutions of the system $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$, then the linear combination

$$c_1\mathbf{x}^{(1)} + c_2\mathbf{x}^{(2)}$$

is also a solution for any constants c_1 and c_2. We can repeatedly apply the result of this theorem to see that every finite linear combination of solutions is itself a solution to $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$.

If $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}$ are linearly independent on I, then $\det(\mathbf{X}(t)) \neq 0$ on I where

$$\mathbf{X}(t) = \begin{pmatrix} x_{11}(t) & \dots & x_{1n}(t) \\ \vdots & \ddots & \vdots \\ x_{n1}(t) & \dots & x_{nn}(t) \end{pmatrix}$$

The Wronskian of $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}$ is defined as

$$W\left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}\right] = \det(\mathbf{X}(t))$$

It follows that $W\left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}\right](t) \neq 0$ on I if $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}$ are linearly independent for each point in I.

Theorem 7.4.3 – Abel's Theorem

If $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}$ are solutions of the system $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$ on $I : \alpha < t < \beta$, then the Wronskian $W\left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}\right](t)$ is either identically zero on I, or else is *never zero* on I.

This result relies on *Abel's Formula* for the Wronskian

$$\frac{dW}{dt} = \underbrace{(p_{11} + p_{22} + \cdots + p_{nm})}_{\text{diagonal sum of } p_{kk}\text{s}} \Rightarrow W(t) = ce^{\int [p_{11}(t)+p_{22}(t)+\cdots+p_{nn}(t)]\,dt}$$

where c is an arbitrary constant. This result allows us to determine whether a given set of solutions $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ are fundamental solutions, through the evaluation of $W\left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}\right](t)$ for *any* point t in $\alpha < t < \beta$.

The significance of Abel's Theorem is that it states it is possible to take the Wronskian of $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ a single, convenient point to determine linear dependence, as opposed to all points in the interval.

Theorem 7.4.4

Let

$$\mathbf{e}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0, \end{pmatrix}, \mathbf{e}^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0, \end{pmatrix}, \ldots, \mathbf{e}^{(n)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1, \end{pmatrix}$$

where each $\mathbf{e}^k$ is independent, and let $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ be the solutions of the system $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$, on $I : \alpha < t < \beta$, which satisfy the initial conditions

$$\mathbf{x}^{(1)}(t_0) = \mathbf{e}^{(1)}, \ldots, \mathbf{x}^{(n)}(t_0) = \mathbf{e}^{(n)}$$

respectively, where t_0 is any point in the interval $\alpha < t < \beta$. Then $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(n)}$ form a fundamental set of solutions of $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$

The theorem above states that a system of equations always has at least one fundamental set of solutions. We will also make use of one final theorem.

Theorem 7.4.5

Consider the system

$$\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$$

where each element of $\mathbf{P}$ is a real-valued, continuous function. If $\mathbf{x} = \mathbf{u}(t) + i\mathbf{v}(t)$ is a complex-valued solution of the system, its real part $\mathbf{u}(t)$ and imaginary part $\mathbf{v}(t)$ are also solutions of this equation.

To recap this segment

1. Any set of n linear independent solutions of the system $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$ constitutes a fundamental set of solutions
2. Under the conditions given in this section, such fundamental sets always exist
3. Every solution of the system $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$ can be expressed as a linear combination of any fundamental set of solutions

Homogeneous Linear Systems with Constant Coefficients

We will now consider a homogeneous system of n first order linear equations with constant, real coefficients

$$\begin{aligned} x_1' &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ x_2' &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ &\vdots \\ x_n' &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{aligned}$$

This system can be rewritten as $\mathbf{x}' = \mathbf{A}\mathbf{x}$, where

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_m(t) \end{pmatrix}, \ \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}$$

Note that this is homogeneous as $g(t)$ terms are equal to zero, and thus do not appear. Note that if $n = 1$, the system reduces to a single first order-equation

$$x' = ax$$

and the solution is $x(t) = ce^{at}$. In this instance, the only critical point when $a \neq 0$ is $x = 0$. Furthermore, $x = 0$ is an *asymptotically stable* solution if $a < 0$, since other solutions approach $x = 0$ in this situation. On the other hand, it is an unstable solution if $a > 0$, since other solutions depart from $x = 0$.

When $n = 2$, the system reduces to

$$\begin{aligned} x_1' &= a_{11}x_1 + a_{12}x_2 \\ x_2' &= a_{21}x_1 + a_{22}x_2 \end{aligned}$$

This is a system that lends itself to visualization in the x_1x_2-plane, also known as the *phase plane*. By evaluating $\mathbf{A}\mathbf{x}$ at various points and determining vector trajectories, we can plot something known as a *phase portrait*, which can help us understand the behavior of the solutions.

For systems of n linear equations, the solution is a little more involved. For $n > 1$, equilibrium solutions are similarly found by solving $\mathbf{A}\mathbf{x} = \mathbf{0}$. We will assume that $\det \mathbf{A} \neq 0$ so that $\mathbf{x} = \mathbf{0}$. Determining whether or not $\mathbf{x} = \mathbf{0}$ is asymptotically stable is an important step as well, as it allows us to determine the long term behavior of the function.

Solving Homogeneous Systems

To construct a general solution to $\mathbf{x}' = \mathbf{A}\mathbf{x}$, we assume a solution of the form

$$\mathbf{x} = \xi e^{rt}$$

where r and the constant vector ξ are to be determined. If we substitute into $\mathbf{x}' = \mathbf{A}\mathbf{x}$, we obtain

$$\begin{aligned} r\xi e^{rt} &= \mathbf{A}\xi e^{rt} \\ r\xi &= \mathbf{A}\xi \\ (\mathbf{A} - r\mathbf{I})\xi &= \mathbf{0} \end{aligned}$$

Recall that the equation of the form$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ has eigenvalues λ and eigenvectors $\mathbf{x}$. So, we must find the eigenvalues and eigenvectors of $\mathbf{A}$ in order to solve the homogeneous system of differential equations. $\mathbf{x} = \boldsymbol{\xi} e^{rt}$ is a solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$ given that r is an eigenvalue and ξ is an eigenvector of the coefficient matrix $\mathbf{A}$.

Homogeneous System Example

Consider the homogeneous equation

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}$$

Substituting $\mathbf{x} = \boldsymbol{\xi} e^{rt}$ in for $\mathbf{x}$ and rewriting system as $(\mathbf{A} - r\mathbf{I})\boldsymbol{\xi} = \mathbf{0}$, we obtain

$$\begin{pmatrix} 1-r & 1 \\ 4 & 1-r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

How can we find r? The same way we find eigenvalues — take the determinant.

$$\begin{aligned} \det(\mathbf{A} - r\mathbf{I}) &= (1-r)^2 - 4 \\ &= 1^2 - 2r + r^2 - 4 \\ &= r^2 - 2r - 3 \\ &= (r-3)(r+1) \end{aligned}$$

Therefore, $r_1 = 3$ and $r_2 = -1$. For $r_1 = 3$, we have the following

$$(\mathbf{A} - r\mathbf{I})\boldsymbol{\xi} = \mathbf{0}$$
$$\begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Therefore, $x_2 = 2\xi_1$ and we obtain the eigenvector

$$\boldsymbol{\xi}^{(1)} = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \xi_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} \Rightarrow \boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

For $r_2 = -1$, we have

$$(\mathbf{A} - r\mathbf{I})\boldsymbol{\xi} = \mathbf{0}$$
$$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

So we have it that $x_2 = -2\xi_1$ and we obtain the eigenvector

$$\boldsymbol{\xi}^{(2)} = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \xi_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} \Rightarrow \boldsymbol{\xi}^{(2)} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

The corresponding solutions $\mathbf{x} = \boldsymbol{\xi} e^{rt}$ of $\mathbf{x}' = \mathbf{A}\mathbf{x}$ are

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}, \mathbf{x}^{(2)}(t) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$$

We can take the Wronskian still to determine linear independence

$$W\left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}\right](t) = \begin{vmatrix} e^{3t} & e^{-t} \\ 2e^{3t} & -2e^{-t} \end{vmatrix} = -4e^{2t} \neq 0$$

Therefore, our solutions are fundamental solutions and the general solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is

$$\begin{aligned} \mathbf{x}(t) &= c_1\mathbf{x}^{(1)}(t) + c_2\mathbf{x}^{(2)}(t) \\ &= c_1\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2\begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} \end{aligned}$$

We can also consider another example in which solutions are negative.

Homogeneous System Example #2

Consider the homogeneous equation

$$\mathbf{x}' = \begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix}\mathbf{x}$$

To find solutions, we assume that $\mathbf{x} = \boldsymbol{\xi} e^{rt}$ and obtain the system

$$\begin{pmatrix} -3-r & \sqrt{2} \\ \sqrt{2} & -2-r \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Taking the determinant, we find that the eigenvalues must satisfy

$$(-3-r)(-2-r) - 2 = r^2 + 5r + 4 = (r+1)(r+4) = 0$$

Therefore $r_1 = -1$ and $r_2 = -4$. For $r_1 = -1$, the system becomes

$$\begin{pmatrix} -2 & \sqrt{2} \\ \sqrt{2} & -1 \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Hence $\xi_2 = \sqrt{2}\xi_1$ and the eigenvector $\boldsymbol{\xi}^{(1)}$ corresponding to the eigenvalue $r_1 = -1$ can be taken as

$$\boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}$$

Similarly for when $r_2 = -4$, we have $\xi_1 = -\sqrt{2}$ so the second eigenvector is

$$\boldsymbol{\xi}^{(2)} = \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix}$$

Therefore we obtain the fundamental set of solutions

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t}, \quad \mathbf{x}^{(2)}(t) = \begin{pmatrix} -2\sqrt{2} \\ 1 \end{pmatrix} e^{-4t}$$

And the general solution

$$\mathbf{x} = c_1\mathbf{x}^{(1)}(t) + c_2\mathbf{x}^{(2)} = c_1\begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t} + c_2\begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} e^{-4t}$$

Complex Eigenvalues

When solving systems of linear homogeneous equations, we can have a case where our eigenvalues (and thus eigenvectors) are complex. We can begin with an example.

Complex Eigenvalues Example

Consider the homogeneous equation

$$\mathbf{x}' = \begin{pmatrix} -1/2 & 1 \\ -1 & -1/2 \end{pmatrix} \mathbf{x}$$

Substituting $\mathbf{x} = \xi e^{rt}$ in for $\mathbf{x}$, we obtain

$$\begin{pmatrix} -1/2 - r & 1 \\ -1 & -1/2 - r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Taking the determinant, we obtain

$$r^2 + r + \frac{5}{4} = 0 \Rightarrow r = -\frac{1}{2} \pm i$$

Therefore, the eigenvalues are $r_1 = -\frac{1}{2} + i$ and $r_2 = -\frac{1}{2} - i$. Substituting in, we can find eigenvectors

$$\xi^{(1)} = \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

For the second eigenvector

$$\xi^{(2)} = \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

However, we can cheat a little bit. By the properties of conjugation, if we solve for one eigenvector, we only need to take the conjugate to compute the second eigenvector. For example

$$\xi^{(2)} = \overline{\xi^{(1)}} = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

However, most times, we still want real solutions. By Theorem 7.4.5, we can choose the real and imaginary parts of either $\mathbf{x}^{(1)}$ or $\mathbf{x}^{(2)}$ as solutions. We can rewrite using Euler's formula to obtain the following solutions $\mathbf{x} = \xi e^{rt}$ of $\mathbf{x}' = \mathbf{A}\mathbf{x}$

$$\mathbf{u}(t) = e^{-t/2} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos t - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin t \right] = e^{-t/2} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}$$
$$\mathbf{v}(t) = e^{-t/2} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin t + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos t \right] = e^{-t/2} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

The Wronskian of these solutions is

$$W[\mathbf{u}, \mathbf{v}](t) = \begin{vmatrix} e^{-t/2} \cos t & e^{-t/2} \sin t \\ -e^{-t/2} \sin t & e^{-t/2} \cos t \end{vmatrix} = e^{-t} \neq 0$$

So $\mathbf{u}(t)$ and $\mathbf{v}(t)$ are real-valued fundamental solutions of $\mathbf{x}' = \mathbf{A}\mathbf{x}$, and the general solution is

$$\mathbf{x} = c_1 \mathbf{u} + c_2 \mathbf{v}$$

Fundamental Matrices

The structure of solutions of systems of linear differential equations can also be explained through the concept of a fundamental matrix. Suppose that $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ form a fundamental set of solutions for the equation $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$. Then the following matrix

$$\mathbf{\Psi}(t) = \begin{pmatrix} x_1^{(1)}(t) & \ldots & x_1^{(n)}(t) \\ \vdots & \ddots & \vdots \\ x_n^{(1)}(t) & \ldots & x_n^{(n)}(t) \end{pmatrix}$$

whose columns are the vectors $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ is a *fundamental matrix* for the system. The matrix is nonsingular since its columns are linearly independent, which means that $\det \mathbf{\Psi} \neq 0$. Note also that since $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ are solutions of $\mathbf{x}' = \mathbf{P}(t)\mathbf{x}$, $\mathbf{\Psi}$ satisfies the matrix differential equation

$$\mathbf{\Psi} = \mathbf{P}(t)\mathbf{\Psi}$$

Solutions of an linear homogeneous system of equations can be written quite compactly using the fundamental matrix. The general solution

$$\mathbf{x} = c_1\mathbf{x}^{(1)}(t) + \cdots + c_n x^{(n)}(t)$$

can be written in terms of $\mathbf{\Psi}$ as

$$\mathbf{x} = \mathbf{\Psi}(t)\mathbf{c}$$

where $\mathbf{c}$ is a constant vector with arbitrary components $c_1, \ldots, c_n$.

Fundamental Matrix Example

Consider the homogeneous equation below

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \mathbf{x}$$

We can take the determinant of $\mathbf{A} - r\mathbf{I}$ similarly to what we have done previously

$$\begin{aligned} \det(\mathbf{A} - r\mathbf{I}) &= \begin{vmatrix} 1-r & 1 \\ 4 & 1-r \end{vmatrix} \\ &= (1-r)^2 - r \\ &= r^2 - 2r - 3 \\ &= (r-3)(r+1) \\ r &= 3, -1 \end{aligned}$$

We then find the eigenvectors/eigenvalues and obtain the following solutions for the system

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}, \ \mathbf{x}^{(2)}(t) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$$

Therefore, a fundamental matrix for the system is

$$\mathbf{\Psi}(t) = \begin{pmatrix} e^{3t} & e^{-t} \\ 2e^{3t} & -2e^{-t} \end{pmatrix}$$

For an initial value problem with the initial condition

$$\mathbf{x}(t_0) = \mathbf{x}^0$$

The constant vector $\mathbf{c}$ must satisfy

$$\mathbf{\Psi}(t_0)\mathbf{c} = \mathbf{x}^0$$

As $\mathbf{\Psi}(t_0)$ is nonsingular

$$\mathbf{c} = \mathbf{\Psi}^{-1}(t_0)\mathbf{x}^0$$

And

$$\mathbf{x} = \mathbf{\Psi}(t)\mathbf{\Psi}^{-1}(t_0)\mathbf{x}^0$$

Repeated Eigenvalues

We will again consider a homogeneous system of n first order linear equations with constant real coefficients

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

If the eigenvalues $r_1, \ldots, r_n$ of $\mathbf{A}$ are real and different, then there are n linearly independent eigenvectors $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(n)}$ and n linearly independent solutions of the form

$$\mathbf{x}^{(1)}(t) = \boldsymbol{\xi}^{(1)}e^{r_1 t}, \ldots, \mathbf{x}^{(n)}(t) = \boldsymbol{\xi}^{(n)}e^{r_n t}$$

If the algebraic multiplicity $m \geq 2$ for some eigenvalues, or in other words some of the eigenvalues are repeated, there may not be n corresponding linearly independent solutions. In that case, we will seek additional solutions. We can illustrate the process with an example.

Repeated Eigenvalues Example

Suppose we want to find the eigenvectors for the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}\mathbf{x}$$

The eigenvalues r and eigenvectors $\boldsymbol{\xi}$ satisfy the equation $(\mathbf{A} - r\mathbf{I})\boldsymbol{\xi} = \mathbf{0}$, or

$$\begin{pmatrix} 1-r & -1 \\ 1 & 3-r \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

To determine r, we take the determinant $\det(\mathbf{A} - r\mathbf{I})$

$$\begin{vmatrix} 1-r & -1 \\ 1 & 3-r \end{vmatrix} = (r-1)(r-3) + 1 = r^2 - 4r + 4 = (r-2)^2$$

Therefore we have the repeated roots $r_1, r_2 = 2, 2$. To find eigenvectors, we solve $(\mathbf{A} - r\mathbf{I})\boldsymbol{\xi} = 0$

$$\begin{pmatrix} 1-r & -1 \\ 1 & 3-r \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We then row reduce the augmented matrix

$$\begin{pmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow{-1R_1} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\xrightarrow{-R_1+R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore

$$\begin{aligned} 1\xi_1 + 1\xi_2 &= 0 \\ 0\xi_2 &= 0 \end{aligned}$$

Therefore $\xi_1 = -\xi_2$. We then choose the simplest eigenvector from

$$\xi^{(1)} = \begin{pmatrix} -\xi_2 \\ \xi_2 \end{pmatrix}$$

which is

$$\xi^{(1)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

So we obtain one solution

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t}$$

Based on methods from chapter 3, we assume that the second solution is in the form $\mathbf{x} = \boldsymbol{\xi} t e^{2t}$. Substituting $\mathbf{x} = \boldsymbol{\xi} t e^{2t}$ into $\mathbf{x}' = \mathbf{A}\mathbf{x}$, we obtain

$$\begin{aligned} \boldsymbol{\xi} e^{2t} + 2\boldsymbol{\xi} t e^{2t} &= \mathbf{A}\boldsymbol{\xi} t e^{2t} \\ \boldsymbol{\xi} e^{2t} + 2\boldsymbol{\xi} t e^{2t} - \mathbf{A}\boldsymbol{\xi} t e^{2t} &= 0 \end{aligned}$$

In order for this equation to be satisfied for all t, it is necessary for the coefficients of te^{2t} and e^{2t} to both be zero. We see that $\boldsymbol{\xi} = \mathbf{0}$ for the e^{2t} term, and hence there is no nonzero solution of the form $\mathbf{x} = \boldsymbol{\xi} t e^{2t}$. However, since both te^{2t} and e^{2t} appear in the above equation, we next consider a solution of the form

$$\mathbf{x} = \boldsymbol{\xi} t e^{2t} + \boldsymbol{\eta} e^{2t}$$

Substituting this into $\mathbf{x}' = \mathbf{A}\mathbf{x}$, we obtain

$$\boldsymbol{\xi} e^{2t} + 2\boldsymbol{\xi} t e^{2t} + 2\boldsymbol{\eta} e^{2t} = \mathbf{A}(\boldsymbol{\xi} t e^{2t} + \boldsymbol{\eta} e^{2t})$$

Equating coefficients yields the two conditions $\mathbf{A}\boldsymbol{\xi} = 2\boldsymbol{\xi}$ and $\mathbf{A}\boldsymbol{\eta} = \boldsymbol{\xi} + 2\boldsymbol{\eta}$ or

$$(\mathbf{A} - 2\mathbf{I})\boldsymbol{\xi} = \mathbf{0}, \quad (\mathbf{A} - 2\mathbf{I})\boldsymbol{\eta} = \boldsymbol{\xi}$$

The first equation is satisfied if $\boldsymbol{\xi}$ is an eigenvector of $\mathbf{A}$ corresponding to the eigenvalue $r = 2$, so we have already got that down. To solve $(\mathbf{A} - 2\mathbf{I})\boldsymbol{\eta} = \boldsymbol{\xi}$, to find $\boldsymbol{\eta}$, we row reduce the augmented matrix from

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}, \ \boldsymbol{\xi} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Which is

$$\left(\begin{array}{cc|c} -1 & 1 & 1 \\ 1 & 1 & -1 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right)$$

So we can determine that

$$\eta_2 = -1 - \eta_1$$

Therefore

$$\boldsymbol{\eta} = \begin{pmatrix} \eta_1 \\ -1 - \eta_1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} + k \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Now we have a second solution $\mathbf{x} = \boldsymbol{\xi} t e^{2t} + \boldsymbol{\eta} e^{2t}$, which is

$$\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{2t} + k \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t}$$

Recalling that our first solution was

$$\mathbf{x}^{(1)}(t) \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t}$$

We see that our second is simply

$$\mathbf{x}^{(2)}(t) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{2t}$$

since the last term of the third term of $\mathbf{x}$ is a multiple of $\mathbf{x}^{(1)}$. Now that we have our two solutions, we can take the Wronskian to determine if they are linearly independent

$$W\left[\mathbf{x}^{(1)}, \mathbf{x}^{(2)}\right](t) = \begin{vmatrix} e^{2t} & te^{2t} \\ -e^{2t} & te^{2t} - e^{2t} \end{vmatrix} = -e^{4t} \neq 0$$

Therefore, both solutions are fundamental solutions, and the general solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t} + c_2 \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{2t} \right]$$

In terms of a general solution, suppose that the repeated eigenvalue $r = \rho$ and that there is only one corresponding eigenvector $\boldsymbol{\xi}$. One solution is

$$\mathbf{x}^{(1)}(t) = \boldsymbol{\xi} e^{\rho t}$$

where $\boldsymbol{\xi}$ satisfies

$$(\mathbf{A} - \rho \mathbf{I})\boldsymbol{\xi} = \mathbf{0}$$

Following the process illustrated in the example above, we find a second solution

$$\mathbf{x}^{(2)}(t) = \boldsymbol{\xi} t e^{\rho t} + \boldsymbol{\eta} e^{\rho t}$$

where $\boldsymbol{\eta}$ is determined from

$$(\mathbf{A} - \rho \mathbf{I})\boldsymbol{\eta} = \boldsymbol{\xi}$$

9 Nonlinear Differential Equations and Stability

There are many differential equations, especially nonlinear ones, which do not lend themselves to finding analytical solutions in a reasonably convenient manner. One way of circumventing this issue is through numerical methods. Another, which we will explore, is a geometrical approach which leads to a qualitative understanding of solutions rather than detailed quantitative information. If we had the system

$$\frac{d}{dt}\mathbf{x} = \mathbf{A}\mathbf{x}$$

we can do this by considering the eigenvectors of the system geometrically on an axis.

Solutions of Second Order Linear Systems

As a quick recap, consider a second order linear homogeneous system with constant coefficients

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

We assume $\mathbf{x} = \xi e^{rt}$ then

$$r\xi e^{rt} = \mathbf{A}\xi e^{rt} \Leftrightarrow (\mathbf{A} - r\mathbf{I})\xi = 0$$

Therefore, provided that r is an eigenvalue and ξ is an eigenvector of the coefficient matrix $\mathbf{A}$, $\mathbf{x} = \xi e^{rt}$ is a solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$. The eigenvalues are the roots of the polynomial equation $\det(\mathbf{A} - r\mathbf{I}) = 0$, and the eigenvectors ξ are determined up to an arbitrary constant from the equation $(\mathbf{A} - r\mathbf{I})\xi = 0$.

Equilibrium Solutions

Solutions $\mathbf{x}$ for which $\mathbf{A}\mathbf{x} = \mathbf{0}$ correspond to *equilibrium solutions* and are called *critical points*. We assume that $\mathbf{A}$ is nonsingular, or $\det(\mathbf{A}) \neq 0$, and hence $\mathbf{x} = \mathbf{0}$ is the only critical point for the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$

A solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is a vector function $\mathbf{x} = \phi(t)$ that satisfies the differential equation, and can be viewed as a parametric representation for a curve in the x_1x_2-plane. This curve can be thought of as a trajectory traversed by a moving particle whose velocity $\frac{d}{dt}\mathbf{x}$ is specified by the differential equation.

The x_1x_2-plane is known as the *phase plane* and the representative set of trajectories is a *phase portrait*.

Characterizing Equations by Trajectory Patterns

When analyzing the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, we must consider several cases, depending on the nature of the eigenvalues of $\mathbf{A}$[1].

[1] Similarly to what we did in sections 7.5-7.8

Real, Unequal Eigenvalues of the Same Sign

When the eigenvalues r_1 and r_2 are both positive or negative, the general solution for $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is

$$\mathbf{x} = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t}$$

Suppose that $r_1 < r_2 < 0$ and that the eigenvectors $\boldsymbol{\xi}^{(1)}$ and $\boldsymbol{\xi}^{(2)}$ are shown below.

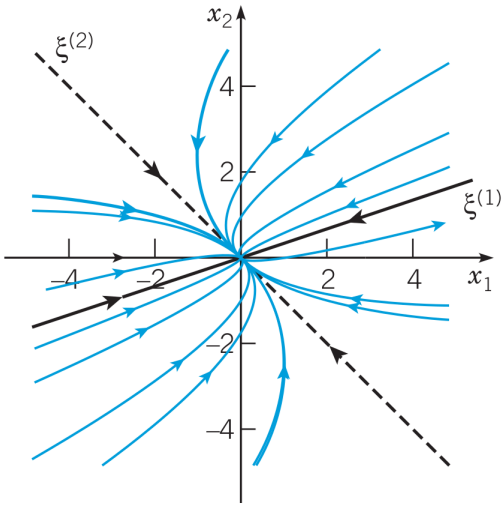


Figure 9.1: Trajectories in phase plane when origin is a node with $r_1 < r_2 < 0$. Solid black and dashed black curves are fundamental solutions $\boldsymbol{\xi}^{(1)}e^{r_1 t}$ and $\boldsymbol{\xi}^{(2)}e^{r_2 t}$

.

It follows then that $\mathbf{x} \longrightarrow \mathbf{0}$ as $t \longrightarrow \infty$, regardless of the values of c_1 and c_2, or rather, all solutions approach the critical point at the origin. This type of critical point is called a *node* or *nodal sink*. If the solution starts at an initial point on the line through $\boldsymbol{\xi}^{(1)}$, then $c_2 = 0$ and the solution remains on this line for all t. Similarly for $\boldsymbol{\xi}^{(2)}$. To make things more clear, the solution can be rewritten as

$$\mathbf{x} = e^{r_2 t}\left[c_1\boldsymbol{\xi}^{(1)}e^{(r_1-r_2)t} + c_2\boldsymbol{\xi}^{(2)}\right]$$

Since $r_1 - r_2 < 0$, the value of the $c_1\boldsymbol{\xi}^{(1)}e^{r_1-r_2}t$ is negligible compared to the $c_2\boldsymbol{\xi}^{(2)}$ term, at least when t is large. So, we see that as $t \longrightarrow \infty$, not only does the trajectory tend toward the origin, but also tends towards the line $\boldsymbol{\xi}^{(2)}$ as well. Therefore, all solutions that do not start exactly on the line through $\boldsymbol{\xi}^{(1)}$ are tangent to $\boldsymbol{\xi}^{(2)}$.

If r_1 and r_2 are both positive, and $0 < r_2 < r_1$, the trajectories have the same pattern as figure 9.1, however the direction will be away from the critical point at the origin. This is because as $t \longrightarrow \infty$, $e^{r_1 t}$ and $e^{r_2 t}$ will approach infinity as well, since our r_i's are positive. In this case, the critical point as called a *nodal source*.

Real, Unequal Eigenvalues of Opposite Sign

Suppose now that $r_1 > 0$ and $r_2 < 0$, with the general solution

$$\mathbf{x} = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t}$$

The corresponding eigenvectors $\boldsymbol{\xi}^{(1)}$ and $\boldsymbol{\xi}^{(2)}$ are shown below:

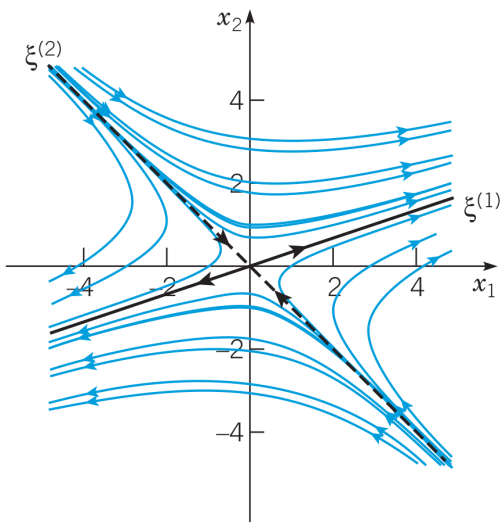


Figure 9.2: Trajectories in phase plane when origin is a saddle point with $r_1 > 0, r_2 < 0$. Solid black and dashed black curves are fundamental solutions $\boldsymbol{\xi}^{(1)}e^{r_1 t}$ and $\boldsymbol{\xi}^{(2)}e^{r_2 t}$

.

If the solution starts at an initial point on the line through $\boldsymbol{\xi}^{(1)}$, then $c_2 = 0$ and the solution remains on this line for all t. Since $r_1 > 0$, it follows that $\|\mathbf{x}\| \to \infty$ as $t \to \infty$. Similarly so, if the initial point is on the line through $\boldsymbol{\xi}^{(2)}$, then $\|\mathbf{x}\| \to 0$ as $t \to \infty$ since $r_2 < 0$. Solutions on other initial points have trajectories that eventually approach infinity due to the positive exponential term. In this case, there is no source/sink, but rather a *saddle point*. The only solutions that approach the critical point at the origin are those that start on the line determined by $\boldsymbol{\xi}^{(2)}$.

Equal Eigenvalues

Suppose now that $r_1 = r_2 = r$. We consider the case in which the repeated eigenvalue is negative. If r is positive, then the trajectories are similar but the direction of motion is reversed. There are two subcases, depending on whether r has two linearly independent eigenvectors or only one.

Two Independent Eigenvectors

If two eigenvectors $\boldsymbol{\xi}^{(1)}$ and $\boldsymbol{\xi}^{(2)}$ are linearly independent, then the general solution is

$$\mathbf{x} = c_1\boldsymbol{\xi}^{(1)}e^{rt} + c_2\boldsymbol{\xi}^{(2)}e^{rt}$$

The ratio of x_2/x_1 is independent of t, but depends on the components of $\boldsymbol{\xi}^{(1)}$ and $\boldsymbol{\xi}^{(2)}$ and on c_1 and c_2. Thus, every trajectory lies is a straight line through the origin. The critical point is known as a *proper node*, or sometimes as a *star point*. If $r < 0$, then trajectories will be pulled into zero. If $r > 0$, trajectories will be pushed out from zero.

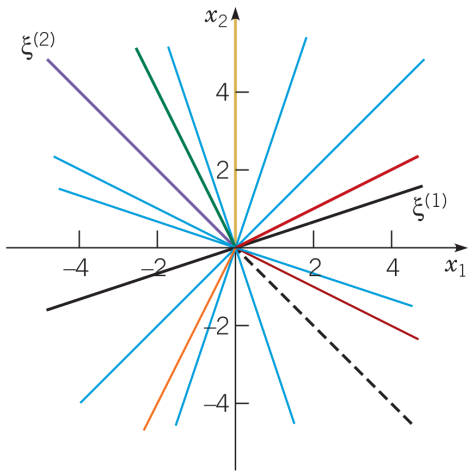


Figure 9.3: Trajectories in phase plane when origin is a proper node with $r_1 = r_2 = r$. Solid black and dashed black curves are fundamental solutions $\boldsymbol{\xi}^{(1)}e^{r_1 t}$ and $\boldsymbol{\xi}^{(2)}e^{r_2 t}$

.

One Independent Eigenvector

If the repeated eigenvalue r has only one linearly independent eigenvector $\boldsymbol{\xi}$, then the general solution is

$$\mathbf{x} = c_1\boldsymbol{\xi}e^{rt} + c_2\left(\boldsymbol{\xi}te^{rt} + \boldsymbol{\eta}e^{rt}\right)$$

For large t, the dominant term is $c_2\boldsymbol{\xi}te^{rt}$, thus every trajectory approaches the origin tangent to the line through eigenvector $\boldsymbol{\xi}$. Conversely, for large negative t the dominant term is again $c_2\boldsymbol{\xi}te^{rt}$, and hence every trajectory is asymptotic to a line parallel to the eigenvector $\boldsymbol{\xi}$. The orientation of each trajectory depends on the relative positions of $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$.

When a double eigenvalue has only one linearly independent eigenvalue, the critical point is called an *improper* or *degenerate node*. One possible situation is shown in the following figure:

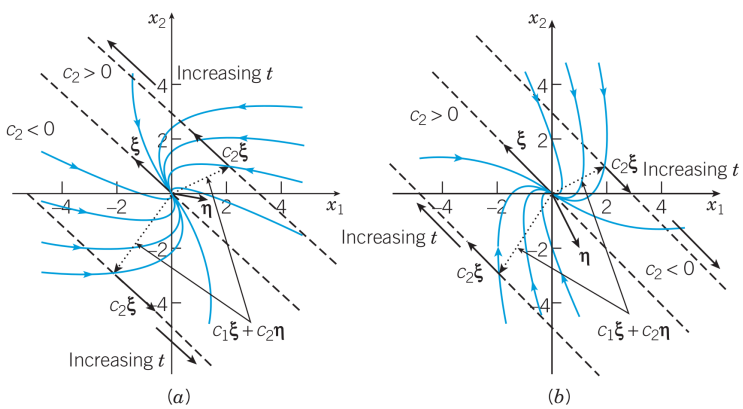


Figure 9.4: Trajectories in phase plane with different generalized eigenvector $\boldsymbol{\eta}$ when origin is an improper node with $r_1 = r_2 < 0$ and one independent eigenvector $\boldsymbol{\xi}$

.

If $r_1 = r_2 > 0$, then the same type of procedure can be followed to sketch the trajectories, except that in this case the trajectories and moving outwards, and the orientation of the trajectories with respect to $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$ are also reversed.

Complex Eigenvalues with Nonzero Real Part

Suppose the eigenvalues are $\lambda \pm i\mu$, where λ and μ are real, with $\lambda \neq 0$ and $\mu > 0$. It is possible to write down the general solution in terms of eigenvalues and eigenvectors. However, in this case, we proceed in a different way. Consider the following system of equations

$$\mathbf{x}' = \begin{pmatrix} \lambda & \mu \\ -\mu & \lambda \end{pmatrix}$$

Or in scalar form

$$\begin{aligned} x_1' &= \lambda x_1 + \mu x_2 \\ x_2' &= -\mu x_1 + \lambda x_2 \end{aligned}$$

We will introduce the polar coordinates r, θ given by

$$r^2 = x_1^2 + x_2^2, \quad \tan\theta = x_2/x_1$$

Differentiating the polar equations with respect to t, we have

$$2r\left(\frac{dr}{dt}\right) = 2x_1\left(\frac{dx_1}{dt}\right) + 2x_2\left(\frac{dx_2}{dt}\right), \quad \sec^2\theta\left(\frac{d\theta}{dt}\right) = \frac{x_1(dx_2/dt) - x_2(dx_1/dt)}{x_1^2}$$

Or in simpler terms

$$rr' = x_1x_1' + x_2x_2', \quad (\sec^2\theta)\theta' = (x_1x_2' - x_2x_1')/x_1^2$$

Substituting the values below into the equation above

$$x_1' = \lambda x_1 + \mu x_2, \quad x_2' = -\mu x_1 + \lambda x_2$$

We find that

$$r' = \lambda r, \quad \theta' = -\mu$$

Solving both of these differential equations, we obtain

$$r = ce^{\lambda t}, \quad \theta = -\mu t + \theta_0, \ \theta_0 = \theta(0)$$

These equations are parametric equations in polar coordinates of the solution trajectories to our system $\mathbf{x}' = \mathbf{A}\mathbf{x}$. Since $\mu > 0$, it follows that θ decreases as t increases, so the direction of motion on a trajectory is clockwise. If $\lambda < 0$, then $r \to 0$ as $t \to \infty$ (will curl into zero), while $r \to \infty$ if $\lambda > 0$ (will curl out from zero). In this case, the trajectories are spirals, which approach or recede from the origin – known as a *spiral point* – depending on the sign of λ.

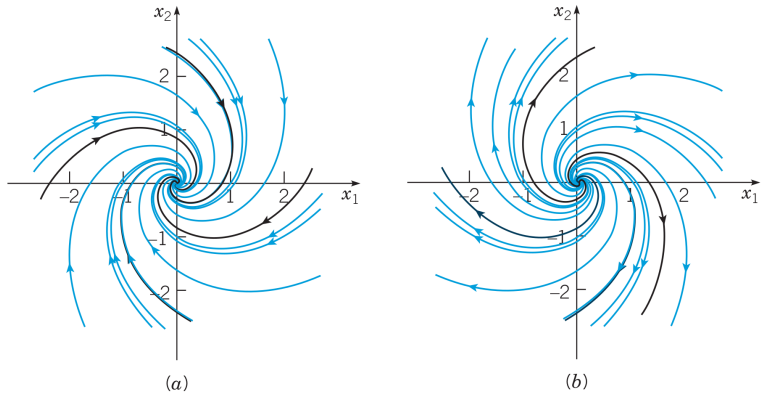


Figure 9.5: Trajectories in phase plane for a linear system with eigenvalues $\lambda \pm i\mu$. The first is a spiral sink when $\lambda < 0$, and the next is a spiral source when $\lambda > 0$.

The terms *spiral sink* and *spiral source*, are used to refer to spiral points whose trajectories approach ($\lambda < 0$) or depart from ($\lambda > 0$) the critical point.

Pure, Imaginary Eigenvalues

Suppose the eigenvalues are $\lambda \pm i\mu$ where $\lambda = 0$ and μ is real. In this case, we have the system

$$\mathbf{x}' = \begin{pmatrix} 0 & \mu \\ -\mu & 0 \end{pmatrix} \mathbf{x}$$

with eigenvalues $\pm i\mu$. Using a process similar to what we did for complex eigenvalues with a nonzero real part above, we obtain

$$r' = 0, \quad \theta' = -\mu$$

Solving these, we obtain

$$r = c, \quad \theta = -\mu t + \theta_0$$

where c and θ_0 are constants. Therefore, we have it that the trajectories are circles centered at the origin. If $\mu > 0$, they are traversed clockwise, and if if $\mu < 0$, they are traversed counterclockwise. The critical point is known as a *center* in this case.

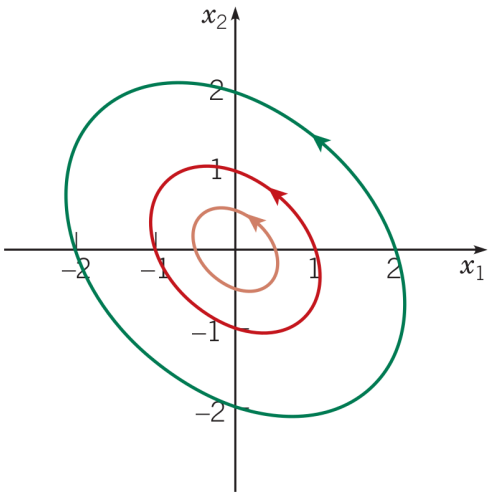


Figure 9.6: Trajectories in phase plane for a linear system with eigenvalues $\pm i\mu$.

Summary

After observing each of the cases, we can make several observations

1. As $t \longrightarrow \infty$, one of three types of behavior is exhibited: the trajectory approaches the critical point $\mathbf{x} = \mathbf{0}$, repeatedly traverses a closed curve that surrounds the critical point, or becomes unbounded.
2. Trajectories do not cross each other, even if it seems to look like many trajectories pass through the critical point, only the equilibrium solution passes through the origin. Solutions that appear to pass through the trajectory only approach the point as $t \longrightarrow \infty$ or $t \longrightarrow -\infty$.
3. In each case, the set of all trajectories is such that one of three situations occurs:
   (a) If eigenvalues are real and negative or complex with a negative real part, trajectories approach the critical point $\mathbf{x} = \mathbf{0}$ as $t \longrightarrow \infty$. The origin is either a nodal sink or a spiral sink.
   (b) If eigenvalues are pure imaginary, all trajectories remain bounded but do not approach the origin as $t \longrightarrow \infty$. The origin is a center.
   (c) If at least one of the eigenvalues is positive or there is a complex eigenvalue with a real part, possibly all trajectories except for $\mathbf{x} = \mathbf{0}$, become unbounded as $t \longrightarrow \infty$. The origin is a nodal source, a spiral source, or a saddle point.

Essentially, to summarize in table form:

Eigenvalues	**Type of Critical Point**	**Stability**
$r_1 > r_2 > 0$	Node	Unstable
$r_1 < r_2 < 0$	Node	Asymptotically stable
$r_2 < 0 < r_1$	Saddle point	Unstable
$r_1 = r_2 > 0$	Proper or improper node	Unstable
$r_1 = r_2 < 0$	Proper or improper node	Asymptotically stable
$r_1, r_2 = \lambda + i\mu$		
$\lambda > 0$	Spiral Point	Unstable
$\lambda < 0$	Spiral Point	Asymptotically stable
$\lambda = 0$	Center	Stable

Table 9.1: Stability Properties of Linear Systems $\mathbf{x} = \mathbf{A}\mathbf{x}$ with $\det(\mathbf{A} - r\mathbf{I}) = 0$ and $\det \mathbf{A} \neq 0$

Note that this only applied to second-order systems $\mathbf{x}' = \mathbf{A}\mathbf{x}$ with $\det \mathbf{A} \neq 0$ whose solutions can be visualized as curves in the phase plane.